

EL&C

2FH4

* quizzes during tutorials
↳ Q's from tutorials

Lec 2 - Vector

tally marks → 11 Q answered.

- notation : $\vec{a}$, $\bar{a}$

Unit Vector $\bar{a}_A$

- components of $\vec{A}$: $(\bar{A}_x, \bar{A}_y, \bar{A}_z)$
 $(A_x \bar{a}_x, A_y \bar{a}_y, A_z \bar{a}_z)$
 $A_x \vec{i} + A_y \vec{j} + A_z \vec{k}$

- magnitude $|\vec{A}|$

- unit $\vec{a}_A = \frac{\vec{A}}{A}$

Ex. $\vec{A} = 3\bar{a}_x - 4\bar{a}_y + 12\bar{a}_z$. magnitude & unit ?

$$|\vec{A}| = \sqrt{3^2 + 4^2 + 12^2} = 13$$

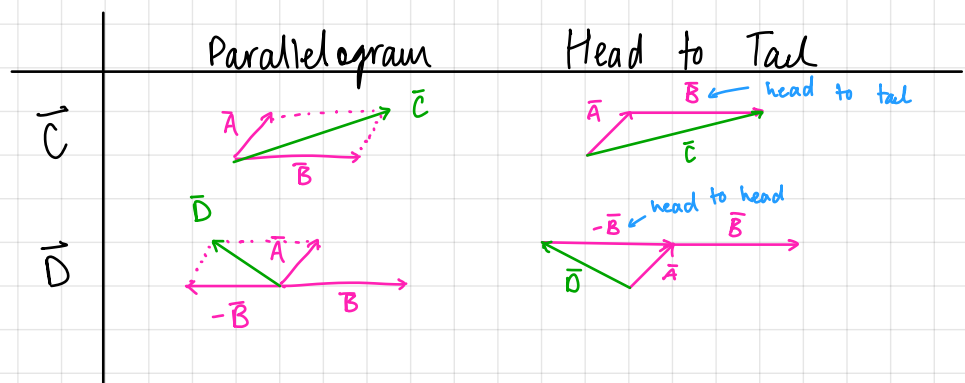
$$\vec{a}_A = \frac{3}{13} \bar{a}_x - \frac{4}{13} \bar{a}_y + \frac{12}{13} \bar{a}_z$$

$$|\vec{a}_A| = 1 = \sqrt{\frac{3^2}{13^2} + \frac{4^2}{13^2} + \frac{12^2}{13^2}}$$

Add & Subtract

- 2 methods
 - ↳ parallelogram
 - ↳ head-to-tail

- $\vec{C} = \vec{A} + \vec{B}$
- $\vec{D} = \vec{A} - \vec{B}$



- $\vec{C} = \vec{A} + \vec{B}$
 $= A_x \bar{a}_x + A_y \bar{a}_y + A_z \bar{a}_z + B_x \bar{a}_x + B_y \bar{a}_y + B_z \bar{a}_z$
 $= (A_x + B_x) \bar{a}_x + (A_y + B_y) \bar{a}_y + (A_z + B_z) \bar{a}_z$

Add & Subtract : Properties

- commutative : $\vec{A} + \vec{B} = \vec{B} + \vec{A}$
- associative : $(\vec{A} + \vec{B}) + \vec{C} = \vec{A} + (\vec{B} + \vec{C})$
- distributive : $k(\vec{A} + \vec{B}) = k\vec{A} + k\vec{B}$

Position & Displacement Vectors

- position vector - from origin $O(0,0,0)$ to $P(x,y,z)$

$$\vec{r}_P = \vec{OP} = \vec{P} - \vec{O}$$

- displacement vector - from $\vec{P}$ to $\vec{Q}$

$$\vec{r}_{PQ} = \vec{PQ} = \vec{r}_Q - \vec{r}_P$$

Ex. $P(2, -1, 2)$ from $\vec{O}$

- a) position vector? $= \vec{r}_P = 2a_x - a_y + 2a_z$
b) magnitude? $= |\vec{r}_P| = \sqrt{2^2 + 1^2 + 2^2} = \sqrt{9} = 3$
c) unit & direction? $\vec{a}_{rP} = \frac{\vec{r}_P}{|\vec{r}_P|} = \frac{2}{3}a_x - \frac{1}{3}a_y + \frac{2}{3}a_z$

Ex. $P(1, 2, 0)$
 $Q(4, -1, 2)$

$$\vec{PQ} = ?$$

$$= \vec{Q} - \vec{P}$$

$$= (4-1)a_x + (-1-2)a_y + (2-0)a_z$$

$$= 3a_x - 3a_y + 2a_z$$

Vector Multiplication

- Dot / Scalar Product
- Cross / Vector Product

Course Agenda

ELECENG 2FH4 - Electromagnetism I

Instructor: Amir Yazdanpour

Email: yazdaa8@mcmaster.ca

Today's plan

- Course overview & expectations
- Structure and organization of the course
- Big-picture view of electromagnetics
- Why vectors and mathematics are essential in EM

Course Overview & Expectations

Lecture:

Monday,	12:30 - 01:20 PM – MDCL 1105
Tuesday,	08:30 - 09:20 AM – ITB 137
Friday,	08:30 - 09:20 AM – ITB AB102

Tutorial:

Monday,	05:30 - 06:20 PM – KTH B135
Wednesday,	05:30 - 06:20 PM – KTH B135

Proposed Schedule Change (Pending Student Feedback):

The **Friday 8:30 AM** lecture may be swapped with the **Monday 5:30 PM** tutorial.

❖ *No change has been made yet.*

Midterm #1: Tuesday, February 10, 06:30 – 08:15 PM

- Three questions, covering material up to the beginning of Chapter 4 ($\approx$ p. 140)
- Two questions on vector algebra and vector calculus
- One question connecting Chapter 4 (electrostatic fields) to vectors

Midterm #2: Monday, March 2, 06:30 – 08:15 PM

- Three questions, covering Chapters 4, 5, 6, and 7 ($\approx$ p. 140–290)

Course Overview & Expectations

Assessment Breakdown

Final Exam: 50%

(Absolute grading, any bonus marks earned during the term may be applied to compensate for missed marks in other components such as quizzes or midterms)

Midterm Exams: 25%

(Two midterms: 10% each, plus an additional 5% assigned to the midterm in which you perform better)

Assignments: 15%

(Five MATLAB assignments, equally weighted)

Quizzes: 10%

(Two quizzes, equally weighted)

Bonus Opportunities (Total: up to 15%)

- ✓ **2%** course evaluation completion with $\geq 75\%$ participation
- ✓ **5%** voluntary Reading Week assignments
- ✓ **3%** the End-of-Term Technical Reflection *(details next slides)*
- ✓ **5%** Voluntary Weekend Assignments / Mini-Projects *(details next slides)*

End-of-Term Technical Reflection

Objective

Reward genuine engagement with course material through a short, structured technical contribution.

Submission Requirements

- ✓ Maximum 1 written page (figures allowed; MATLAB code excluded from page limit)
- ✓ Must include at least one of the following:
 - A MATLAB simulation
 - A numerical example with plots
 - A clear, well-labeled figure supporting the explanation
- ✓ **One submission per student**

Content Criteria

The submission must satisfy **at least two** of the following:

- Clearly explains one electromagnetics or vector concept in your own words
- Connects a course concept to a physical or engineering application
- Identifies a common misconception and explains why it is incorrect
- Demonstrates insight through a simple simulation, calculation, or visualization
- Reflects on what was learned and how understanding improved

End-of-Term Technical Reflection

Evaluation Guidelines

- ✓ Emphasis on [clarity, correctness, and insight](#), not length
 - ✓ MATLAB code does not need to be extensive or optimized
 - ✓ Bonus is awarded based on quality of engagement, not technical difficulty
 - ✓ Submissions must reflect original understanding
- ❖ *Think of this as a one-page technical note, short, clear, and supported by a simple MATLAB plot or simulation.*

Voluntary Weekend Assignments / Mini-Projects (5% Bonus Total)

- A total of 5 voluntary weekend assignments or mini-projects will be offered during the term.
- Each assignment will be released on Friday evening via Avenue.
- A **clearly specified submission window** will be provided for each assignment.
- The **Avenue dropbox will close automatically** at the stated deadline.

Solution & Self-Correction Requirement

- ✓ Full solutions will be released after the submission deadline.
- ✓ To earn bonus credit, students must:
 1. Compare your submission with the posted solution
 2. Identify errors or misconceptions
 3. Explain why the mistake occurred
 4. Provide a corrected explanation in your own words
- ❖ *Copying the posted solution without explanation will not receive bonus credit.*

Course Communication & Support Guidelines

Official Communication

- ✓ All official course announcements are made through **Avenue**.
- ✓ Students are responsible for checking Avenue regularly.

Contacting the Instructor

- ✓ Contact me only through your McMaster email account.
- ✓ Email subject line format:
2FH4 – [brief description of your inquiry]
- ❖ *Messages sent via Avenue messaging or Microsoft Teams will not be monitored.*

TA Assignment & Support Structure

Designated TA System

- ✓ Each student is assigned a designated TA.
- ✓ All questions related to:
 - Assignments
 - MATLAB work
 - Quizzes
 - Marking issuesmust be addressed to **your assigned TA first**.
- ✓ Only contact the TA assigned to you.
 - ❖ *Do not email multiple TAs for the same issue.*

TA Assignment Information

- ✓ A file listing assigned TA names, contact information, and office hours will be uploaded on Avenue.

Escalation & Office Hours

Escalation Policy

- If an issue is **not resolved** after contacting your TA:
 1. Email the instructor
 2. Forward all prior communication with the TA in your message

Office Hours

- All office hours (instructor and TAs) will be held online via Microsoft Teams.
- Office hour schedules and links will be posted on Avenue.

Professional Communication

- Please include in emails:
 1. Full name
 2. Student number
 3. Course code (2FH4)

Deadlines & Submissions

- All assignment deadlines will be as posted on Avenue.
- The Avenue Dropbox will close promptly at the deadline.

Texts & Supplementary Resources

Recommended Texts

1. Matthew N. O. Sadiku, *Elements of Electromagnetics*, 7th edition, Oxford University Press.
 2. M. H. Bakr, *MATLAB Experiments Manual for EE2FH4*, McMaster University Courseware, 2026
(available through Avenue)
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Supplementary Video Lectures

- ✓ Prof. Bakr, who taught this course for many years, has a YouTube channel with lecture videos covering each topic in this course.
- ✓ These videos are **highly recommended** as a supplementary resource.

What We Will Build First

Foundations of Electromagnetics

In this course, we will first learn the mathematical language needed to describe fields:

- Vectors and vector algebra
- Coordinate systems and transformations
- Vector calculus (gradient, divergence, curl)

Using these tools, we will study:

1. **Electrostatics**, electric fields, charge, and electric energy
2. **Magnetostatics**, magnetic fields, currents, and magnetic energy

How It All Comes Together

From Fields to Electromagnetic Waves

When electric and magnetic fields vary in time:

- A changing electric field produces a magnetic field
- A changing magnetic field produces an electric field

Together, they form a new physical object:

Electromagnetic Waves

In this course, you will learn:

- ✓ How electromagnetic waves are generated
- ✓ How they propagate through space and materials
- ✓ How energy flows through an electromagnetic wave

Vector Algebra

Today's plan

- Distinguish scalars from vectors
- Use unit vectors to represent directions
- Perform vector addition and subtraction
- Understand position and displacement vectors
- Apply dot and cross products
- Resolve vectors into components

Vector Algebra – Scalars and Vectors

Scalar: The term *scalar* refers to a quantity whose **value** can be represented by a single real number, which may be positive or negative.

Vector: A *vector* quantity has both a **magnitude** and a **direction** in space.

- ❖ *In this course, we are concerned only with two- and three-dimensional space, although vectors may be defined in n -dimensional space in more advanced applications.*
- ❖ *We adopt the convention that magnitude refers to absolute value; therefore, the magnitude of any vector quantity is always positive.*

Vector Algebra – Scalars and Vectors

Scalar

Length,
Area,
Volume

Speed

Work &
power

Mass,
Density,
Pressure

Temperature,
Energy,
Entropy

Vector

Force &
Acceleration

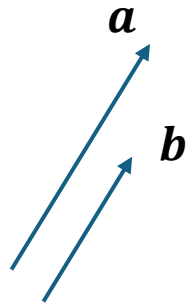
Velocity

Weight

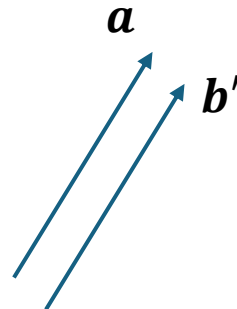
Electrical &
Magnetic
field

Current
density

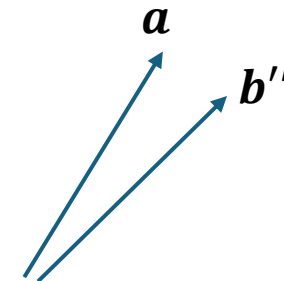
Momentum



$$a \neq b$$



$$a = b$$



$$a \neq b$$

Attention:

- ❖ *To indicate a vector quantity in typed text, it should be written in boldface or with an arrow or bar over the symbol, for example: $\mathbf{A}$, $\vec{A}$, $\bar{A}$.*
- ❖ *When written by hand, the arrow or bar must be included to distinguish the vector from a scalar.*

Vector Algebra

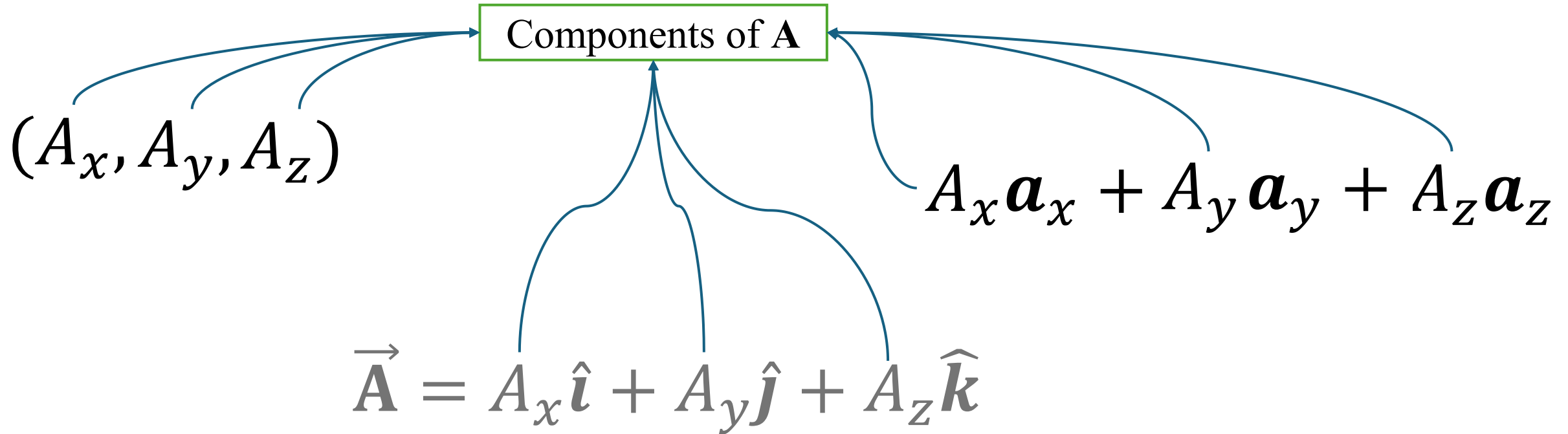
- ✓ Distinguish scalars from vectors
- **Use unit vectors to represent directions**
- Perform vector addition and subtraction
- Understand position and displacement vectors
- Apply dot and cross products
- Resolve vectors into components

Vector Algebra – Unit Vector

- As mentioned, a *vector* has both **magnitude** and **direction**.
- A vector can be represented using any of the notations we learned, $\mathbf{A}$, $\vec{\mathbf{A}}$, $\overline{\mathbf{A}}$.
- The magnitude of a vector is denoted by A or $|\mathbf{A}|$, $|\vec{\mathbf{A}}|$, $|\overline{\mathbf{A}}|$, and is always a **positive scalar** quantity.
- Vectors require a coordinate system to be defined. There are many different coordinate systems, three important ones are **rectangular** (Cartesian), cylindrical, and spherical.

Vector Algebra – Unit Vector – Vectors in Rectangular System

- When we want to represent a vector in a rectangular system there are different notation we can use:



Vector Algebra – Unit Vector – Magnitude of a Vector

- The following formula gives the **magnitude** of a *vector* expressed in rectangular components:

$$A = |\mathbf{A}| = \sqrt{A_x^2 + A_y^2 + A_z^2}$$

- ❖ *The magnitude of a vector is coordinate-independent.*

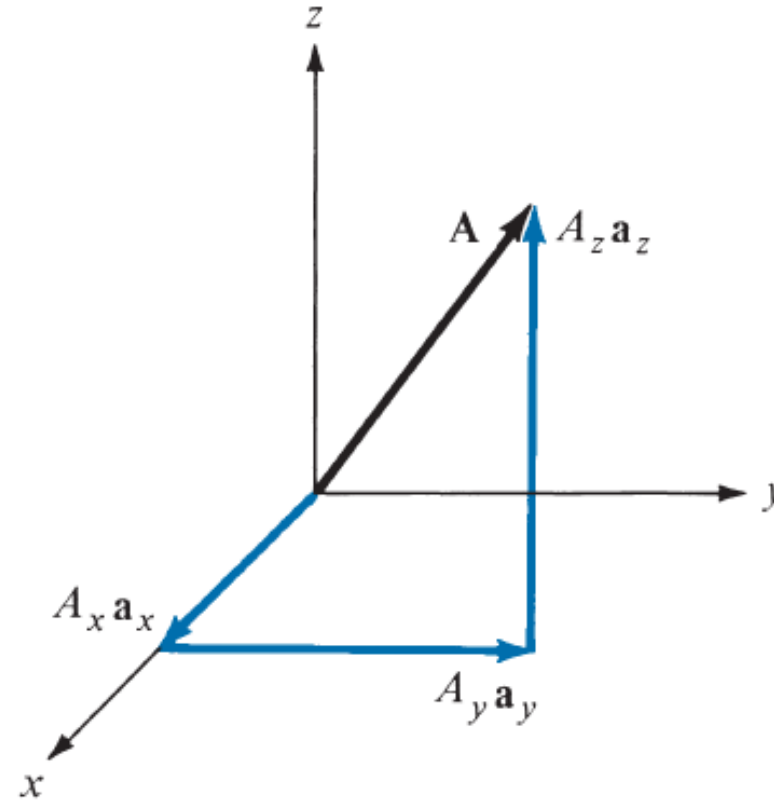
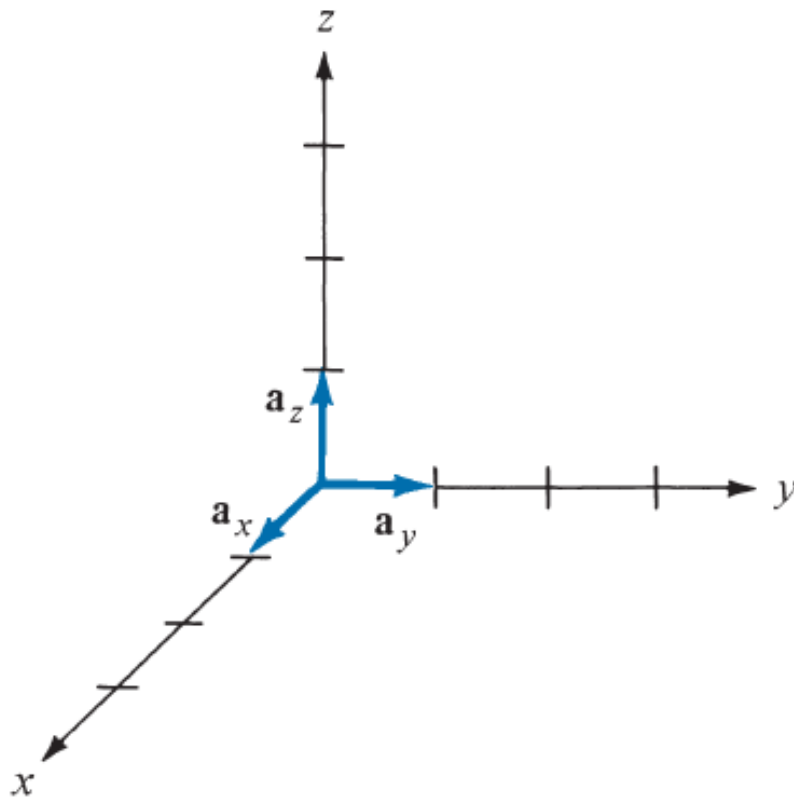
Vector Algebra – Unit Vector

➤ Now that we know the **magnitude** of a vector, we can define its **unit vector**.

$$\mathbf{a}_A = \frac{\mathbf{A}}{|\mathbf{A}|} = \frac{A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z}{\sqrt{A_x^2 + A_y^2 + A_z^2}}$$

$$\mathbf{a}_A = \frac{A_x}{|\mathbf{A}|} \mathbf{a}_x + \frac{A_y}{|\mathbf{A}|} \mathbf{a}_y + \frac{A_z}{|\mathbf{A}|} \mathbf{a}_z$$

Vector Algebra – Unit Vector



Vector Algebra – Unit Vector – Example 1

- Given $\mathbf{A} = 3\mathbf{a}_x - 4\mathbf{a}_y + 12\mathbf{a}_z$, find the magnitude and the unit vector of $\mathbf{A}$.

Vector Algebra

- ✓ Distinguish scalars from vectors
- ✓ Use unit vectors to represent directions
- **Perform vector addition and subtraction**
- Understand position and displacement vectors
- Apply dot and cross products
- Resolve vectors into components

Vector Algebra – Vector Addition and Subtraction, Geometric Interpretation

- Assume that we have two vectors $\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$ & $\mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$, adding them will give us a third vector, $\mathbf{C} = C_x \mathbf{a}_x + C_y \mathbf{a}_y + C_z \mathbf{a}_z$. Subtracting $\mathbf{A}$ & $\mathbf{B}$ will give us another vector, $\mathbf{D} = D_x \mathbf{a}_x + D_y \mathbf{a}_y + D_z \mathbf{a}_z$.

Parallelogram

Head-to-Tail

Vector Algebra – Vector Addition and Subtraction, Algebraic Expressions

- Assume that we have two vectors $\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$ & $\mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$, adding them will give us a third vector, $\mathbf{C} = C_x \mathbf{a}_x + C_y \mathbf{a}_y + C_z \mathbf{a}_z$. Subtracting $\mathbf{A}$ & $\mathbf{B}$ will give us another vector, $\mathbf{D} = D_x \mathbf{a}_x + D_y \mathbf{a}_y + D_z \mathbf{a}_z$.

Vector Algebra – Vector Addition and Subtraction, Properties

1. Vector addition is *commutative*
2. Vector addition is *associative*
3. Vector addition is *distributive*

Vector Algebra

- ✓ Distinguish scalars from vectors
- ✓ Use unit vectors to represent directions
- ✓ Perform vector addition and subtraction
- **Understand position and displacement vectors**
 - Apply dot and cross products
 - Resolve vectors into components

Vector Algebra – Position and Displacement Vectors

Position Vector: the position vector, $\mathbf{r}_P$, of point P is defined as **directed distance** from the origin $O(0,0,0)$ to $P(x,y,z)$:

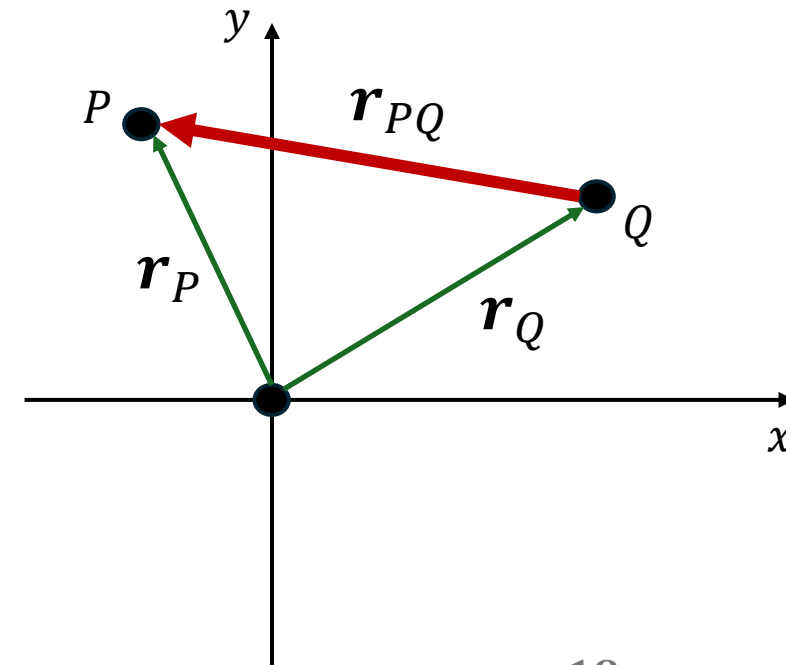
$$\mathbf{r}_P = \overrightarrow{OP} = x\mathbf{a}_x + y\mathbf{a}_y + z\mathbf{a}_z$$

- ❖ *The position vector points from the origin to the point; its magnitude equals the shortest distance, but the vector itself includes direction.*

Vector Algebra – Position and Displacement Vectors

When we have two points $P(x_p, y_p, z_p)$ & $Q(x_Q, y_Q, z_Q)$, the **distance vector** from P to Q is the displacement from one of these points to the other one.

$$\begin{aligned}\overrightarrow{PQ} &= \mathbf{r}_{PQ} = \mathbf{r}_Q - \mathbf{r}_P \\ &= (x_Q - x_P)\mathbf{a}_x + (y_Q - y_P)\mathbf{a}_y + (z_Q - z_P)\mathbf{a}_z\end{aligned}$$



Red arrow is the displacement vector

Vector Algebra – Position and Displacement Vectors – Example 2

A point charge is located at the origin and creates an electric field that points radially outward. Consider point $P(2, -1, 2)$.

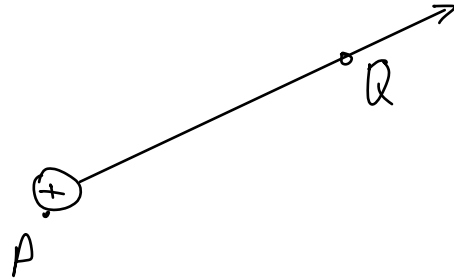
- a) Find the position vector of point P .
- b) Find the magnitude of the position vector.
- c) Find unit vector in the direction of the position vector.

Vector Algebra – Position and Displacement Vectors – Example 3

A point charge $+Q$ is located at point $P(1, 2, 0)$ find the direction of the electric field at point $Q(4, -1, 2)$.

$$P(1, 2, 0)$$

$$Q(4, -1, 2)$$



$$\begin{aligned} \mathbf{r}_{PQ} &= \mathbf{r}_Q - \mathbf{r}_P \\ &= (4, -1, 2) - (1, 2, 0) \\ &= (3, -3, 2) \end{aligned}$$

Vector Algebra

- ✓ Distinguish scalars from vectors
- ✓ Use unit vectors to represent directions
- ✓ Perform vector addition and subtraction
- ✓ Understand position and displacement vectors
- **Apply dot and cross products**
- Resolve vectors into components

Vector Algebra – Vector Multiplication

- There are two types of multiplication defined for vectors,
 1. Dot product $\mathbf{A} \cdot \mathbf{B}$
 2. Cross product $\mathbf{A} \times \mathbf{B}$
- The result of each of these multiplications has a different physical meaning.
- The other name for **dot product** is **scalar product** → because the result is a scalar quantity.
- The other name for **cross product** is **vector product** → because the result is a vector quantity.
- ❖ *Dot product tells us how **aligned** two vectors are; cross product tells us how **perpendicular** they are.*

Vector Algebra / Coordinate Systems

Today's plan

- Dot (scalar) product
- Cross (vector) product
- Scalar triple product
- Vector triple product
- Components of a vector
- Circular cylindrical coordinates

- **Dot (scalar) product**
 - **Definitions**
 - **Properties and Applications**
- Cross (vector) product
- Scalar triple product
- Vector triple product
- Components of a vector
- Circular cylindrical coordinates

Vector Algebra – The Dot Product, Definition

- Dot product = scalar product = inner product
- Geometric definition of dot product

$$\mathbf{A} \cdot \mathbf{B} = |\mathbf{A}||\mathbf{B}|\cos\theta_{AB}$$

- Operational definition of dot product,

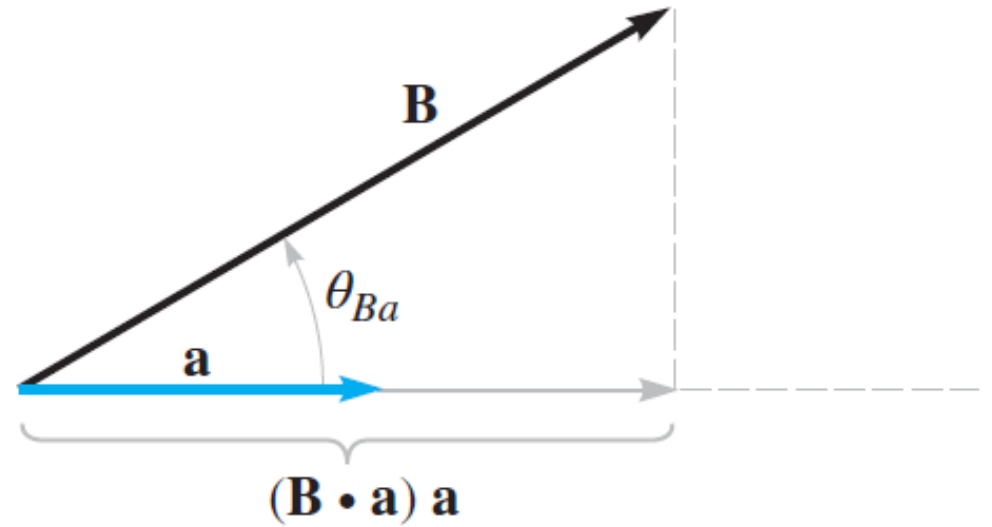
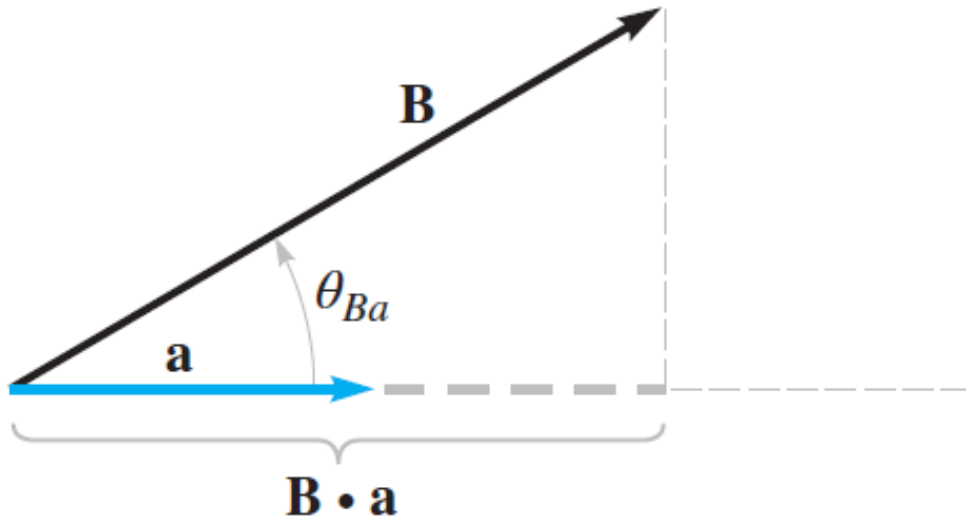
$$\mathbf{A} = A_x\mathbf{a}_x + A_y\mathbf{a}_y + A_z\mathbf{a}_z \quad \& \quad \mathbf{B} = B_x\mathbf{a}_x + B_y\mathbf{a}_y + B_z\mathbf{a}_z$$

- $\mathbf{A} \cdot \mathbf{B} =$

Vector Algebra – The Dot Product, Properties & Applications

1. Commutative
2. Distributive
3. Dot product of a vector by itself is its magnitude squared
4. As the application of dot product is that we can use it to find the components of a vector in the direction of unit vector.

Vector Algebra – The Dot Product, Properties & Applications



Vector Algebra – The Dot Product – Example 1

Consider the vector field $\vec{G} = y\mathbf{a}_x - 2.5x\mathbf{a}_y + 3\mathbf{a}_z$ at the point $Q(4, 2, 5)$ find,

- $\vec{G}$ at Q .
- The scalar component of $\vec{G}$ at Q in the direction of $\mathbf{a}_N = \frac{1}{3}(2\mathbf{a}_x + \mathbf{a}_y - 2\mathbf{a}_z)$.
- The vector component of $\vec{G}$ at Q in the direction of $\mathbf{a}_N$.
- The angle $\theta_{G\mathbf{a}_N}$.

$$\begin{aligned} \text{a) } \vec{G}_Q &= (y, -2.5x, 3) \quad @ (4, 2, 5) \\ &= (2, -2.5(4), 3) \\ &= (2, -10, 3) \end{aligned}$$

$$\begin{aligned} \text{c) } \text{proj} &= (\vec{G}_Q \cdot \mathbf{a}_N) \mathbf{a}_N \\ &= -4 \left(\frac{1}{3} \right) (2, 1, -2) \\ &= \frac{-4}{3} (2, 1, -2) \\ &= \left(-\frac{8}{3}, -\frac{4}{3}, \frac{8}{3} \right) \end{aligned}$$

$$\begin{aligned} \text{b) } \vec{G}_Q \cdot \mathbf{a}_N &= \frac{1}{3} (2, 1, -2) \cdot (2, -10, 3) \\ &= \frac{1}{3} (4 - 10 - 6) \\ &= \frac{1}{3} (-12) \\ &= -4 \end{aligned}$$

$$\begin{aligned} \text{d) } \vec{G}_Q \cdot \mathbf{a}_N &= |\vec{G}_Q| |\mathbf{a}_N| \cos \theta_{G\mathbf{a}_N} \\ -4 &= \sqrt{2^2 + 10^2 + 3^2} \cancel{\sqrt{113}} \cos \theta \\ -4 &= \sqrt{113} \cos \theta \\ \theta &= \cos^{-1} \left(\frac{-4}{\sqrt{113}} \right) \end{aligned}$$

Vector Algebra / Coordinate Systems

- ✓ Dot (scalar) product
- **Cross (vector) product**
 - Scalar triple product
 - Vector triple product
 - Components of a vector
 - Circular cylindrical coordinates

Vector Algebra – The Cross Product, Definition

- Cross product = vector product = outer product
- Geometric definition of cross product

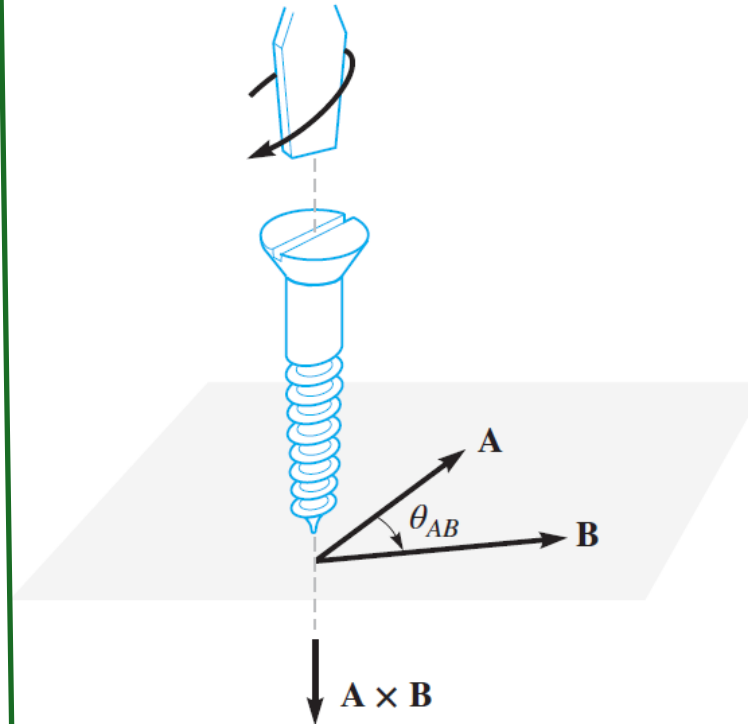
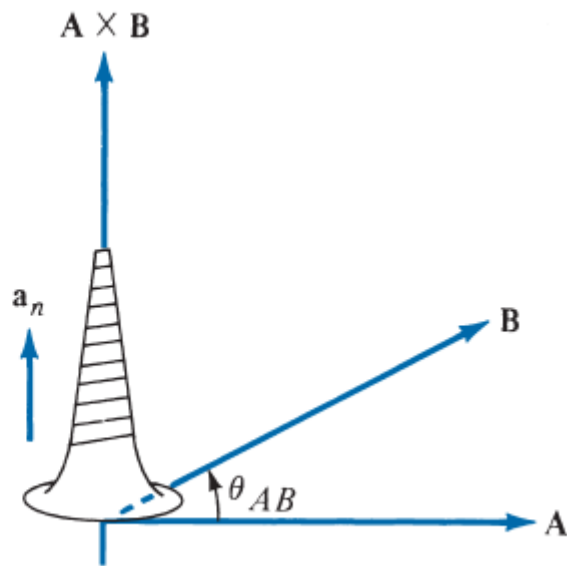
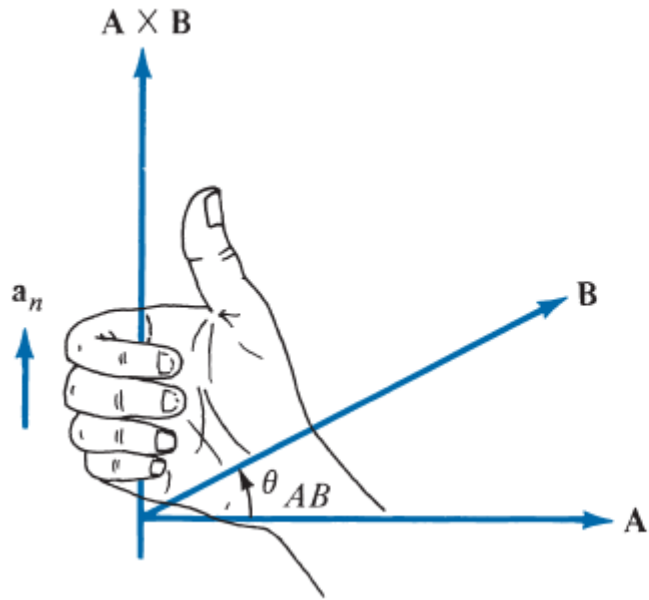
$$\mathbf{A} \times \mathbf{B} = \mathbf{a}_n |\mathbf{A}| |\mathbf{B}| \sin \theta_{AB}$$

- Operational definition of cross product,

$$\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z \quad \& \quad \mathbf{B} = B_x \mathbf{a}_x + B_y \mathbf{a}_y + B_z \mathbf{a}_z$$

- $\mathbf{A} \times \mathbf{B} =$

Vector Algebra – The Cross Product, Definition



Vector Algebra – The Cross Product, Definition

$$\mathbf{A} \times \mathbf{B} = (A_y B_z - A_z B_y) \mathbf{a}_x + (A_z B_x - A_x B_z) \mathbf{a}_y + (A_x B_y - A_y B_x) \mathbf{a}_z$$

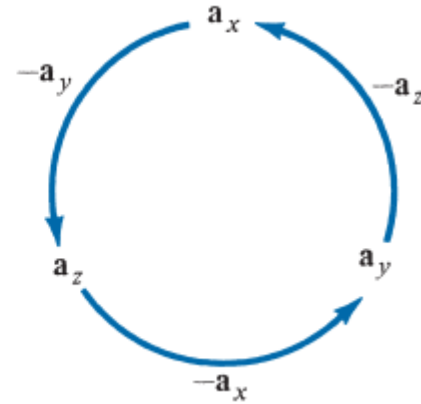
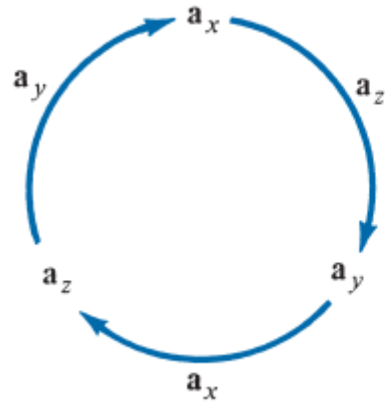
- Example $\mathbf{A} = 2\mathbf{a}_x - 3\mathbf{a}_y + \mathbf{a}_z$ & $\mathbf{B} = -4\mathbf{a}_x - 2\mathbf{a}_y + 5\mathbf{a}_z$

$$\mathbf{A} \times \mathbf{B} =$$

Vector Algebra – The Cross Product, Properties

1. Anticommutative
2. Distributive
3. Not associative
4. Scaling
5. A vector cross multiplication by itself is zero

Vector Algebra – The Cross Product, Properties



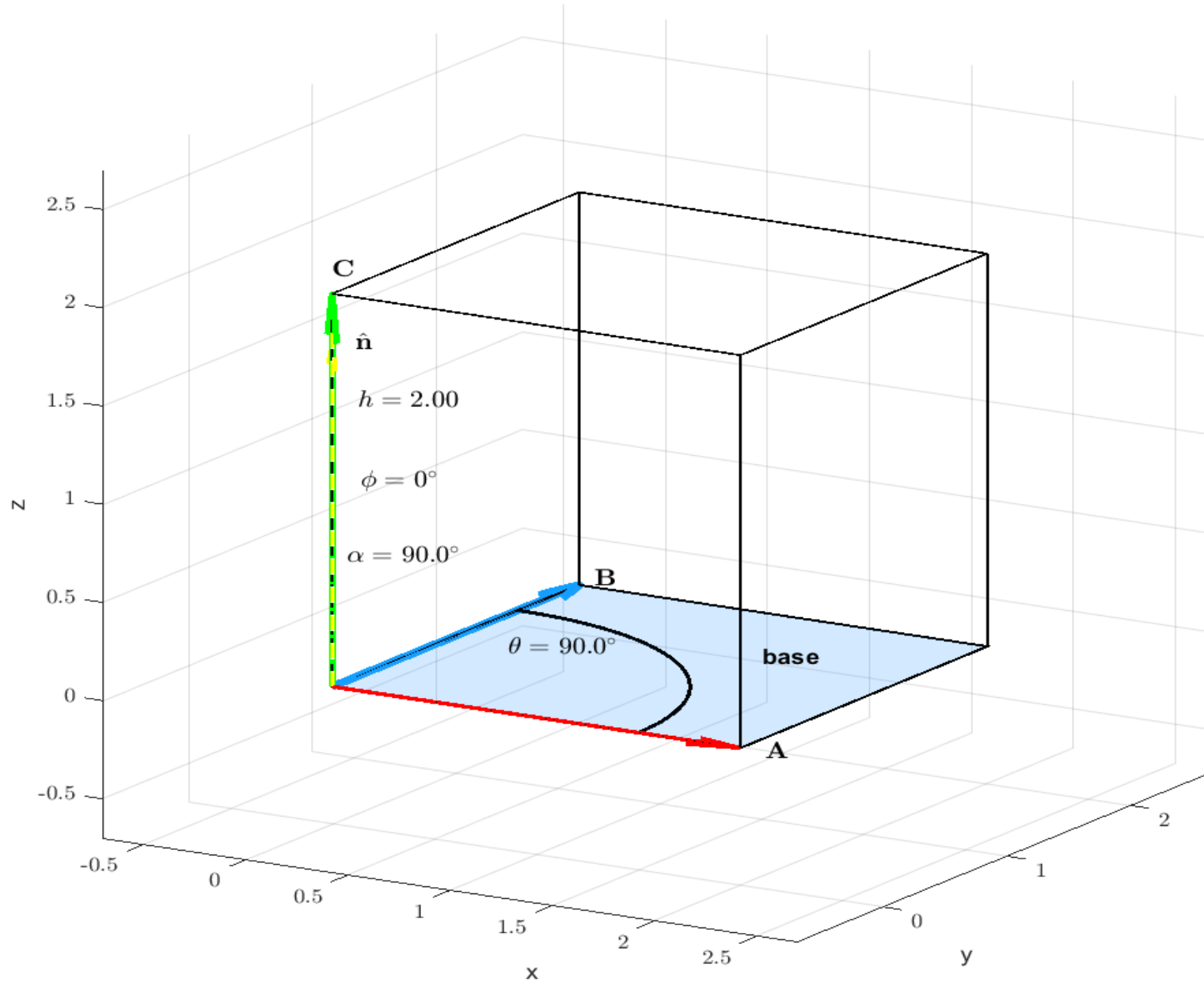
Vector Algebra / Coordinate Systems

- ✓ Dot (scalar) product
- ✓ Cross (vector) product
- **Scalar triple product**
 - Vector triple product
 - Components of a vector
 - Circular cylindrical coordinates

Vector Algebra – Scalar Triple Product

$$\mathbf{A} \cdot (\mathbf{B} \times \mathbf{C}) =$$

Vector Algebra – Scalar Triple Product

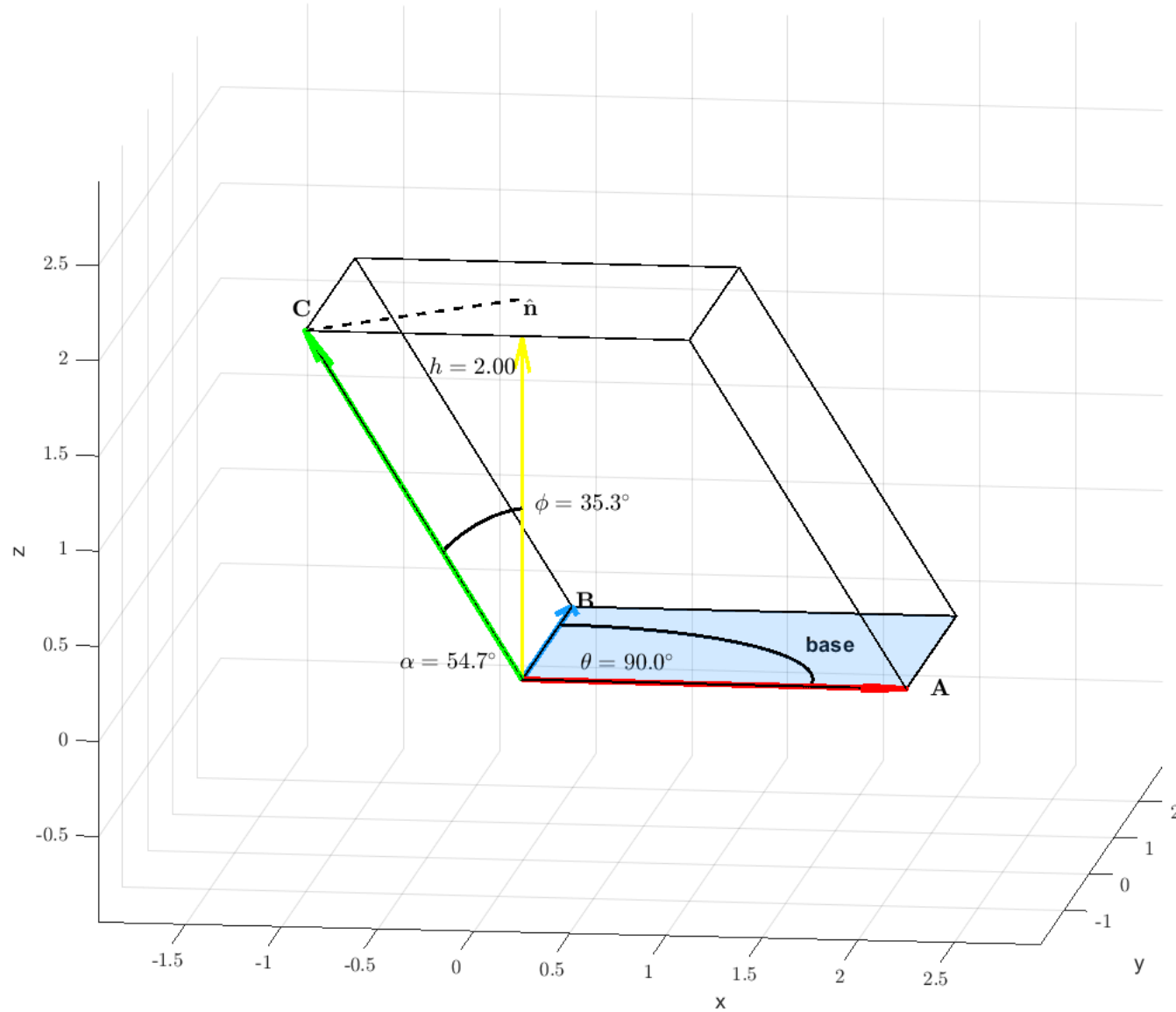


Vector Algebra – Scalar Triple Product

Triple Scalar Product = Volume

$$|\mathbf{C} \cdot (\mathbf{A} \times \mathbf{B})| = 8.000$$

$$|\mathbf{A} \times \mathbf{B}| = 4.000, \quad h = |\mathbf{C}| \cos \phi = 2.000$$

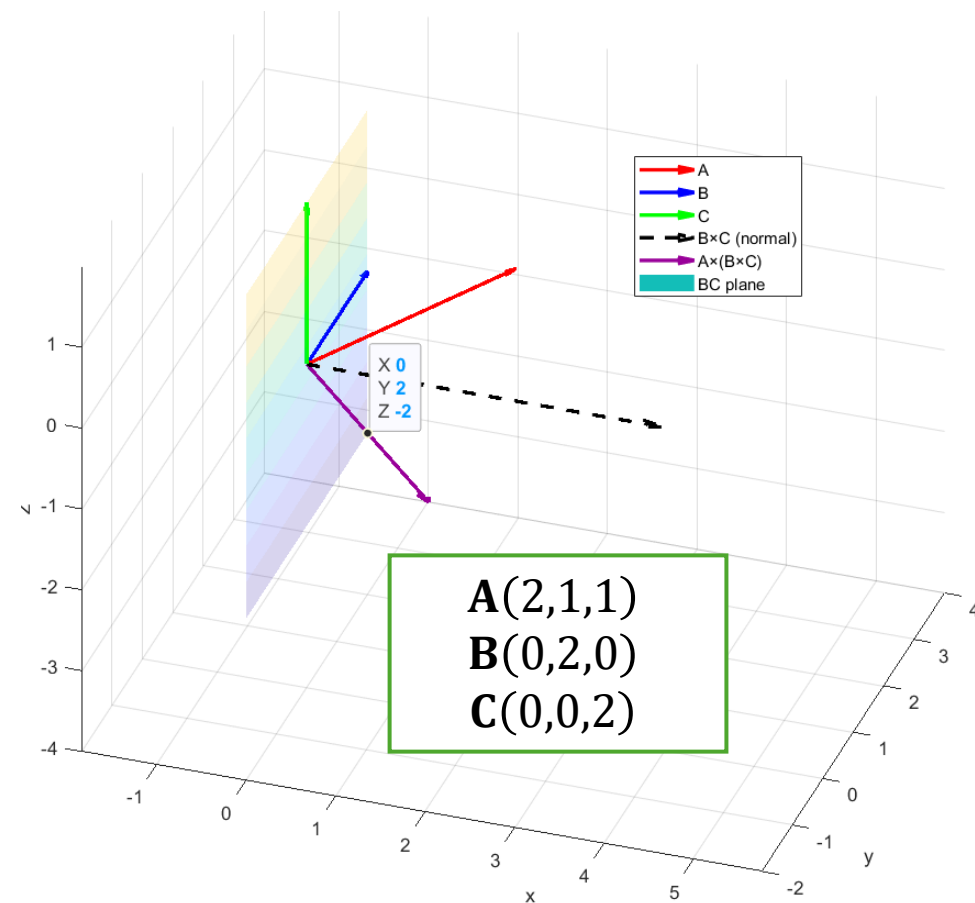
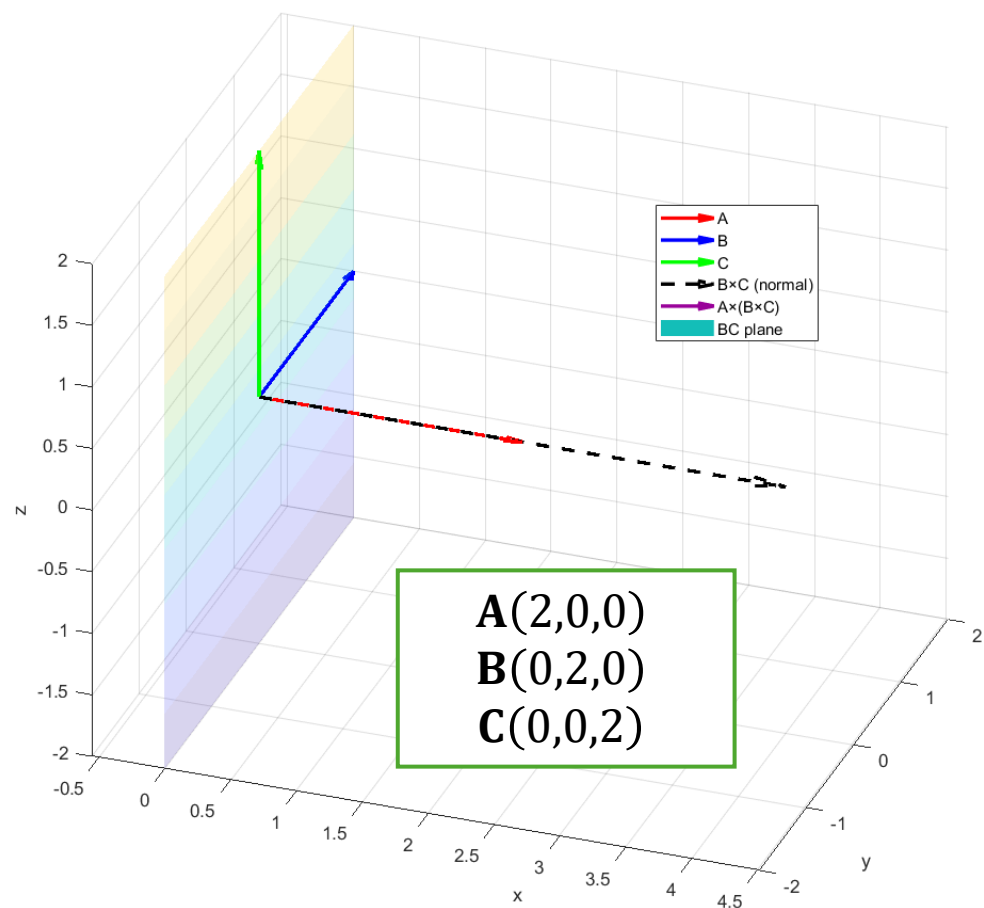


Vector Algebra / Coordinate Systems

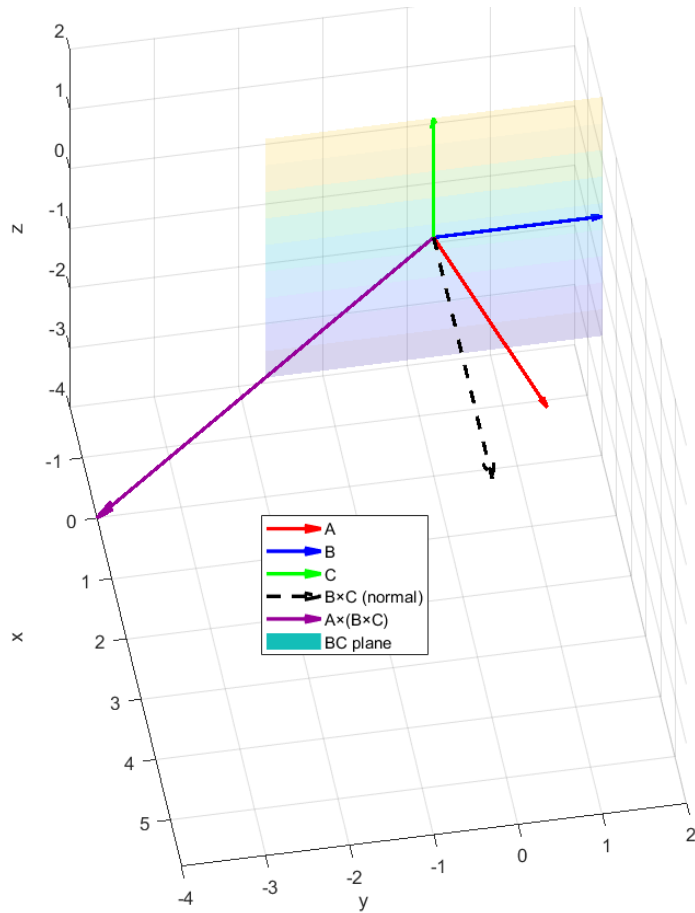
- ✓ Dot (scalar) product
- ✓ Cross (vector) product
- ✓ Scalar triple product
- **Vector triple product**
 - Components of a vector
 - Circular cylindrical coordinates

Vector Algebra – Cross Triple Product

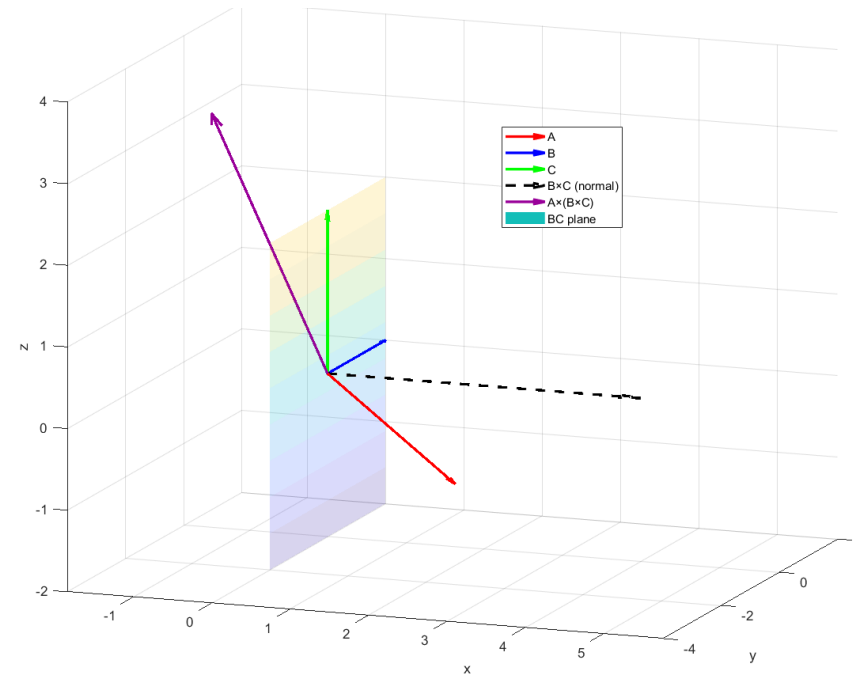
$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$



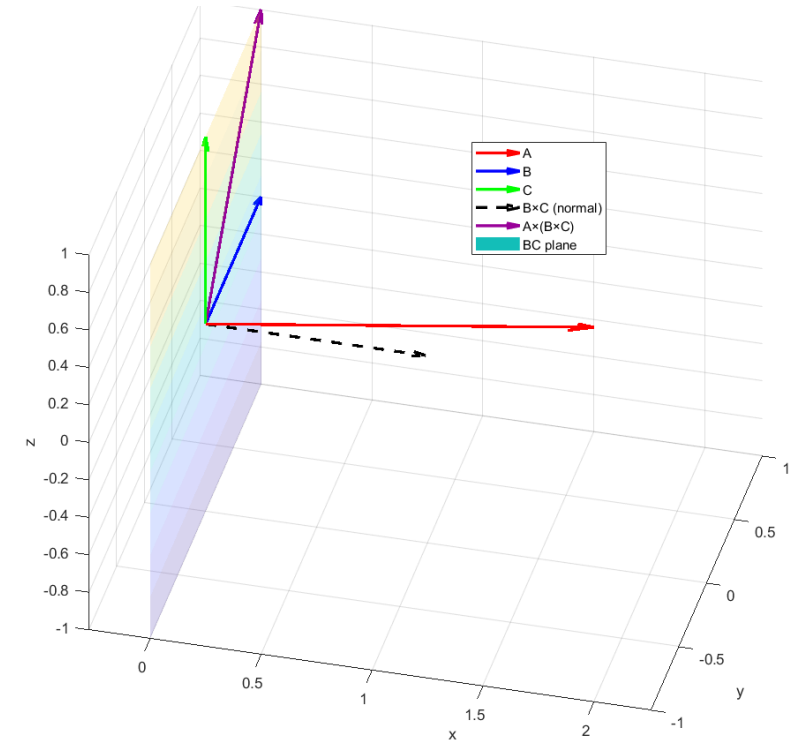
Vector Algebra – Cross Triple Product



$\mathbf{A}(2, 1, -1)$
 $\mathbf{B}(0, 1, 0)$
 $\mathbf{C}(0, 0, 1)$



$\mathbf{A}(2, -1, -1)$
 $\mathbf{B}(0, 2, 0)$
 $\mathbf{C}(0, 0, 2)$



$\mathbf{A}(2, -1, 1)$
 $\mathbf{B}(0, 1, 0)$
 $\mathbf{C}(0, 0, 1)$

Vector Algebra – Cross Triple Product –Example 2

Give the following vectors verify the BAC-CAB rule,

$$\vec{A} = \mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z$$

$$\vec{B} = -\mathbf{a}_x + 2\mathbf{a}_z$$

$$\vec{C} = 2\mathbf{a}_x + \mathbf{a}_y - \mathbf{a}_z$$

Vector Algebra – Cross Triple Product –Example 2 (Cont'd)

Vector Algebra / Coordinate Systems

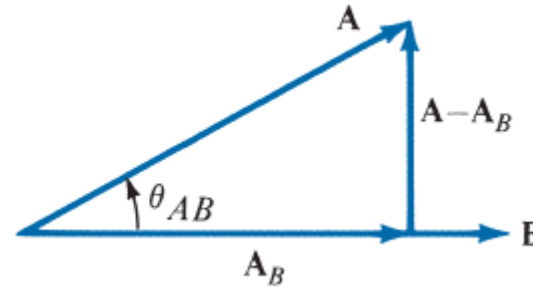
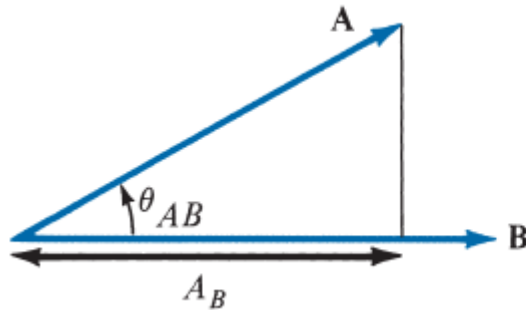
- ✓ Dot (scalar) product
- ✓ Cross (vector) product
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- ✓ Vector triple product
- **Components of a vector**
- Circular cylindrical coordinates

Vector Algebra – Components of a Vector

- Scalar product can be used to find the projection of **A** along **B**. For this purpose, we need to do dot multiplication between **A** and the unit vector in the direction of **B**, **a_B**.

$$A_B = A \cos \theta_{AB} = |\mathbf{A}| |\mathbf{a}_B| \cos \theta_{AB} = \mathbf{A} \cdot \mathbf{a}_B$$

$$\mathbf{A}_B = A_B \mathbf{a}_B = (\mathbf{A} \cdot \mathbf{a}_B) \mathbf{a}_B$$



Vector Algebra / Coordinate Systems

- ✓ Dot (scalar) product
- ✓ Cross (vector) product
- ✓ Scalar triple product
- ✓ Vector triple product
- ✓ Components of a vector
- Circular cylindrical coordinates

Coordinate Systems / Coordinate Systems

Today's plan

- Vector triple product (1.7 – p. 11)
- Components of a vector (1.8 – p. 16)
- Circular cylindrical coordinates (2.3 – p. 32)
- Spherical Coordinates (2.4 – p. 35)
- Constant-Coordinate Surfaces (2.5 – p. 44)

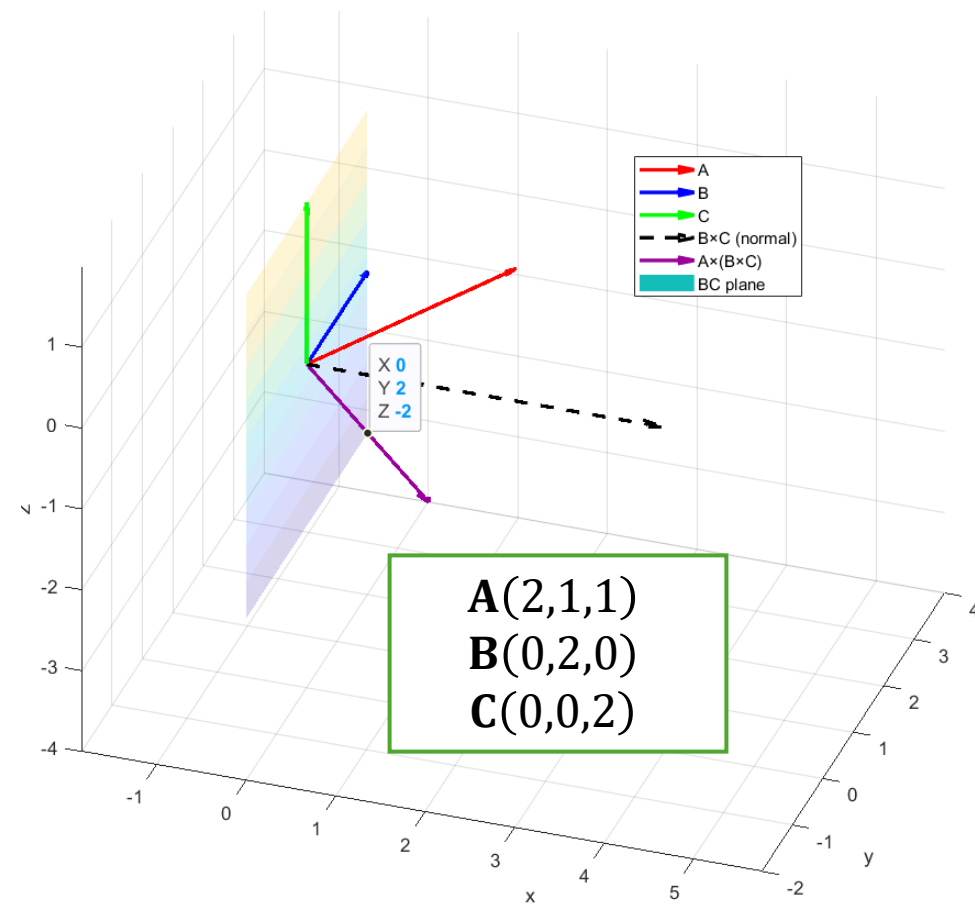
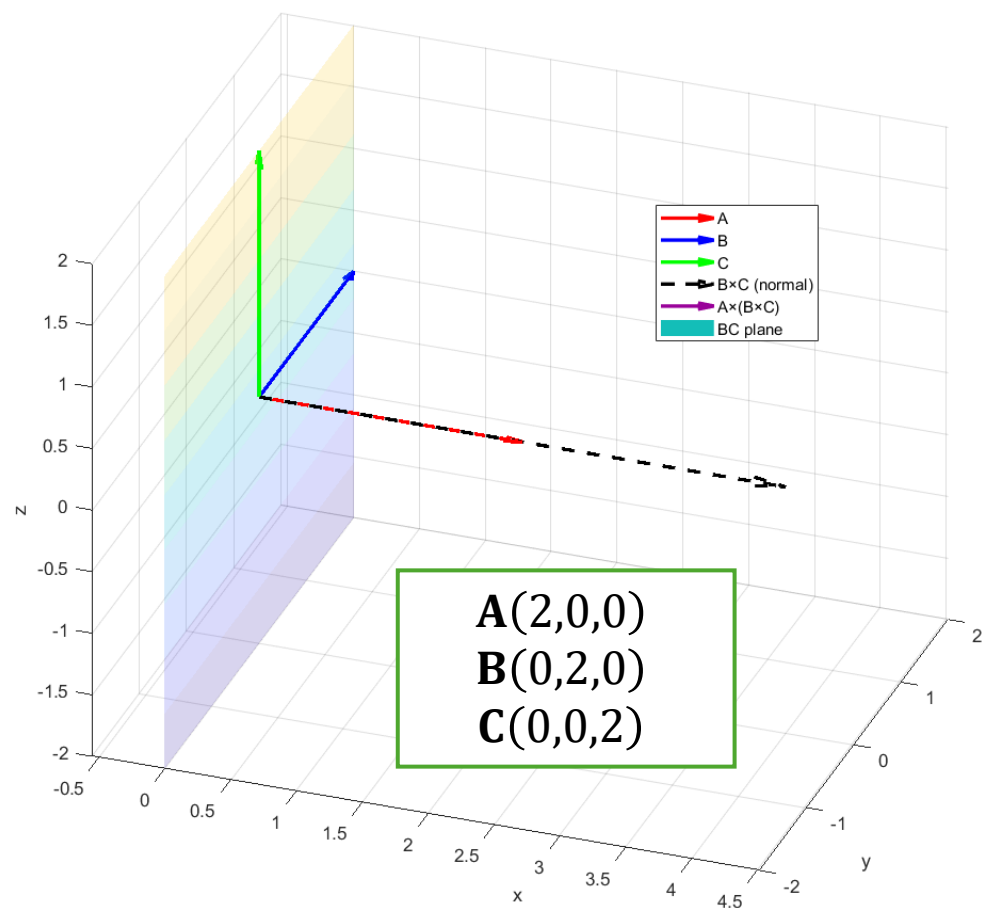
Vector Algebra / Coordinate Systems

- **Vector triple product**
- Components of a vector
- Circular cylindrical coordinates
- Spherical Coordinates
- Constant-Coordinate Surfaces

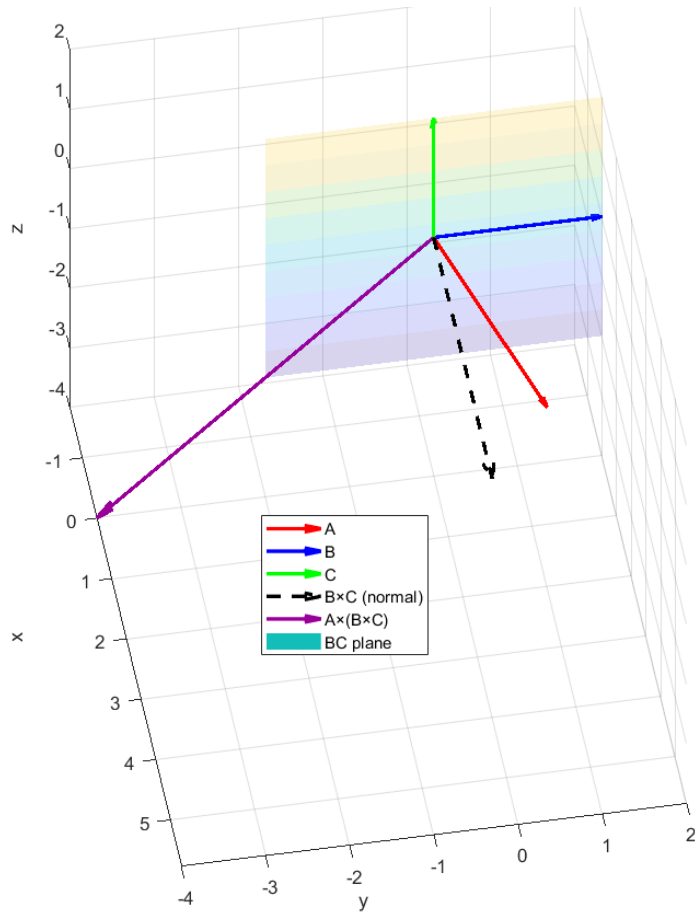
Vector Algebra – Cross Triple Product

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = \mathbf{B}(\mathbf{A} \cdot \mathbf{C}) - \mathbf{C}(\mathbf{A} \cdot \mathbf{B})$$

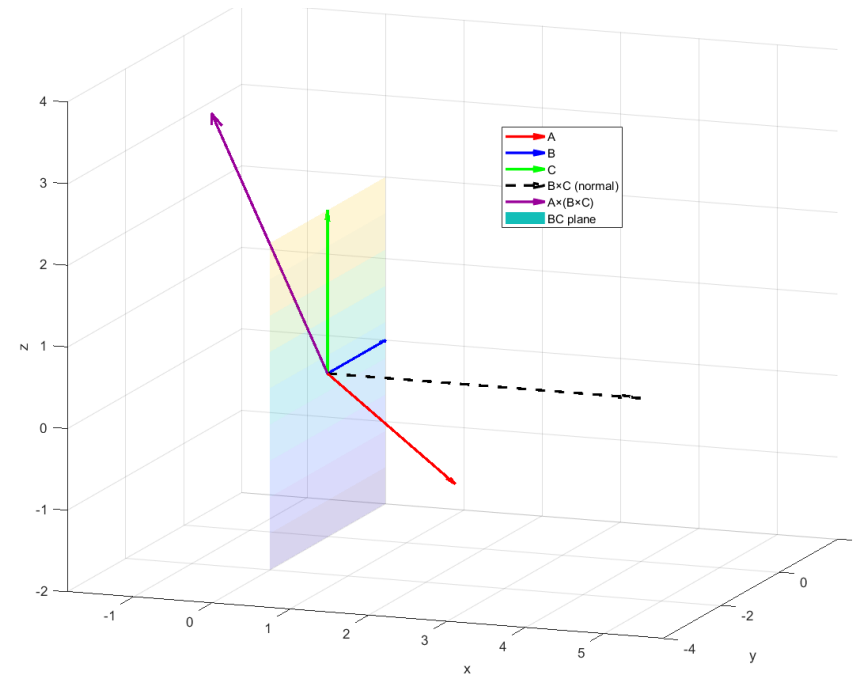
$$= \mathbf{0}$$



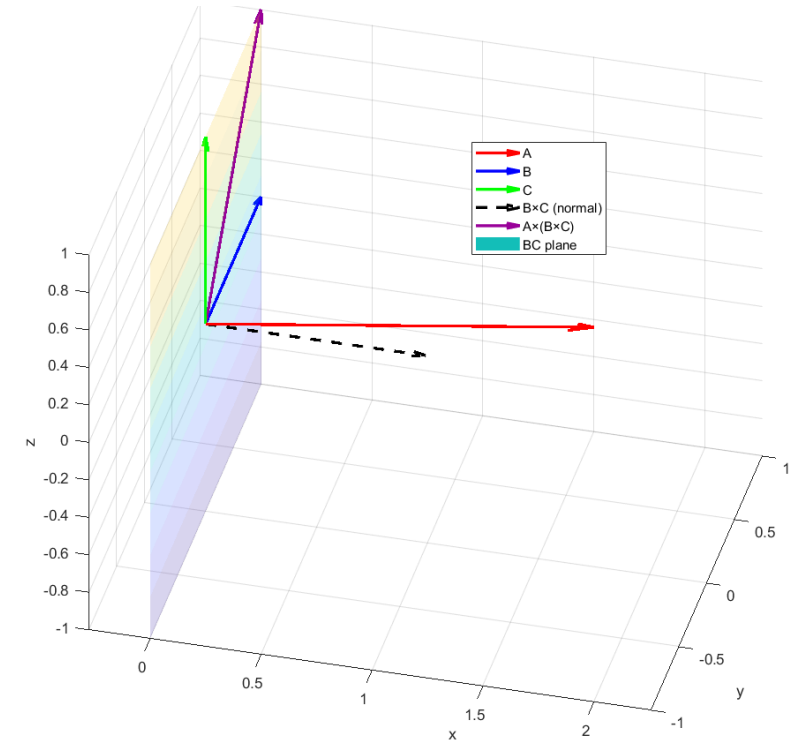
Vector Algebra – Cross Triple Product



$A(2, 1, -1)$
 $B(0, 1, 0)$
 $C(0, 0, 1)$



$A(2, -1, -1)$
 $B(0, 2, 0)$
 $C(0, 0, 2)$



$A(2, -1, 1)$
 $B(0, 1, 0)$
 $C(0, 0, 1)$

Vector Algebra – Cross Triple Product – Example 1

Give the following vectors verify the BAC-CAB rule,

$$\vec{A} = \mathbf{a}_x + 2\mathbf{a}_y + 3\mathbf{a}_z$$

$$\vec{B} = -\mathbf{a}_x + 2\mathbf{a}_z$$

$$\vec{C} = 2\mathbf{a}_x + \mathbf{a}_y - \mathbf{a}_z$$

LH

$$A \times (B \times C)$$

$$B \times C = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ -1 & 0 & 2 \\ 2 & 1 & -1 \end{vmatrix}$$

$$\vec{D} = -2\mathbf{a}_x + 3\mathbf{a}_y - \mathbf{a}_z$$

$$\vec{A} \times \vec{D} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ 1 & 2 & 3 \\ -2 & 3 & -1 \end{vmatrix}$$

$$A \times B \times C = -11\mathbf{a}_x - 5\mathbf{a}_y + 7\mathbf{a}_z$$

= RH

$$= B(A \cdot C) - C(A \cdot B)$$

$$\begin{aligned} A \cdot C &= 1 \cdot 2 + 2 \cdot 1 + 3 \cdot (-1) \\ &= 2 + 2 - 3 \\ &= 1 \end{aligned}$$

$$\begin{aligned} A \cdot B &= 1 \cdot (-1) + 2 \cdot 0 + 3 \cdot 2 \\ &= -1 + 0 + 6 \\ &= 5 \end{aligned}$$

$$= 1B + 5C$$

$$= -\mathbf{a}_x + 2\mathbf{a}_z + 10\mathbf{a}_x + 5\mathbf{a}_y - 5\mathbf{a}_z$$

$$= 9\mathbf{a}_x + 5\mathbf{a}_y - 3\mathbf{a}_z$$

oops

Vector Algebra – Cross Triple Product –Example 2 (Cont'd)

Coordinate Systems

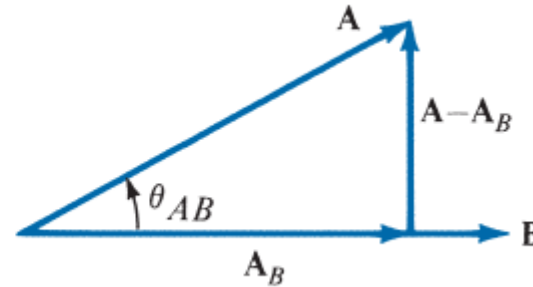
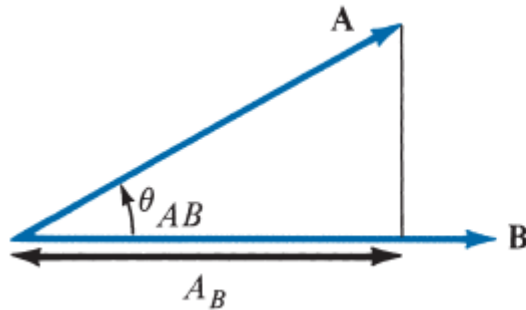
- ✓ Vector triple product
- **Components of a vector**
 - Circular cylindrical coordinates
 - Spherical Coordinates
 - Constant-Coordinate Surfaces

Vector Algebra – Components of a Vector

- Scalar product can be used to find the projection of **A** along **B**. For this purpose, we need to do dot multiplication between **A** and the unit vector in the direction of **B**, **a_B**.

$$A_B = A \cos \theta_{AB} = |\mathbf{A}| |\mathbf{a}_B| \cos \theta_{AB} = \mathbf{A} \cdot \mathbf{a}_B$$

$$\mathbf{A}_B = A_B \mathbf{a}_B = (\mathbf{A} \cdot \mathbf{a}_B) \mathbf{a}_B$$



Vector Algebra – Components of a Vector – Example 2

Let $\mathbf{A} = 2\hat{x} - 1\hat{y} + 2\hat{z}$ $\mathbf{B} = 1\hat{x} + 2\hat{y} + 2\hat{z}$

Find

- a) Scalar projection of $\mathbf{A}$ along $\mathbf{B}$
- b) Vector Projection
- c) Perpendicular component

a) $A_B = \mathbf{A} \cdot \mathbf{a}_B$

$$\mathbf{a}_B = \frac{\mathbf{B}}{|\mathbf{B}|} = \frac{\mathbf{B}}{3} = \frac{1}{3}\mathbf{a}_x + \frac{2}{3}\mathbf{a}_y + \frac{2}{3}\mathbf{a}_z$$

$$\begin{aligned} A_B &= \mathbf{A} \cdot \mathbf{a}_B \\ &= \frac{2}{3} - \frac{2}{3} + \frac{4}{3} \\ &= \frac{4}{3} \end{aligned}$$

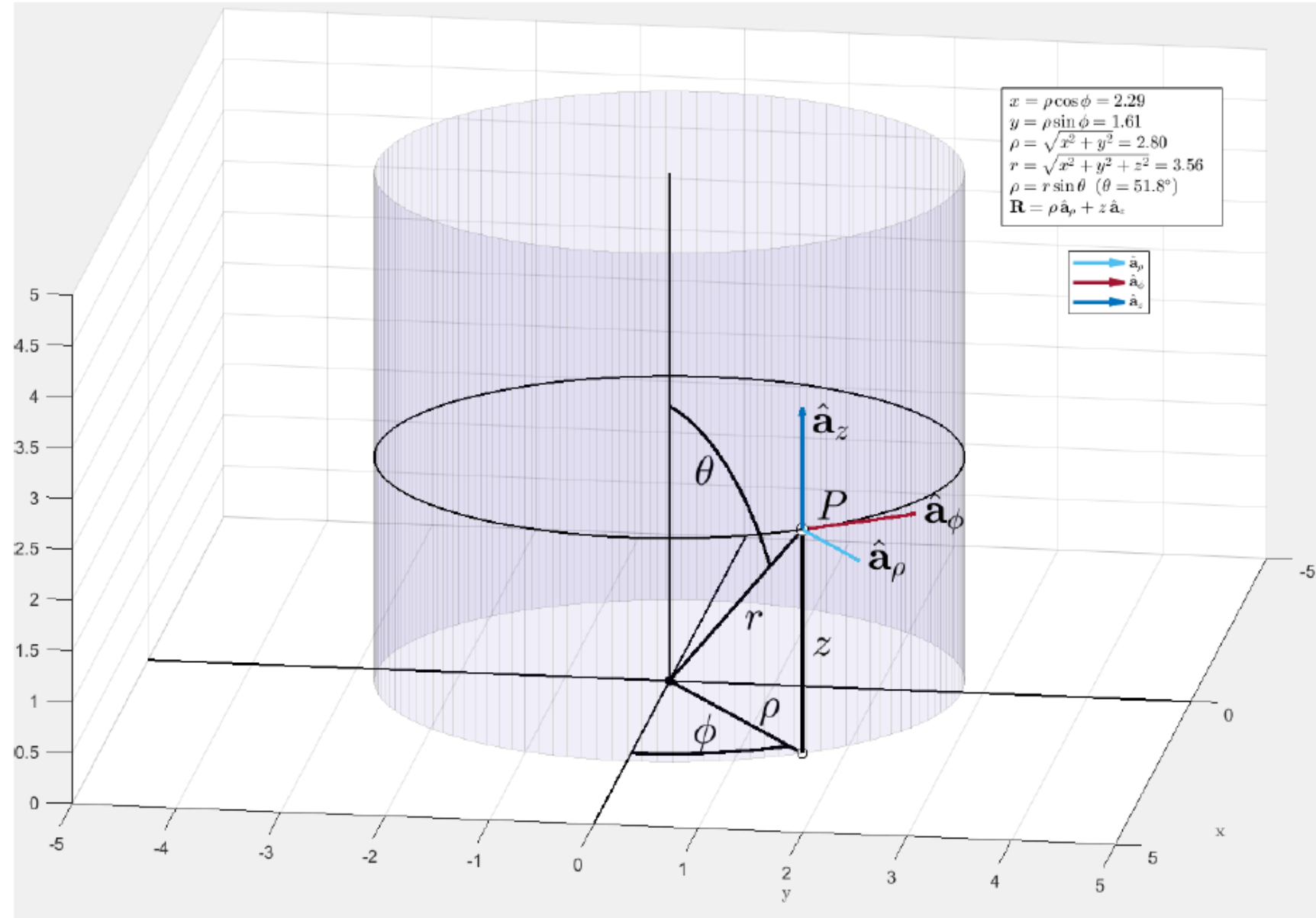
b) $(\vec{A} \cdot \vec{a}_B) \vec{a}_B = \frac{4}{3}\mathbf{a}_x + \frac{8}{3}\mathbf{a}_y + \frac{8}{3}\mathbf{a}_z = A_{\parallel B}$

c) $\begin{aligned} \vec{A}_{\perp} &= \vec{A} - \vec{A}_{\parallel B} \\ &= \left(2 - \frac{4}{3}\right)\mathbf{a}_x + \left(-1 - \frac{8}{3}\right)\mathbf{a}_y + \left(2 - \frac{8}{3}\right)\mathbf{a}_z \end{aligned}$

Vector Algebra / Coordinate Systems

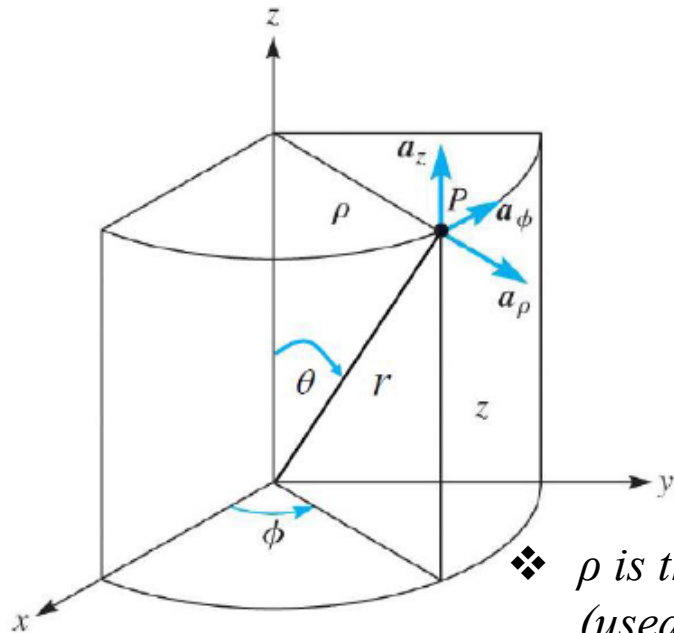
- ✓ Vector triple product
- ✓ Components of a vector
- **Circular cylindrical coordinates**
- Spherical Coordinates
- Constant-Coordinate Surfaces

Coordinate Systems – Cylindrical Coordinates



Coordinate Systems – Cylindrical Coordinates (Cont'd)

- Cylindrical coordinates are best suited for rotational symmetry around the z-axis
- A point P is specified by:
 - 1) ρ : distance from the z-axis
 - 2) ϕ : angle in the x — y plane, measured from $+x$
 - 3) z : height along the z-axis



$$x = \rho \cos \phi, \quad y = \rho \sin \phi$$

$$\rho = \sqrt{x^2 + y^2}, \quad r = \|\mathbf{R}\| = \sqrt{x^2 + y^2 + z^2}$$

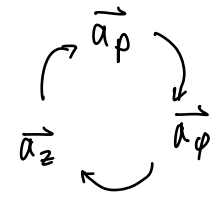
$$\rho = r \sin \theta$$

❖ ρ is the cylindrical radial distance, while $r = \|\mathbf{R}\|$ is the magnitude of the position vector (used in spherical coordinates).

Cylindrical Coordinates – Position Vector and Unit Vectors

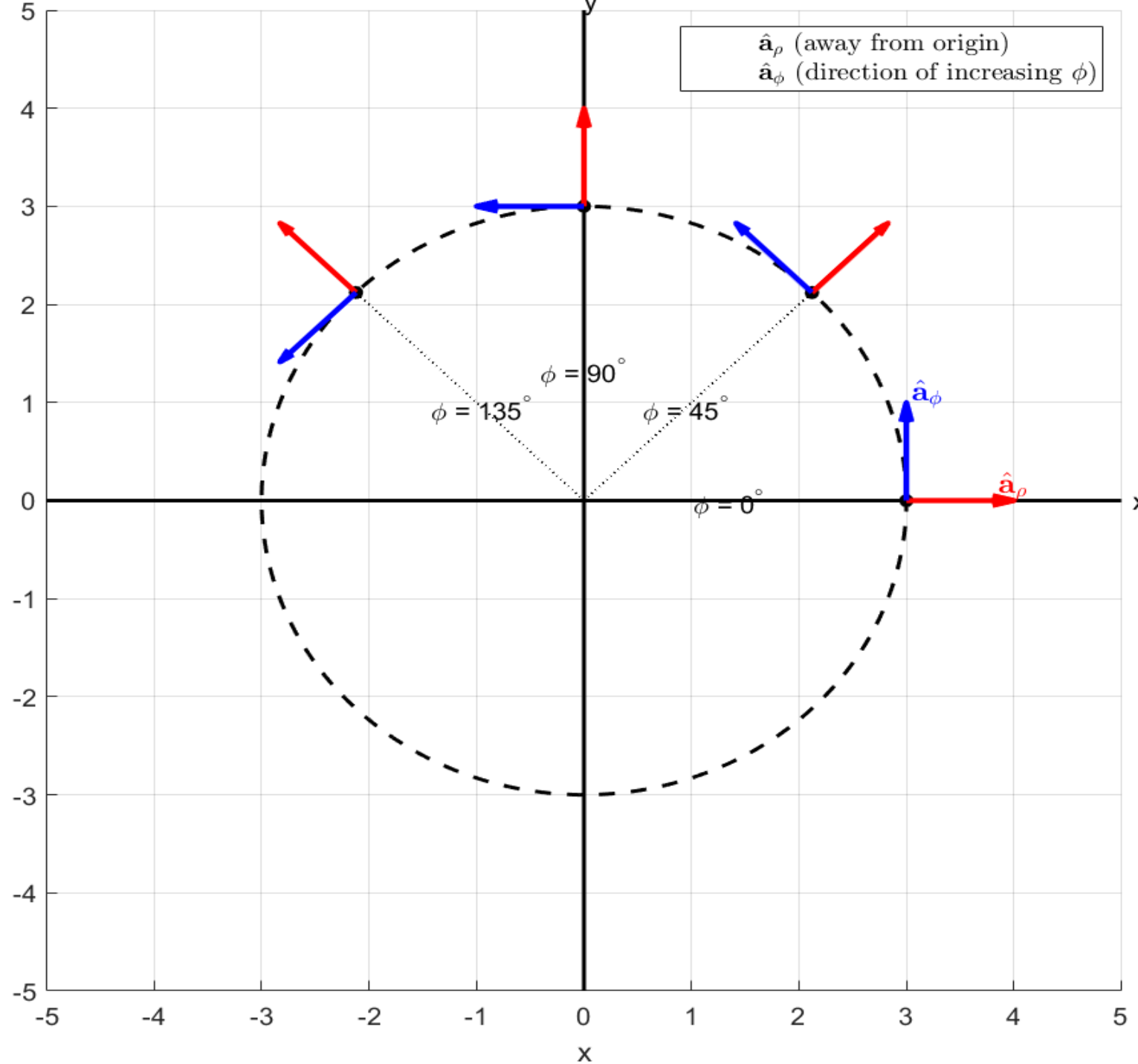
$$\mathbf{R} = \rho \mathbf{a}_\rho + z \mathbf{a}_z$$

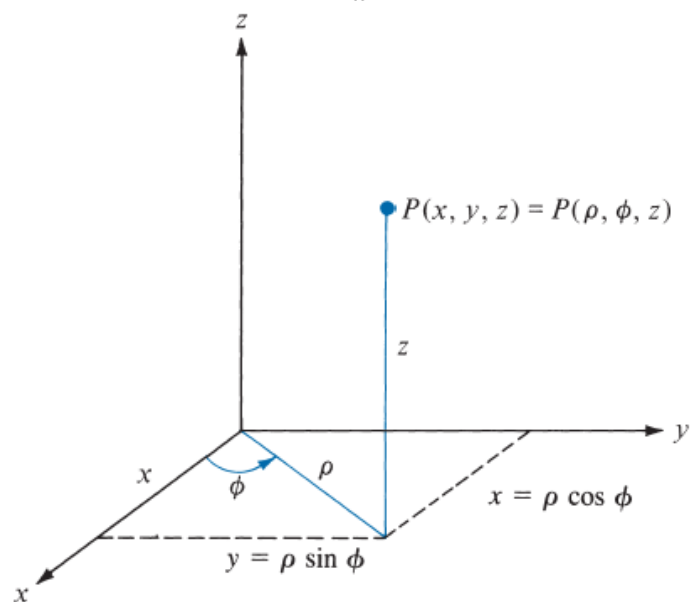
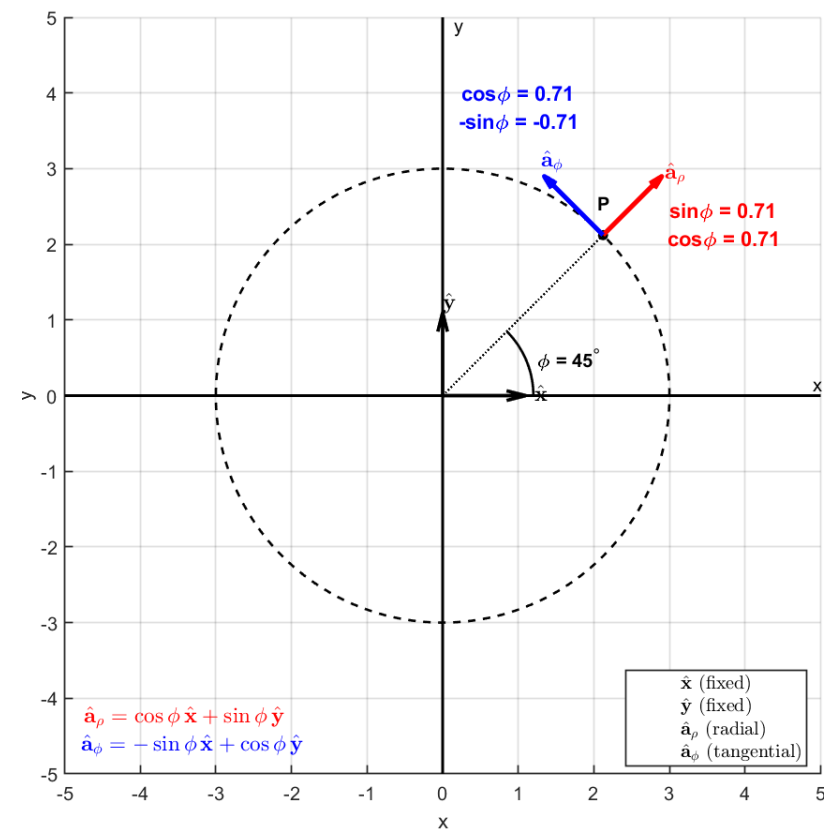
$$\vec{a}_\rho \times \vec{a}_\phi = \vec{a}_z$$



- The position vector always points from the origin to the point
- There is no $\mathbf{a}_\phi$ term
- The unit vector $\mathbf{a}_\rho$ depends on position, $\mathbf{a}_\rho = \cos\phi \mathbf{a}_x + \sin\phi \mathbf{a}_y$. Hence $\mathbf{a}_\rho$ changes direction as ϕ changes, even if $\rho = \text{constant}$.
- The unit vector $\mathbf{a}_\phi$ is perpendicular to the plane formed by $\mathbf{R}$ and $\mathbf{a}_z$

❖ *There is no $\mathbf{a}_\phi$ term because changing ϕ does not move the point away from the origin.*





$$0 \leq \rho < \infty$$

$$0 \leq \phi < 2\pi$$

$$-\infty < z < \infty$$

$$A_\rho \mathbf{a}_\rho + A_\phi \mathbf{a}_\phi + A_z \mathbf{a}_z$$

$$\mathbf{a}_\rho \cdot \mathbf{a}_\rho = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = \mathbf{a}_z \cdot \mathbf{a}_z = 1$$

$$\mathbf{a}_\rho \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_z = \mathbf{a}_z \cdot \mathbf{a}_\rho = 0$$

$$\mathbf{a}_\rho \times \mathbf{a}_\phi = \mathbf{a}_z$$

$$\mathbf{a}_\phi \times \mathbf{a}_z = \mathbf{a}_\rho$$

$$\mathbf{a}_z \times \mathbf{a}_\rho = \mathbf{a}_\phi$$

➤ Cylindrical to Rectangular

$$x = \rho \cos \phi, \quad y = \rho \sin \phi, \quad z = z$$

➤ Rectangular to Cylindrical

$$\rho = \sqrt{x^2 + y^2}, \quad \phi = \tan^{-1} \frac{y}{x}, \quad z = z$$

❖ form a right-handed coordinate system.

➤ Position vector and magnitude

$$\mathbf{R} = x \mathbf{a}_x + y \mathbf{a}_y + z \mathbf{a}_z$$

$$\mathbf{R} = \rho \mathbf{a}_\rho + z \mathbf{a}_z$$

$$|\mathbf{R}| = \sqrt{x^2 + y^2 + z^2} = \sqrt{\rho^2 + z^2}$$

➤ Cylindrical to Rectangular and vice-versa unit vectors

$$\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y$$

$$\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$$

$$\mathbf{a}_x = \cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi$$

* possible exam question

Coordinate Systems – Rectangular to Cylindrical

➤ Given $\mathbf{A} = A_x \mathbf{a}_x + A_y \mathbf{a}_y + A_z \mathbf{a}_z$

1. Identify the angle $\phi = \tan^{-1} \frac{y}{x}$,

2. Write rectangular unit vectors in cylindrical form

$$\mathbf{a}_x = \cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi$$

$$\mathbf{a}_z = \mathbf{a}_z$$

3. Substitute into $\mathbf{A}$, $\mathbf{A} = A_x(\cos \phi \mathbf{a}_\rho - \sin \phi \mathbf{a}_\phi) + A_y(\sin \phi \mathbf{a}_\rho + \cos \phi \mathbf{a}_\phi) + A_z \mathbf{a}_z$

4. Collect terms

Coordinate Systems – Cylindrical to Rectangular

➤ Given

$$\mathbf{A} = A_\rho \hat{\mathbf{a}}_\rho + A_\phi \hat{\mathbf{a}}_\phi + A_z \hat{\mathbf{a}}_z$$

1. Write cylindrical unit vectors in rectangular form

$$\hat{\mathbf{a}}_\rho = \cos \phi \hat{\mathbf{a}}_x + \sin \phi \hat{\mathbf{a}}_y$$

$$\hat{\mathbf{a}}_\phi = -\sin \phi \hat{\mathbf{a}}_x + \cos \phi \hat{\mathbf{a}}_y$$

$$\hat{\mathbf{a}}_z = \hat{\mathbf{a}}_z$$

2. Substitute

$$\mathbf{A} = A_\rho(\cos \phi \hat{\mathbf{a}}_x + \sin \phi \hat{\mathbf{a}}_y) + A_\phi(-\sin \phi \hat{\mathbf{a}}_x + \cos \phi \hat{\mathbf{a}}_y) + A_z \hat{\mathbf{a}}_z$$

3. Collect terms

$$\mathbf{A} = A_x \hat{\mathbf{a}}_x + A_y \hat{\mathbf{a}}_y + A_z \hat{\mathbf{a}}_z$$

$$A_x = A_\rho \cos \phi - A_\phi \sin \phi$$

$$A_y = A_\rho \sin \phi + A_\phi \cos \phi$$

Coordinate Systems – Matrix Form

➤ Rectangular → Cylindrical

$$\begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & \sin \phi & 0 \\ -\sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

➤ Cylindrical → Rectangular

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \cos \phi & -\sin \phi & 0 \\ \sin \phi & \cos \phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} A_\rho \\ A_\phi \\ A_z \end{bmatrix}$$

Vector Algebra / Coordinate Systems

- ✓ Vector triple product
- ✓ Components of a vector
- ✓ Circular cylindrical coordinates
- **Spherical Coordinates**
- Constant-Coordinate Surfaces

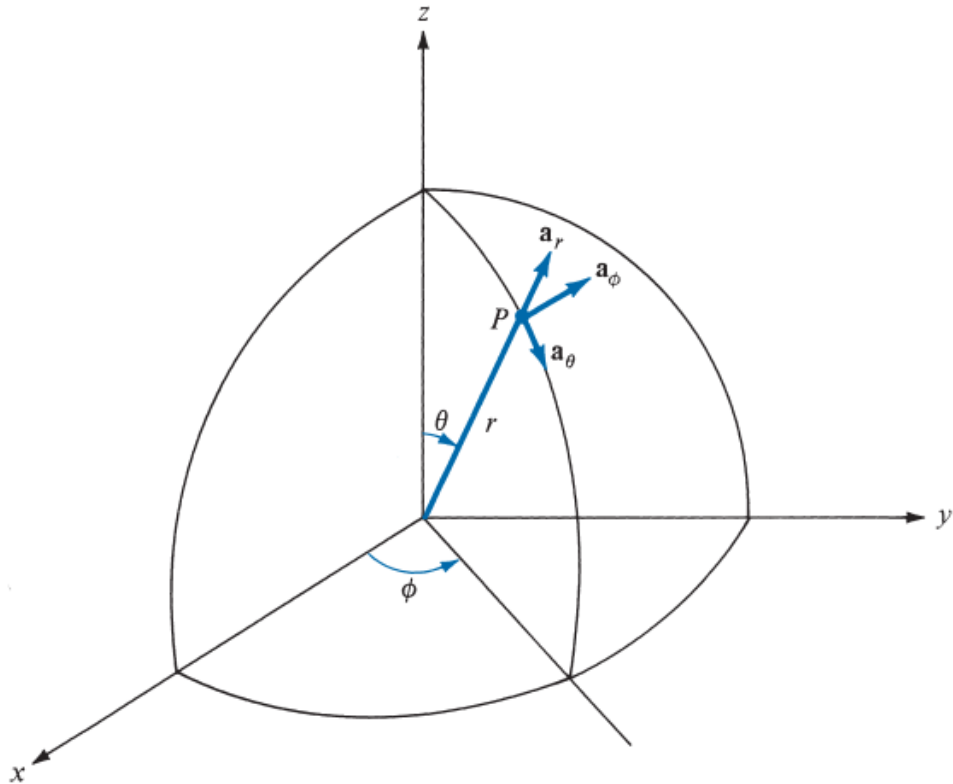
Coordinate Systems – Spherical Coordinates

- A vector can be shown as $A_r \mathbf{a}_r + A_\theta \mathbf{a}_\theta + A_\phi \mathbf{a}_\phi$
- The dot products among the unit vectors: $\mathbf{a}_r \cdot \mathbf{a}_r = \mathbf{a}_\theta \cdot \mathbf{a}_\theta = \mathbf{a}_\phi \cdot \mathbf{a}_\phi = 1$
 $\mathbf{a}_r \cdot \mathbf{a}_\theta = \mathbf{a}_\theta \cdot \mathbf{a}_\phi = \mathbf{a}_\phi \cdot \mathbf{a}_r = 0$
- For the cross products we have:

$$\mathbf{a}_r \times \mathbf{a}_\theta = \mathbf{a}_\phi$$

$$\mathbf{a}_\theta \times \mathbf{a}_\phi = \mathbf{a}_r$$

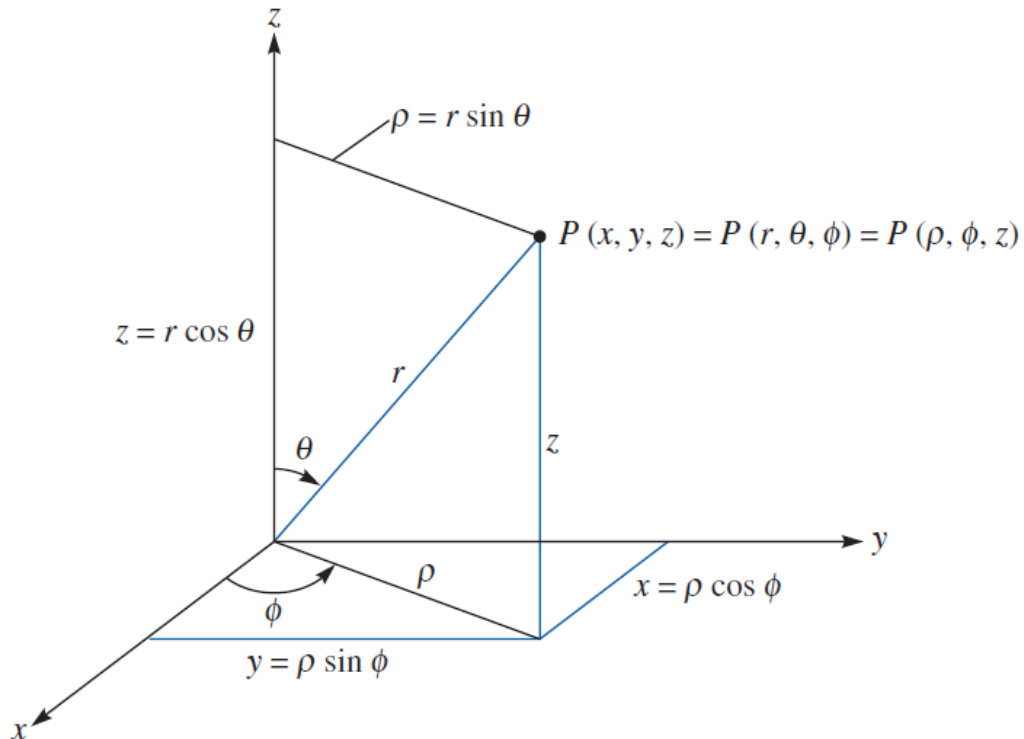
$$\mathbf{a}_\phi \times \mathbf{a}_r = \mathbf{a}_\theta$$



Coordinate Systems – Spherical Coordinates

$$r = \sqrt{x^2 + y^2 + z^2}, \quad \theta = \tan^{-1} \frac{\sqrt{x^2 + y^2}}{z}, \quad \phi = \tan^{-1} \frac{y}{x}$$

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta$$

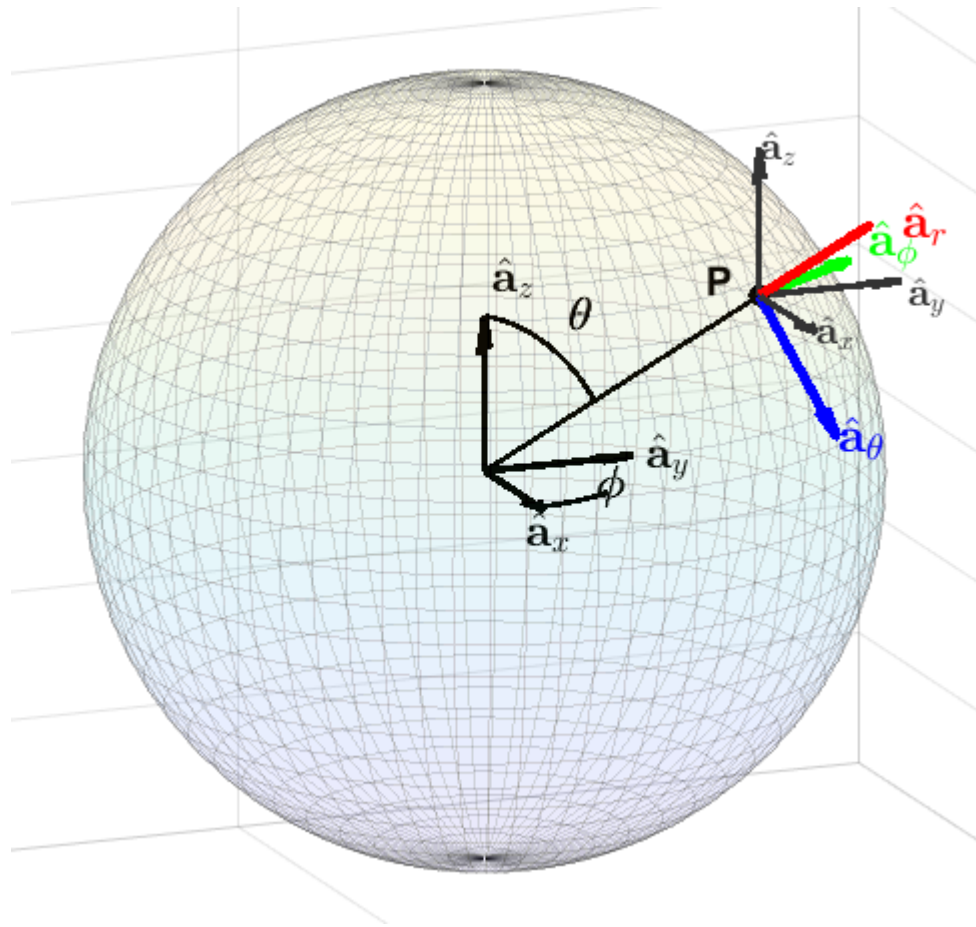


$$\mathbf{a}_x = \sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \theta \sin \phi \mathbf{a}_r + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi$$

$$\mathbf{a}_z = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_\theta$$

Coordinate Systems – Spherical Coordinates, Unit Vectors



$$\mathbf{a}_x = \sin \theta \cos \phi \mathbf{a}_r + \cos \theta \cos \phi \mathbf{a}_\theta - \sin \phi \mathbf{a}_\phi$$

$$\mathbf{a}_y = \sin \theta \sin \phi \mathbf{a}_r + \cos \theta \sin \phi \mathbf{a}_\theta + \cos \phi \mathbf{a}_\phi$$

$$\mathbf{a}_z = \cos \theta \mathbf{a}_r - \sin \theta \mathbf{a}_\theta$$

$$\mathbf{a}_r = \sin \theta \cos \phi \mathbf{a}_x + \sin \theta \sin \phi \mathbf{a}_y + \cos \theta \mathbf{a}_z$$

$$\mathbf{a}_\theta = \cos \theta \cos \phi \mathbf{a}_x + \cos \theta \sin \phi \mathbf{a}_y - \sin \theta \mathbf{a}_z$$

$$\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$$

Coordinate Systems – Matrix Form

➤ Rectangular → Spherical

$$\begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \sin \theta \sin \phi & \cos \theta \\ \cos \theta \cos \phi & \cos \theta \sin \phi & -\sin \theta \\ -\sin \phi & \cos \phi & 0 \end{bmatrix} \begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix}$$

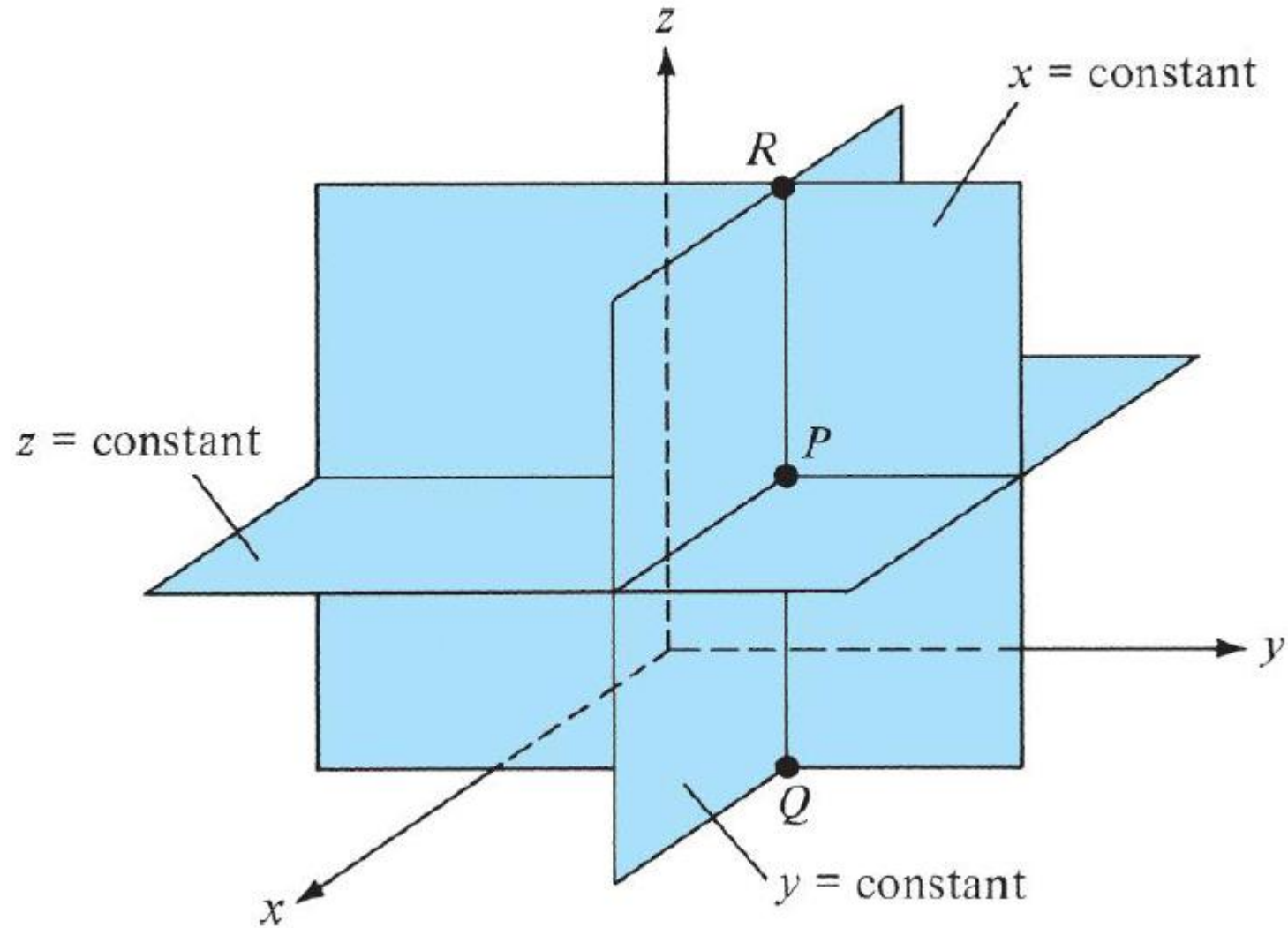
➤ Spherical → Rectangular

$$\begin{bmatrix} A_x \\ A_y \\ A_z \end{bmatrix} = \begin{bmatrix} \sin \theta \cos \phi & \cos \theta \cos \phi & -\sin \phi \\ \sin \theta \sin \phi & \cos \theta \sin \phi & \cos \phi \\ \cos \theta & -\sin \theta & 0 \end{bmatrix} \begin{bmatrix} A_r \\ A_\theta \\ A_\phi \end{bmatrix}$$

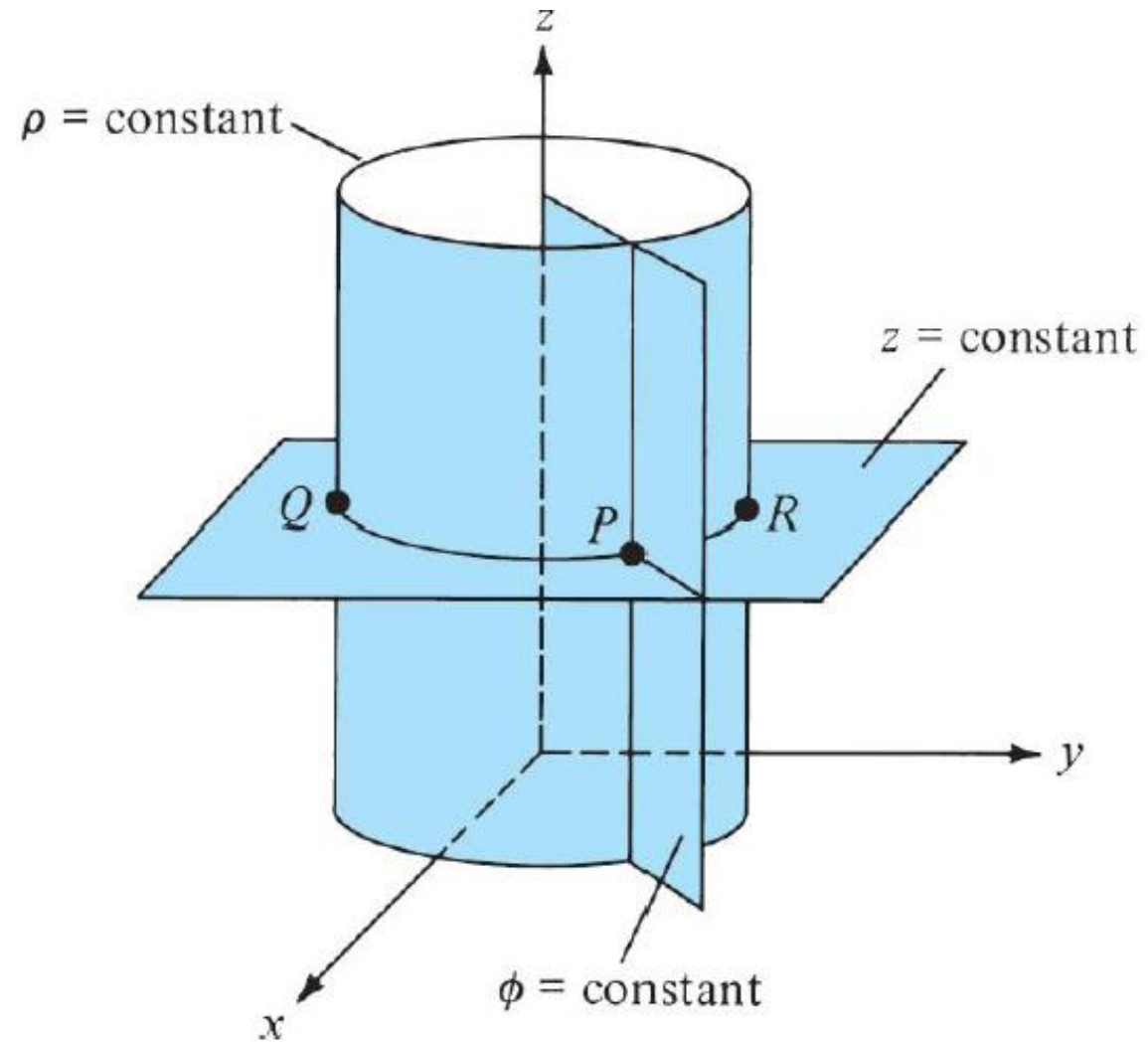
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- ✓ Spherical Coordinates
- **Constant-Coordinate Surfaces**

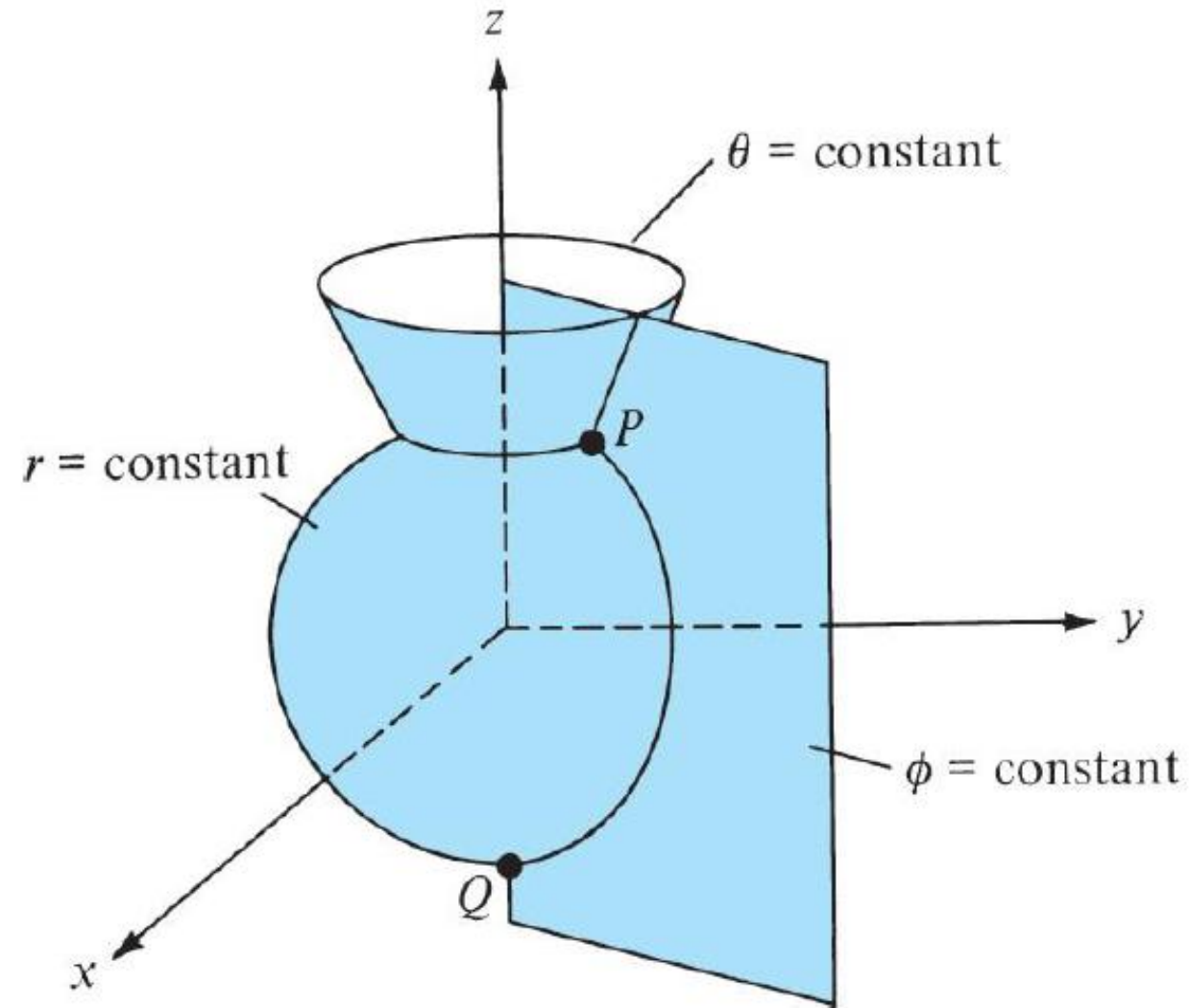
Coordinate Systems – Constant Surfaces in Rectangular



Coordinate Systems – Constant Surfaces in Cylindrical



Coordinate Systems – Constant Surfaces in Spherical



Vector Algebra / Coordinate Systems

- ✓ Vector triple product
- ✓ Components of a vector
- ✓ Circular cylindrical coordinates
- ✓ Spherical Coordinates
- ✓ Constant-Coordinate Surfaces

Demonstrate the ambiguity that results when the cross product is used to find the angle between two vectors by finding the angle between $\mathbf{A} = 3\mathbf{a}_x - 2\mathbf{a}_y + 4\mathbf{a}_z$ and $\mathbf{B} = 2\mathbf{a}_x + \mathbf{a}_y - 2\mathbf{a}_z$. Does this ambiguity exist when the dot product is used?

Find (a) the vector component of $\mathbf{F} = 10\mathbf{a}_x - 6\mathbf{a}_y + 5\mathbf{a}_z$ that is parallel to $\mathbf{G} = 0.1\mathbf{a}_x + 0.2\mathbf{a}_y + 0.3\mathbf{a}_z$; (b) the vector component of $\mathbf{F}$ that is perpendicular to $\mathbf{G}$; (c) the vector component of $\mathbf{G}$ that is perpendicular to $\mathbf{F}$.

Three vectors extending from the origin are given as $\mathbf{r}_1 = (7, 3, -2)$, $\mathbf{r}_2 = (-2, 7, -3)$, and $\mathbf{r}_3 = (0, 2, 3)$. Find (a) a unit vector perpendicular to both $\mathbf{r}_1$ and $\mathbf{r}_2$; (b) a unit vector perpendicular to the vectors $\mathbf{r}_1 - \mathbf{r}_2$ and $\mathbf{r}_2 - \mathbf{r}_3$; (c) the area of the triangle defined by $\mathbf{r}_1$ and $\mathbf{r}_2$; (d) the area of the triangle defined by the heads of $\mathbf{r}_1$, $\mathbf{r}_2$, and $\mathbf{r}_3$.

a) $\mathbf{r}_1 \times \mathbf{r}_2$

$$= \begin{vmatrix} a_x & a_y & a_z \\ 7 & 3 & -2 \\ -2 & 7 & -3 \end{vmatrix}$$

$$= (-9 - 14, \dots, \dots)$$

$$= (5, 25, 55)$$

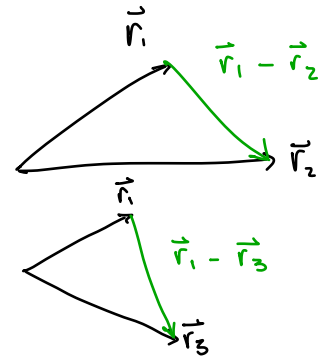
$$\begin{aligned} a_{\mathbf{r}_1 \times \mathbf{r}_2} &= \frac{\vec{r}_1 \times \vec{r}_2}{|\vec{r}_1 \times \vec{r}_2|} \\ &= \frac{1}{\sqrt{5^2 + 25^2 + 55^2}} \mathbf{r}_1 \times \mathbf{r}_2 \\ &= \frac{1}{35\sqrt{3}} \vec{r}_1 \times \vec{r}_2 \\ &= \frac{1}{7\sqrt{3}} \vec{a}_x + \frac{5}{7\sqrt{3}} \vec{a}_y + \frac{11}{7\sqrt{3}} \vec{a}_z \end{aligned}$$

b) $\mathbf{r}_1 - \mathbf{r}_2$

$$\mathbf{r}_1 - \mathbf{r}_3$$

c) $A_A = \frac{1}{2} |\vec{r}_1 \times \vec{r}_2|$
 $= \frac{35}{2} \sqrt{3}$

d) $A_\Delta = \frac{1}{2} |(\vec{r}_1 - \vec{r}_2) \times (\vec{r}_1 - \vec{r}_3)|$



Express in cylindrical components: (a) the vector from $C(3, 2, -7)$ to $D(-1, -4, 2)$; (b) a unit vector at D directed toward C ; (c) a unit vector at D directed toward the origin.

A vector field is expressed as $\mathbf{F} = Az \mathbf{a}_z$ where A is a constant. Evaluate the components of this field that are (a) normal and (b) tangent to a spherical surface of radius a . Express all results in spherical coordinates and components.

A vector field is defined in spherical coordinates by

$$\mathbf{E}(r, \theta) = \frac{A}{r^3} (2 \cos \theta \mathbf{a}_r + \sin \theta \mathbf{a}_\theta),$$

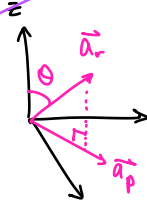
where A is a constant and $r > 0$.

- Determine the surface on which the vector field is everywhere perpendicular to the xy -plane. Express the field on this surface in cylindrical coordinates.
- Determine the coordinate axis along which the field is everywhere perpendicular to the xy -plane and express the field there in cylindrical coordinates.
- Determine the surface on which the vector field is everywhere parallel to the xy -plane.

$$\vec{a}_r = \sin \theta \vec{a}_\rho + \cos \theta \vec{a}_z$$

↓ derivative

$$\vec{a}_\theta = \cos \theta \vec{a}_\rho - \sin \theta \vec{a}_z$$



$$\vec{E} = \frac{A}{r^3} \left(3 \sin \theta \cos \theta \vec{a}_\rho + (3 \cos^3 \theta - 1) \vec{a}_z \right)$$

a) $\perp$ to xy plane

$$3 \sin \theta \cos \theta = 0$$

$$\theta = \left(0, \frac{\pi}{2}, \pi \right)$$

$$\rho = r \sin \theta$$

$$\theta = \frac{\pi}{2} \rightarrow z=0, r=\rho, 3 \cos^3 \theta - 1 = -1$$

$$\vec{E}|_{z=0} = \frac{-A}{\rho^3} \vec{a}_z$$

* 2.1 Convert the following Cartesian points to cylindrical and spherical coordinates:

(a) $P(2, 5, 1)$

(b) $Q(-3, 4, 0)$

(c) $R(6, 2, -4)$

a) $P(2, 5, 1)$

cylindrical:

$$\rho = \sqrt{2^2 + 5^2}$$

$$= \sqrt{29}$$

$$\phi = \tan^{-1}\left(\frac{y}{x}\right)$$

$$= \tan^{-1}\left(\frac{5}{2}\right) \quad * \& \text{ in which quadratic}$$

$$\vec{P}_c = (\sqrt{29}, \tan^{-1}\left(\frac{5}{2}\right), 1)$$

$\uparrow$ $\uparrow$ $\uparrow$
 $\#$ θ $\#$

spherical:

$$r = \sqrt{2^2 + 5^2 + 1^2}$$

$$= \sqrt{30}$$

$$\phi = \tan^{-1}\left(\frac{5}{2}\right)$$

$$\theta = \cos^{-1}\left(\frac{z}{r}\right)$$

$$= \cos^{-1}\left(\frac{1}{\sqrt{30}}\right)$$

$$\vec{P}_s = (\sqrt{30}, \tan^{-1}\left(\frac{5}{2}\right), \cos^{-1}\left(\frac{1}{\sqrt{30}}\right))$$

$\uparrow$ $\uparrow$ $\uparrow$
 $\#$ θ θ

* rectangular $\rightarrow (\#, \#, \#)$
 cylind $\rightarrow (\#, \theta, \#)$
 spher $\rightarrow (\#, \theta, \theta)$

2.2 Express the following points in Cartesian coordinates:

(a) $P_1(2, 30^\circ, 5)$

(b) $P_2(1, 90^\circ, -3)$

(c) $P_3(10, \pi/4, \pi/3)$

(d) $P_4(4, 30^\circ, 60^\circ)$

c) $\vec{P}_3(10, \pi/4, \pi/3)$
 $\begin{matrix} \uparrow & \uparrow & \uparrow \\ \phi & \theta & \theta \end{matrix}$

$$\vec{x} = 10 \sin \pi/4 \cos \pi/3 = \frac{5\sqrt{2}}{2}$$

$$y = 10 \sin \pi/4 \sin \pi/3 = \frac{5\sqrt{6}}{2}$$

$$z = 10 \cos \pi/4 = 5\sqrt{2}$$

2.3 The rectangular coordinates at point P are $(x = 2, y = 6, z = -4)$. (a) What are its cylindrical coordinates? (b) What are its spherical coordinates?

2.4 The cylindrical coordinates of point Q are $\rho = 5$, $\phi = 120^\circ$, $z = 1$. Express Q as rectangular and spherical coordinates.

2.5 Given point $T(10, 60^\circ, 30^\circ)$ in spherical coordinates, express T in Cartesian and cylindrical coordinates.

2.6 (a) If $V = xz - xy + yz$, express V in cylindrical coordinates.

(b) If $U = x^2 + 2y^2 + 3z^2$, express U in spherical coordinates.

2.7 Convert the following vectors to cylindrical and spherical systems:

$$(a) \quad \mathbf{F} = \frac{x\mathbf{a}_x + y\mathbf{a}_y + 4\mathbf{a}_z}{\sqrt{x^2 + y^2 + z^2}}$$

$$(b) \quad \mathbf{G} = (x^2 + y^2) \left[\frac{x\mathbf{a}_x}{\sqrt{x^2 + y^2 + z^2}} + \frac{y\mathbf{a}_y}{\sqrt{x^2 + y^2 + z^2}} + \frac{z\mathbf{a}_z}{\sqrt{x^2 + y^2 + z^2}} \right]$$

2.8 Let $\mathbf{B} = \sqrt{x^2 + y^2} \mathbf{a}_x + \frac{y}{\sqrt{x^2 + y^2}} \mathbf{a}_y + z \mathbf{a}_z$. Transform $\mathbf{B}$ to cylindrical coordinates.

2.9 Given vector $\mathbf{A} = 2\mathbf{a}_\rho + 3\mathbf{a}_\phi + 4\mathbf{a}_z$, convert $\mathbf{A}$ into Cartesian coordinates at point $(2, \pi/2, -1)$.

2.10 Express the following vectors in rectangular coordinates:

(a) $\mathbf{A} = \rho \sin \phi \mathbf{a}_\rho + \rho \cos \phi \mathbf{a}_\phi - 2z \mathbf{a}_z$

(b) $\mathbf{B} = 4r \cos \phi \mathbf{a}_r + r \mathbf{a}_\theta$

2.17 At point $P(2,0,-1)$, calculate the value of the following dot products:

(a) $\mathbf{a}_\rho \cdot \mathbf{a}_x$, (b) $\mathbf{a}_\phi \cdot \mathbf{a}_y$, (c) $\mathbf{a}_r \cdot \mathbf{a}_z$

2.26 At point $(0, 4, -1)$, express $\mathbf{a}_\rho$ and $\mathbf{a}_\phi$ in Cartesian coordinates.

2.29 Given that $\mathbf{B} = \rho^2 \sin \phi \mathbf{a}_\rho + (z - 1) \cos \phi \mathbf{a}_\phi + z^2 \mathbf{a}_z$, find $\mathbf{B} \cdot \mathbf{a}_x$ at $(4, \pi/4, -1)$.

2.30 A vector field in “mixed” coordinate variables is given by

$$\mathbf{G} = \frac{x \cos \phi}{\rho} \mathbf{a}_x + \frac{2yz}{\rho^2} \mathbf{a}_y + \left(1 - \frac{x^2}{\rho^2}\right) \mathbf{a}_z$$

Express $\mathbf{G}$ completely in the spherical system.

2.33 If $\mathbf{H} = \rho^2 \cos \phi \mathbf{a}_\rho - \rho \sin \phi \mathbf{a}_\phi$, find $\mathbf{H} \cdot \mathbf{a}_x$ at point $P(2, 60^\circ, -1)$.

Vector Calculus

Today's plan

- Differential Length, Area, and Volume (3.2 – p. 59)
- Line, Surface, and Volume Integrals (3.3 – p. 66)

Vector Calculus

- **Differential Length, Area, and Volume**
 - **Rectangular Coordinate System**
 - Cylindrical Coordinate Systems
 - Spherical Coordinate Systems
- Line, Surface, and Volume Integrals

Vector Calculus – Rectangular Coordinate

- Differential displacement or length element,

$$d\mathbf{l} = dx \mathbf{a}_x + dy \mathbf{a}_y + dz \mathbf{a}_z$$

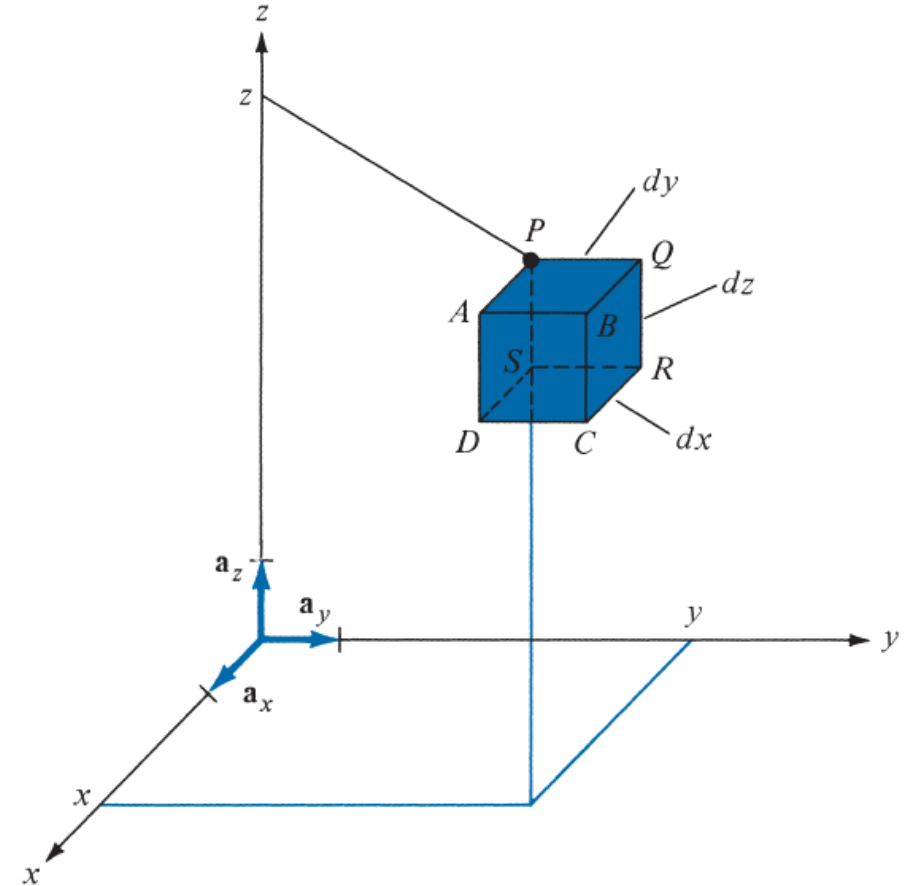
- Differential normal surface area,

$$d\mathbf{S} = \mathbf{n} dS$$

- Differential volume,

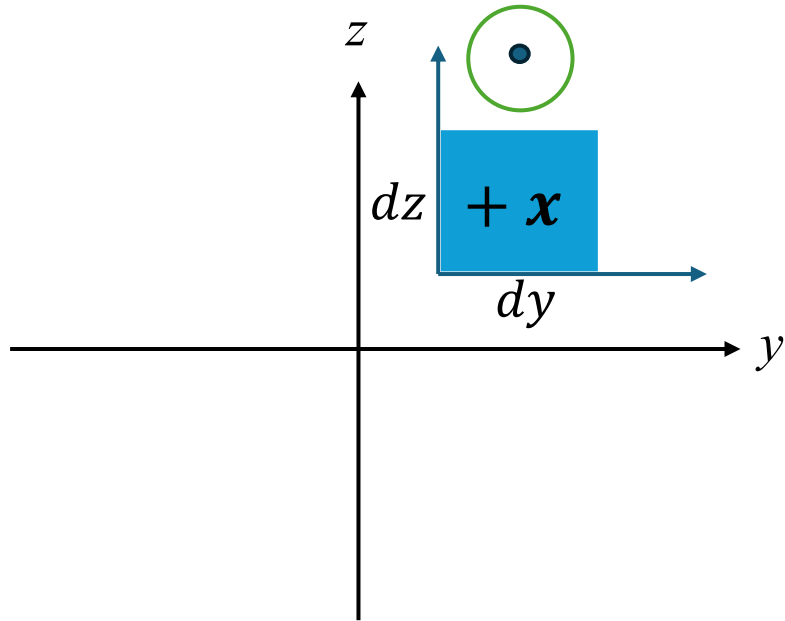
$$dv = dx dy dz$$

$$dx dy dz \mathbf{a}_x \cdot \mathbf{a}_k$$

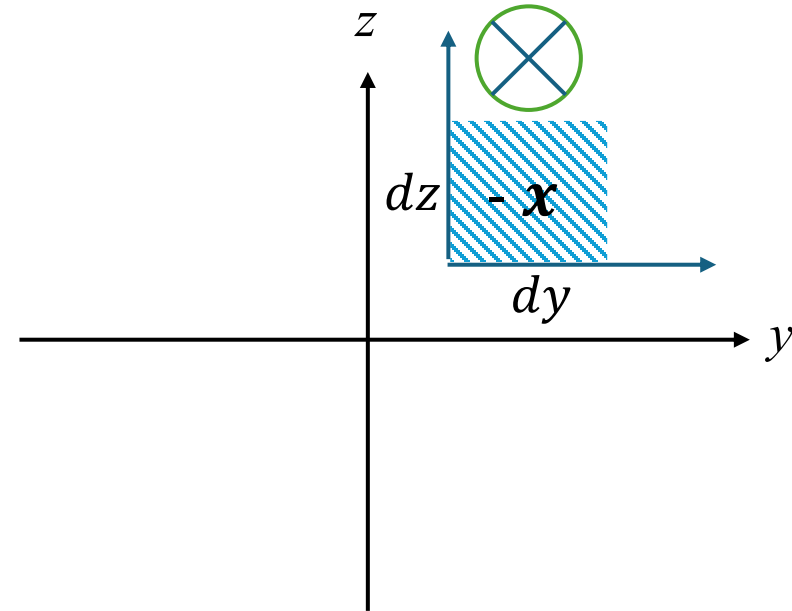


Vector Calculus – Rectangular Coordinate

➤ Differential displacement or length element,



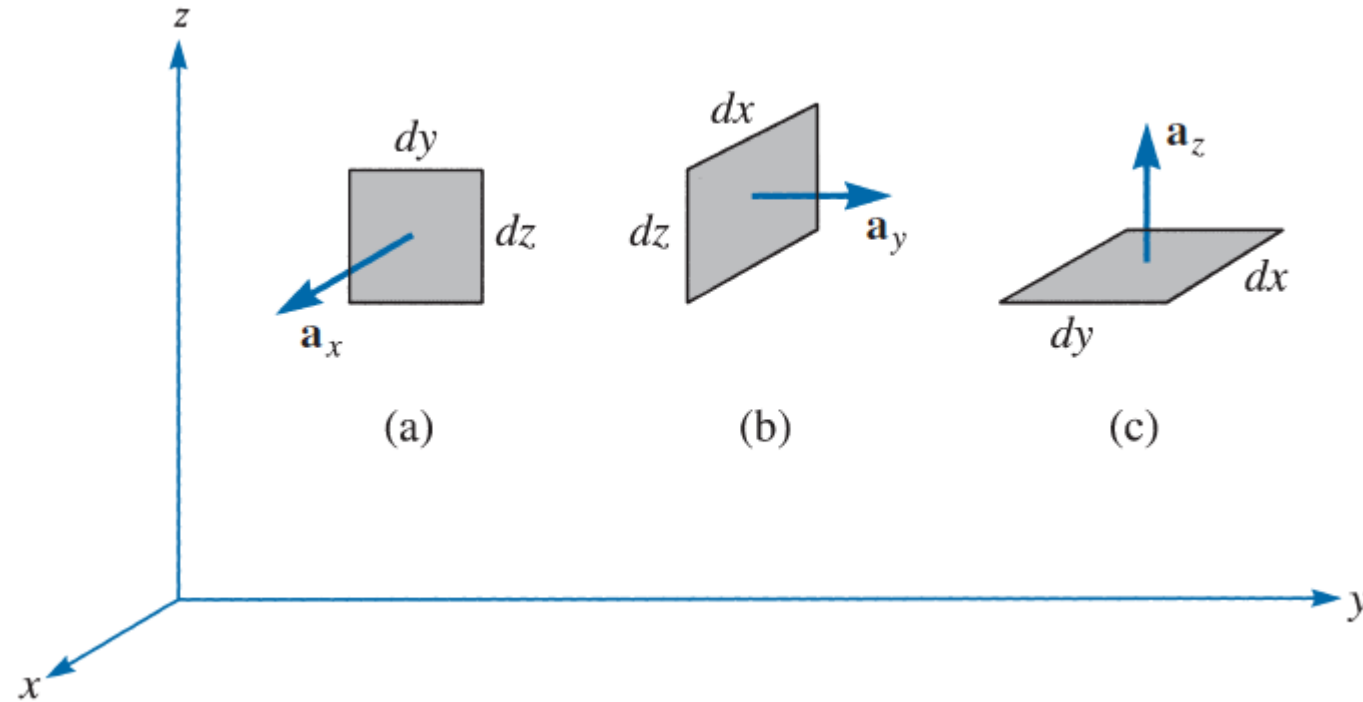
$$\begin{aligned} d\mathbf{S} &= dy\mathbf{a}_y \times dz\mathbf{a}_z \\ &= dydz \, \vec{a}_y \times \vec{a}_z \\ &= dydz \, \vec{a}_x \end{aligned}$$



$$\begin{aligned} d\mathbf{S} &= dz\mathbf{a}_z \times dy\mathbf{a}_y \\ &= dzdy \, \mathbf{a}_z \times \mathbf{a}_y \\ &= dzdy \, -\vec{a}_x \end{aligned}$$

Vector Calculus – Differential Normal Surface Areas in Rectangular Coordinates

- Differential displacement or length element,

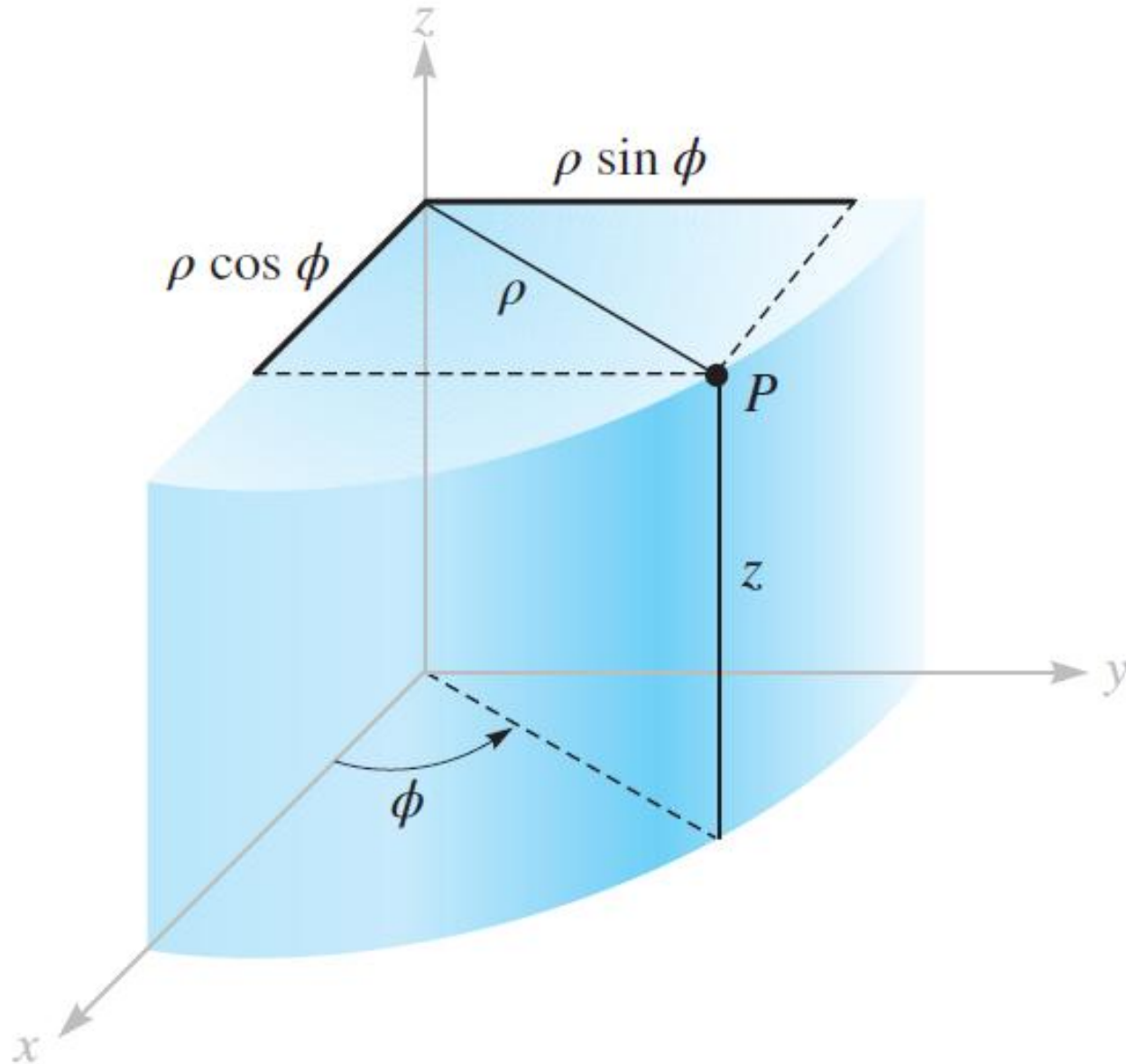


(a) $d\mathbf{S} = dy dz \mathbf{a}_x$, (b) $d\mathbf{S} = dx dz \mathbf{a}_y$, (c) $d\mathbf{S} = dx dy \mathbf{a}_z$.

Vector Calculus

- **Differential Length, Area, and Volume**
 - Rectangular Coordinate System
 - **Cylindrical Coordinate Systems**
 - Spherical Coordinate Systems
- Line, Surface, and Volume Integrals

Vector Calculus

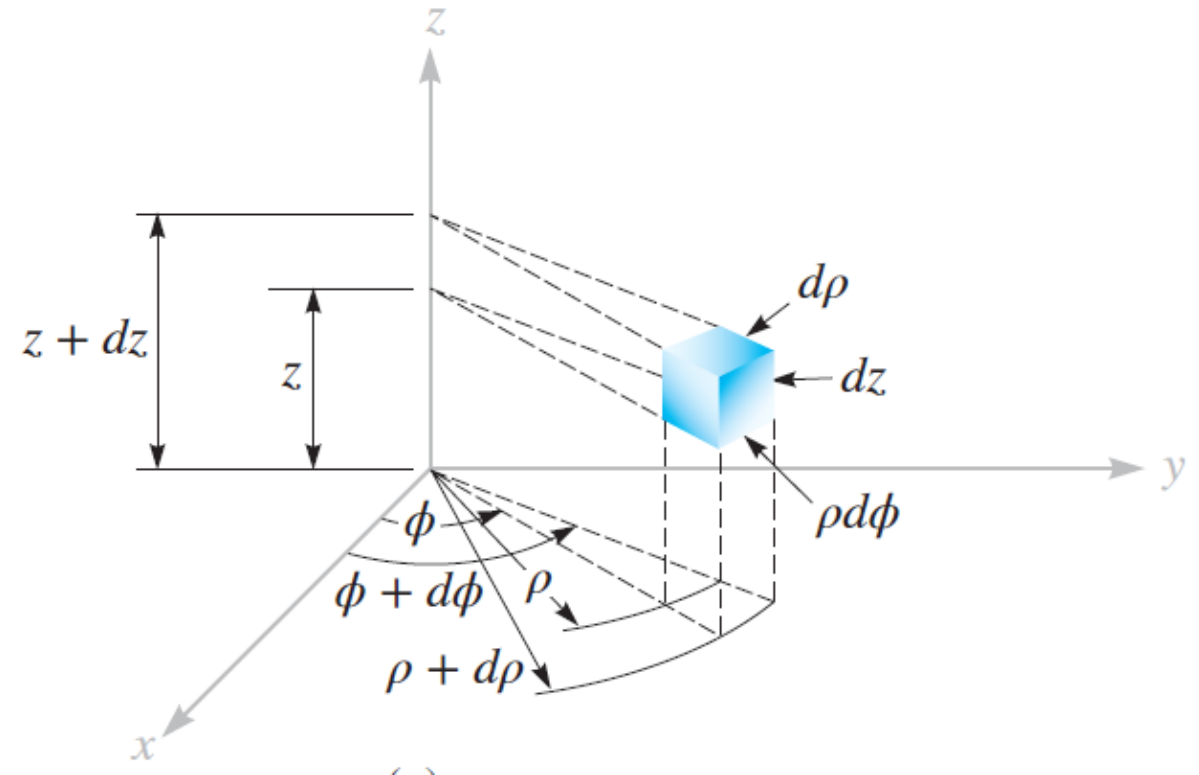
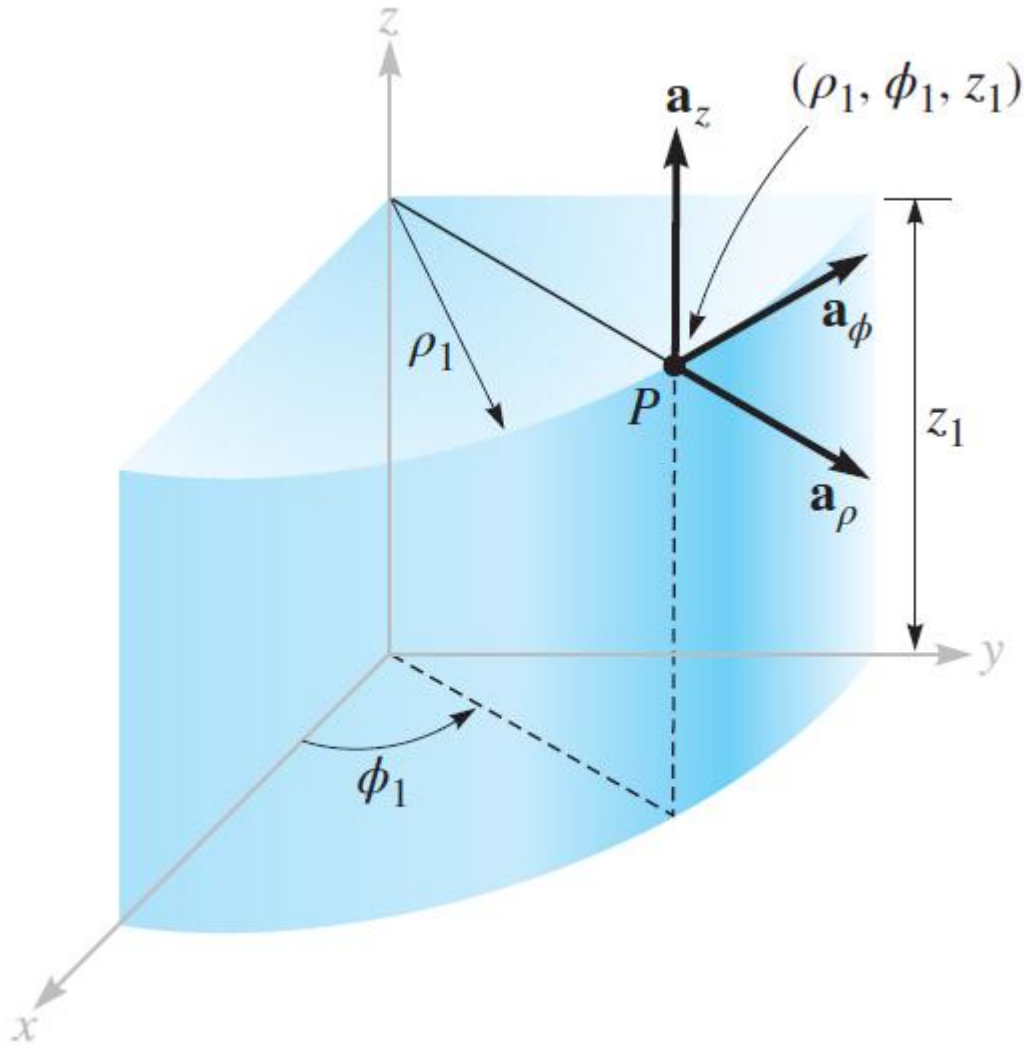


$$\rho = \sqrt{x^2 + y^2}$$

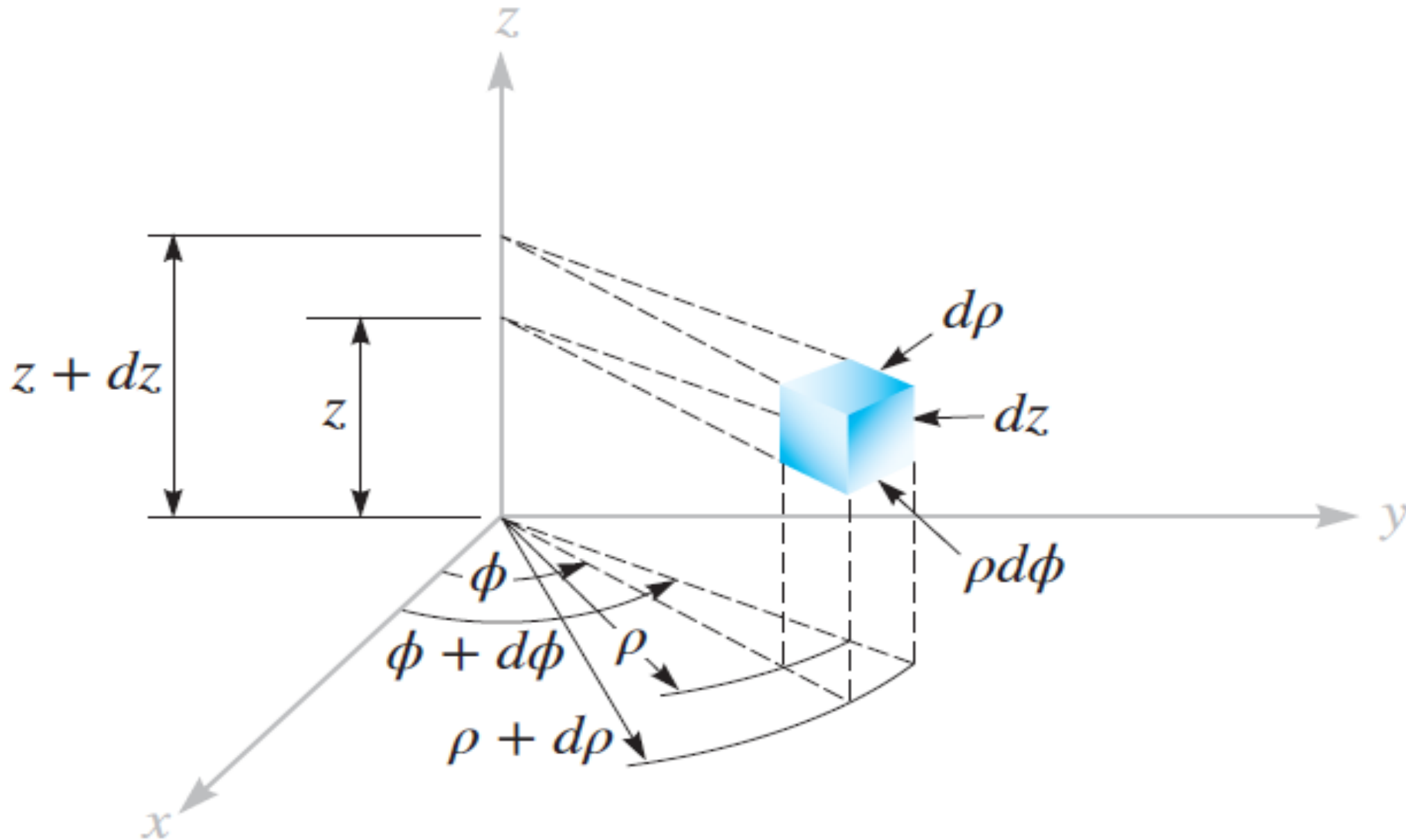
$$\phi = \tan^{-1} \frac{y}{x}$$

$$z = z$$

Vector Calculus – Cylindrical Coordinate



Vector Calculus – Cylindrical Coordinate



Vector Calculus – Cylindrical Coordinate

Vector Calculus – Cylindrical Coordinate

- Differential displacement or length element,

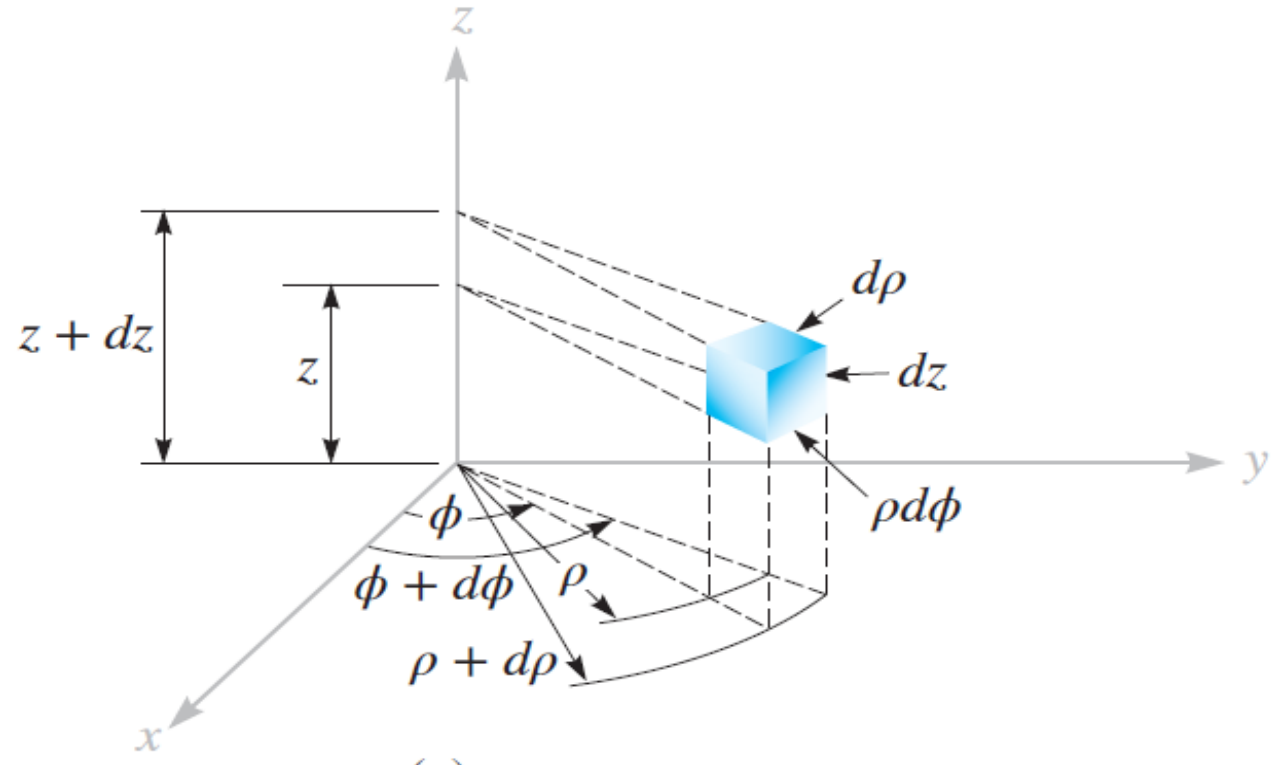
$$d\mathbf{l} = d\rho \mathbf{a}_\rho + \rho d\phi \mathbf{a}_\phi + dz \mathbf{a}_z$$

- Differential normal surface area,

$$d\mathbf{S} = \mathbf{n} dS$$

- Differential volume element,

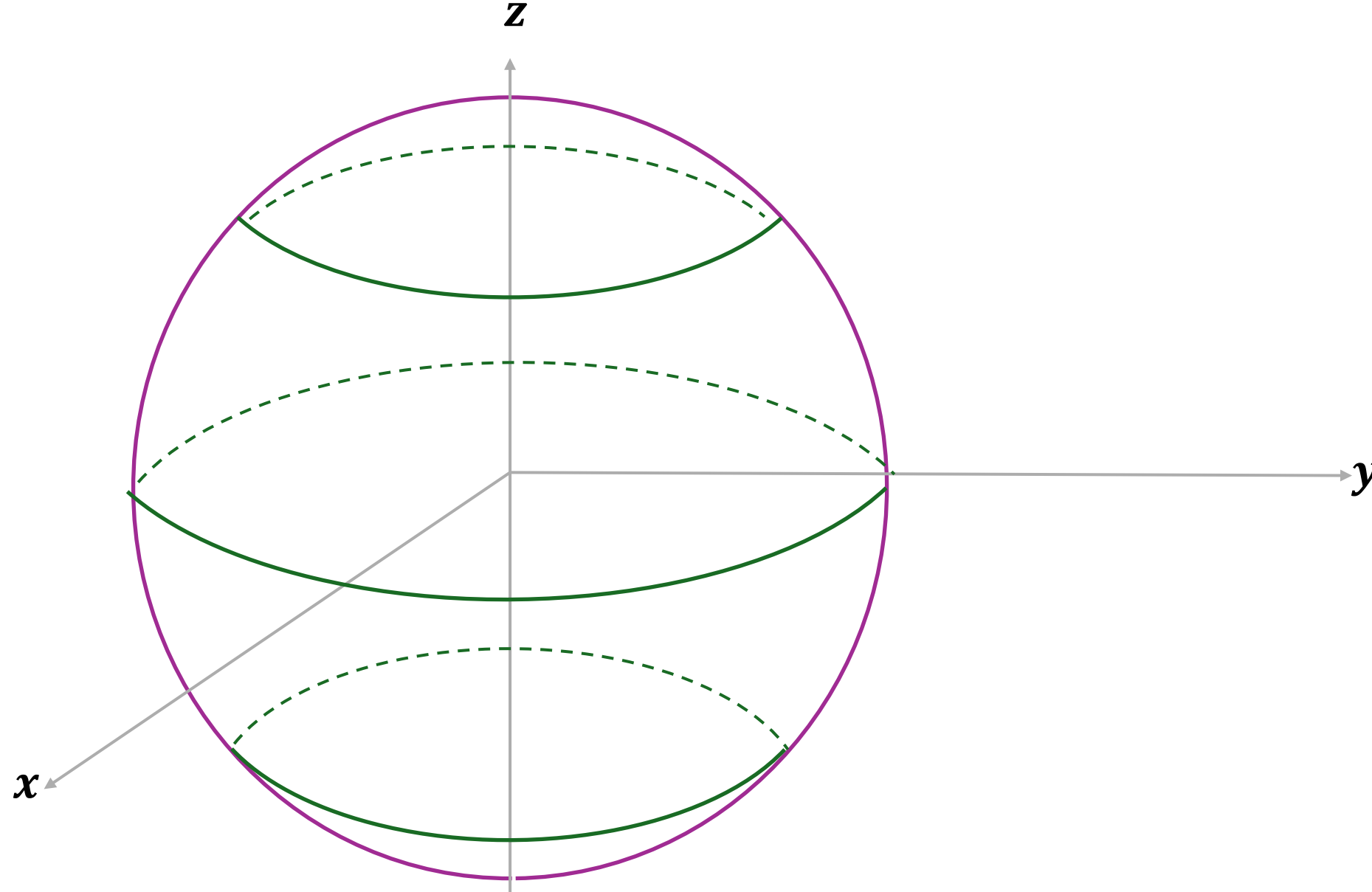
$$dv = \rho d\rho d\phi dz$$



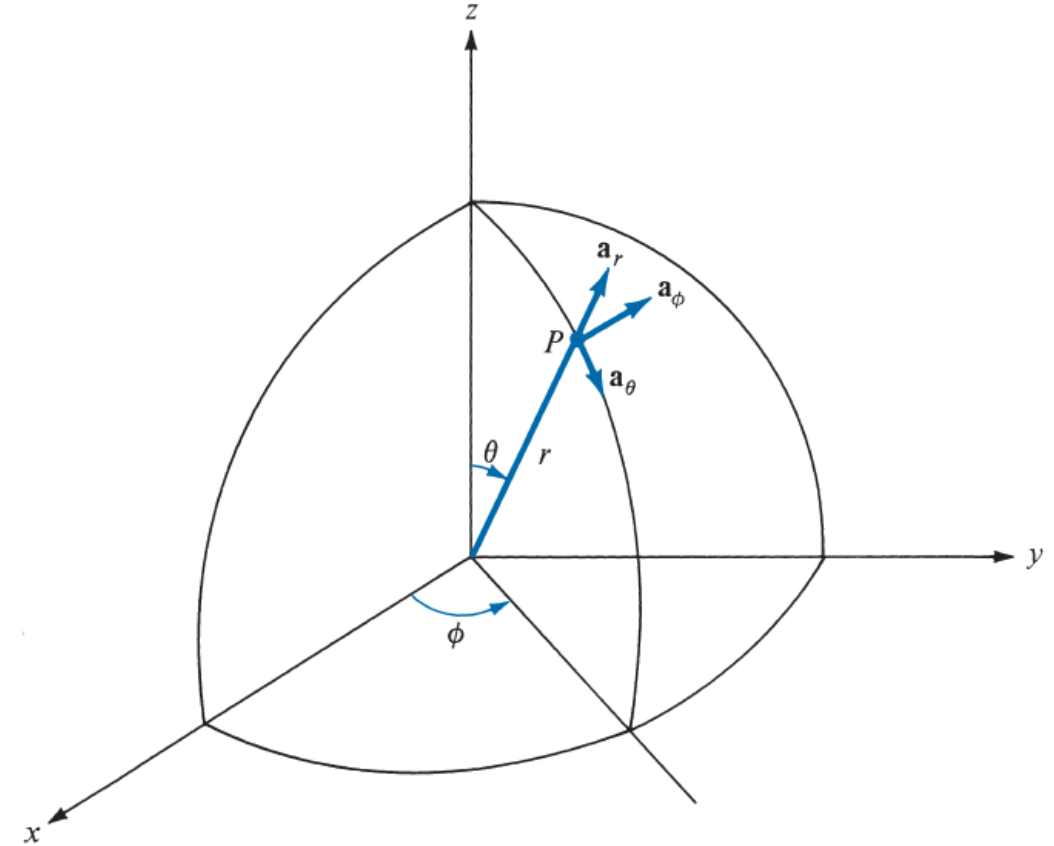
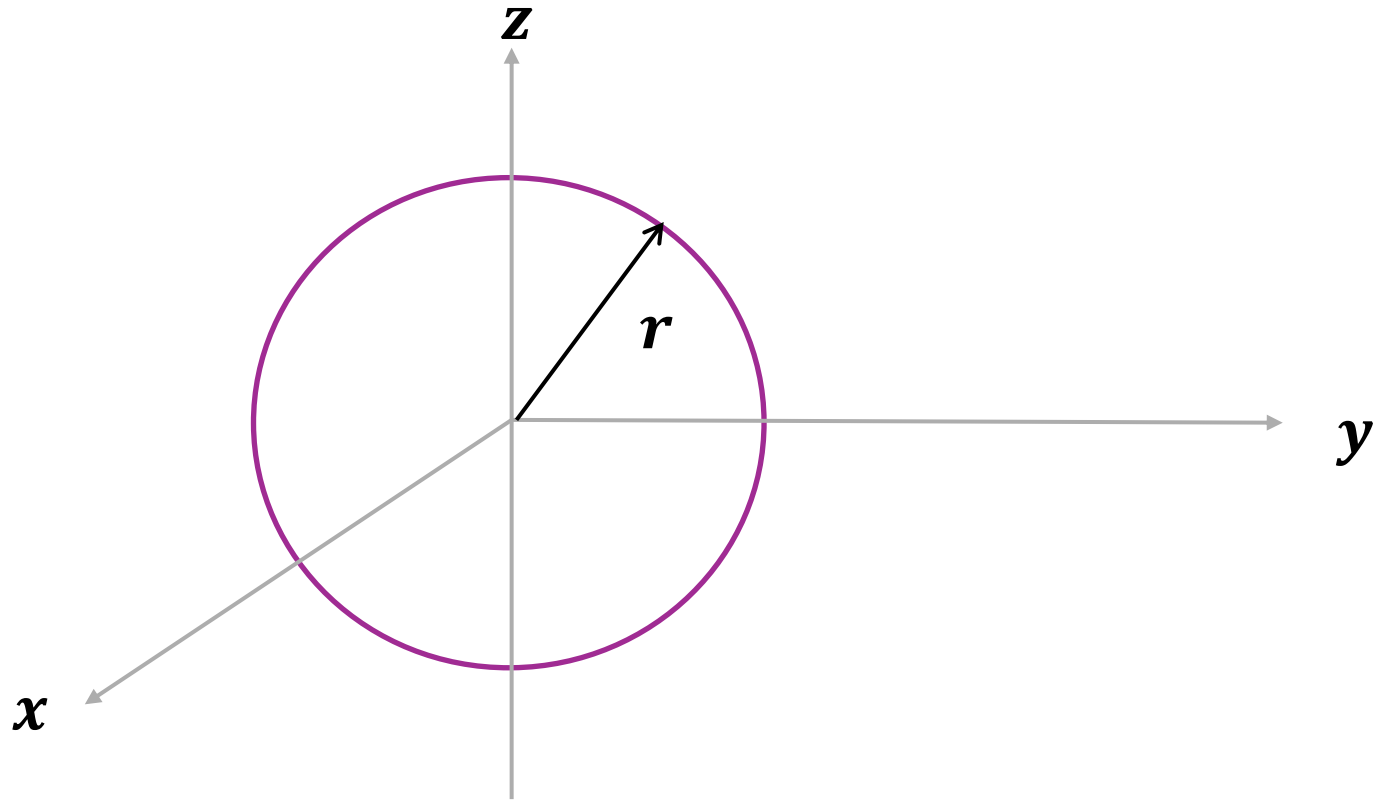
Vector Calculus

- **Differential Length, Area, and Volume**
 - Rectangular Coordinate System
 - Cylindrical Coordinate Systems
 - **Spherical Coordinate Systems**
- Line, Surface, and Volume Integrals

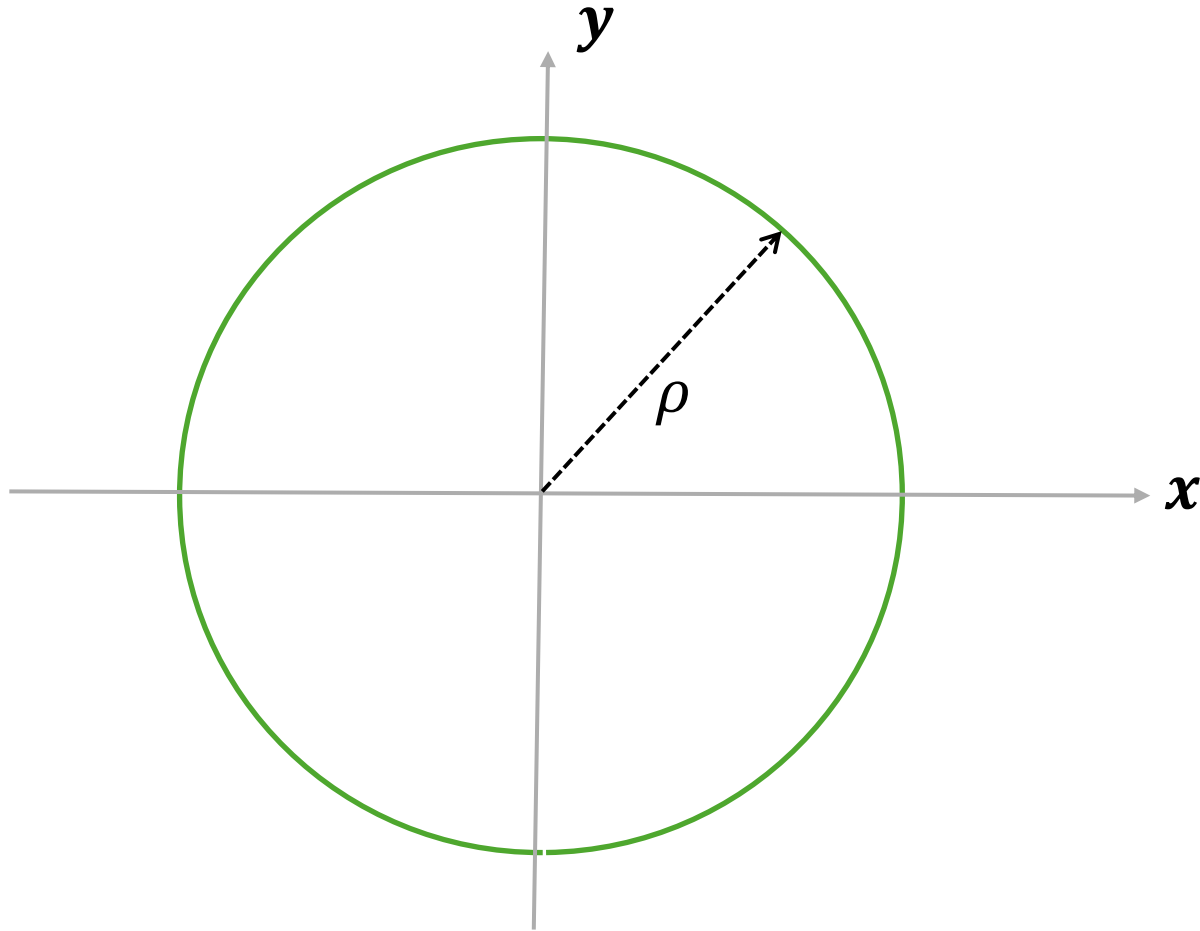
Vector Calculus – Spherical Coordinate Systems



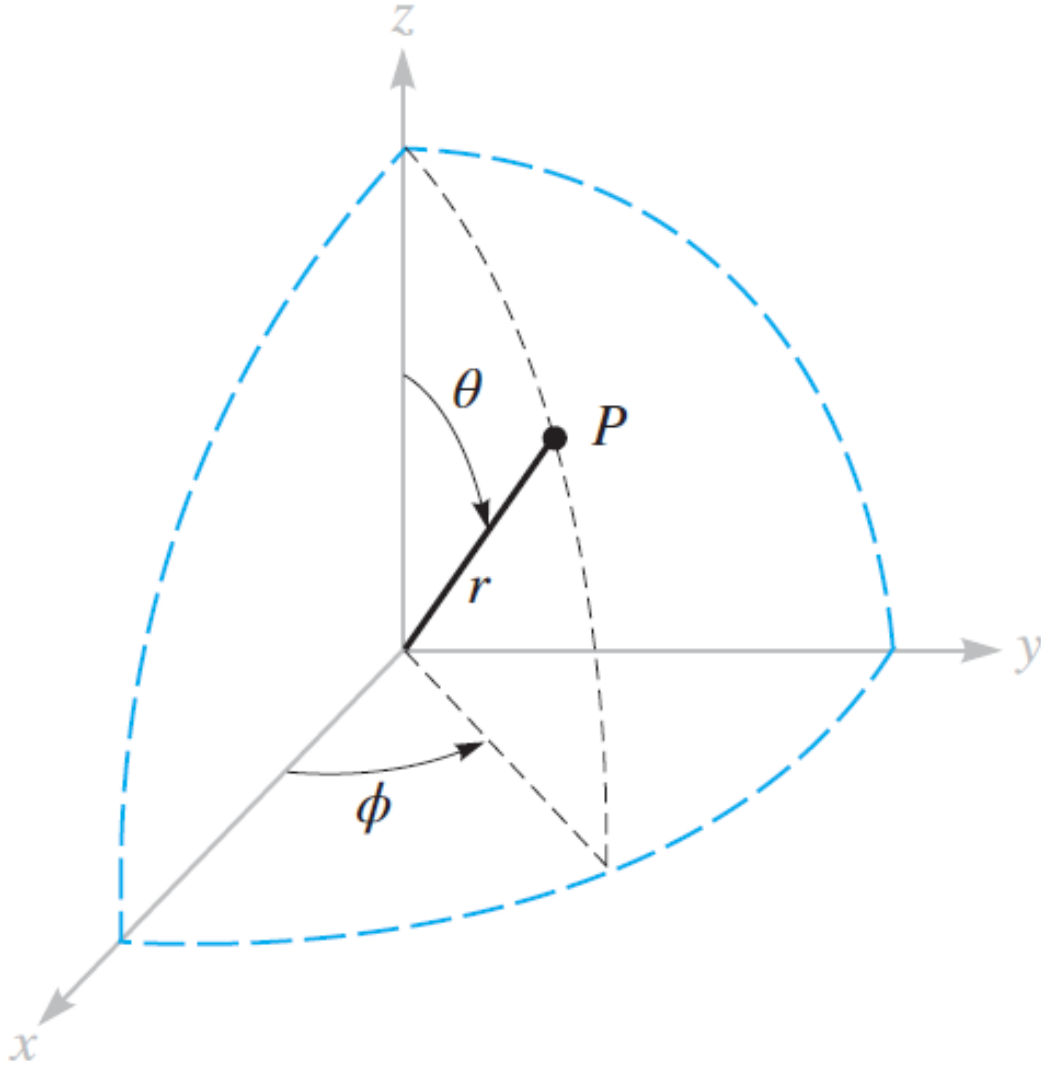
Vector Calculus – Spherical Coordinate Systems



Vector Calculus – Spherical Coordinate Systems

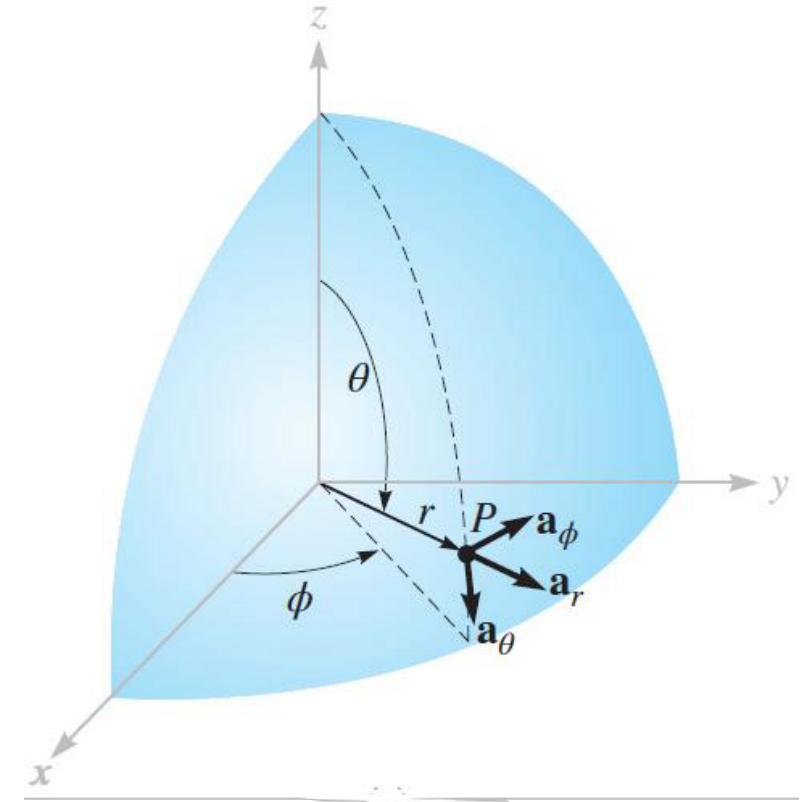
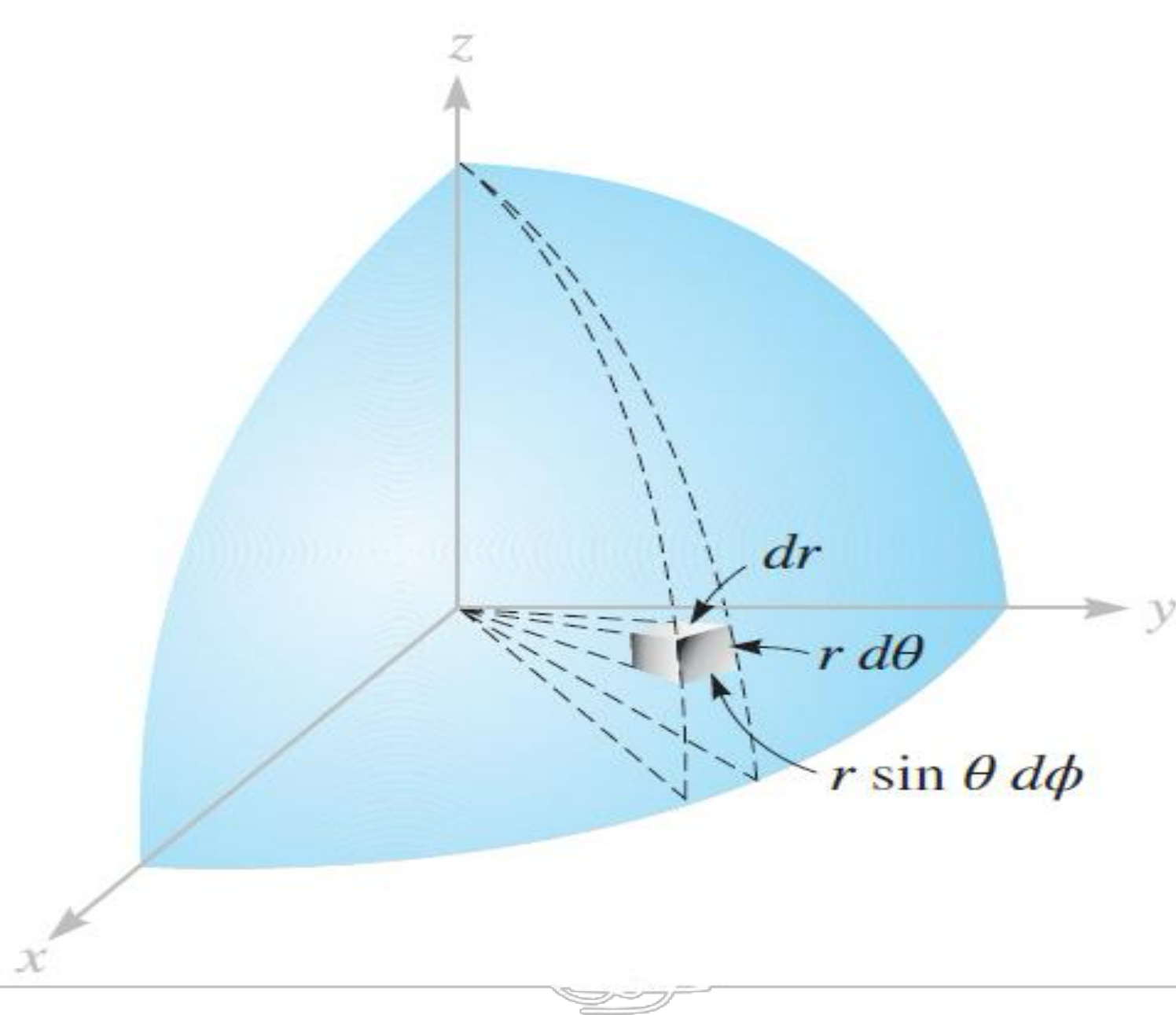


Vector Calculus – Spherical Coordinate Systems



$$r = \sqrt{x^2 + y^2 + z^2}$$
$$\theta = \cos^{-1} \frac{z}{\sqrt{x^2 + y^2 + z^2}}$$
$$\phi = \tan^{-1} \frac{y}{x}$$

Vector Calculus – Spherical Coordinate Systems



Vector Calculus – Spherical Coordinate Systems

- Differential displacement or length element,

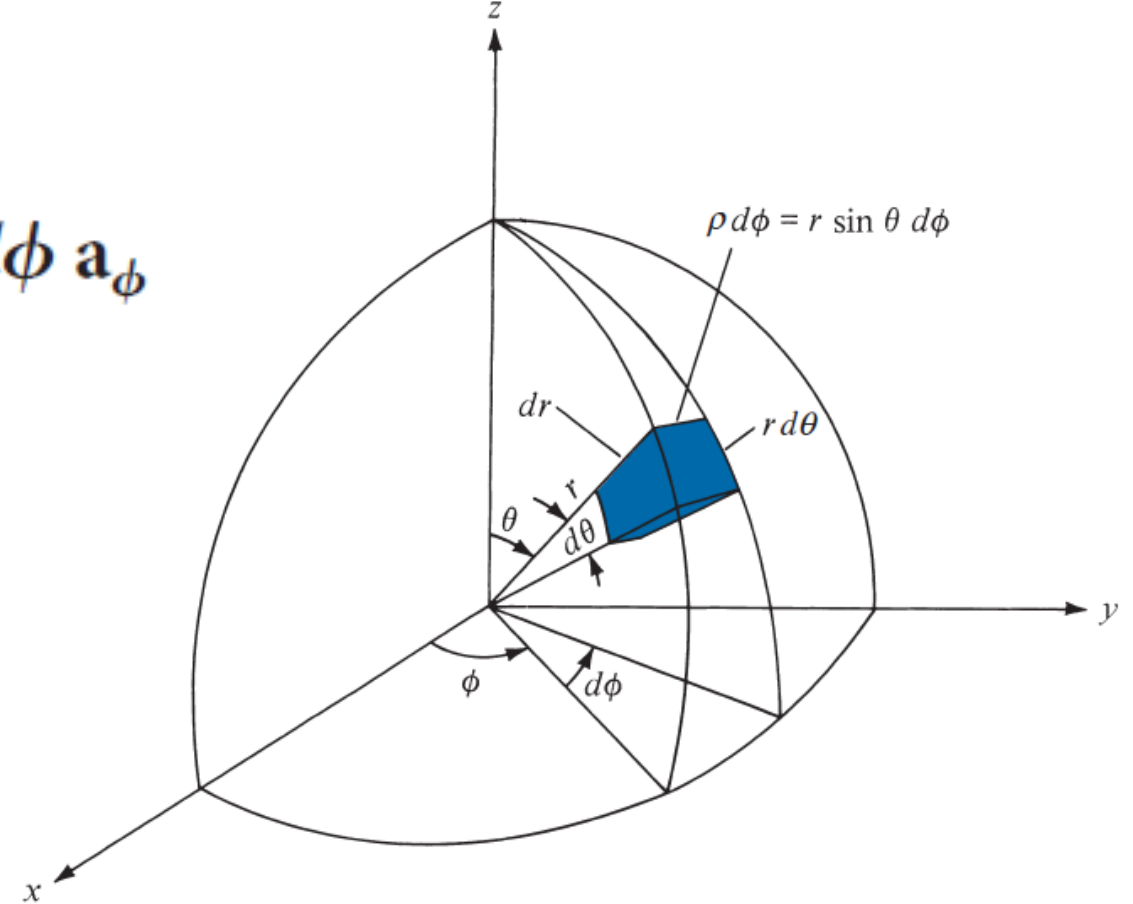
$$d\mathbf{l} = dr \mathbf{a}_r + r d\theta \mathbf{a}_\theta + r \sin \theta d\phi \mathbf{a}_\phi$$

- Differential normal surface area,

$$d\mathbf{S} = \mathbf{n} dS$$

- Differential volume element,

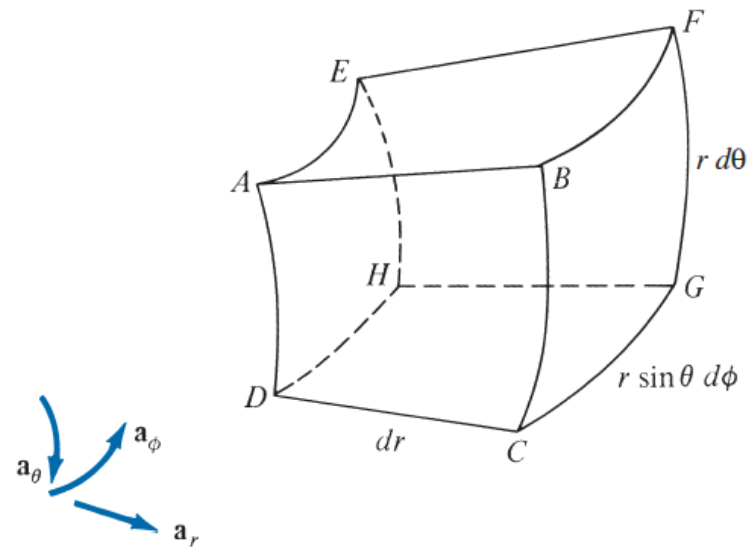
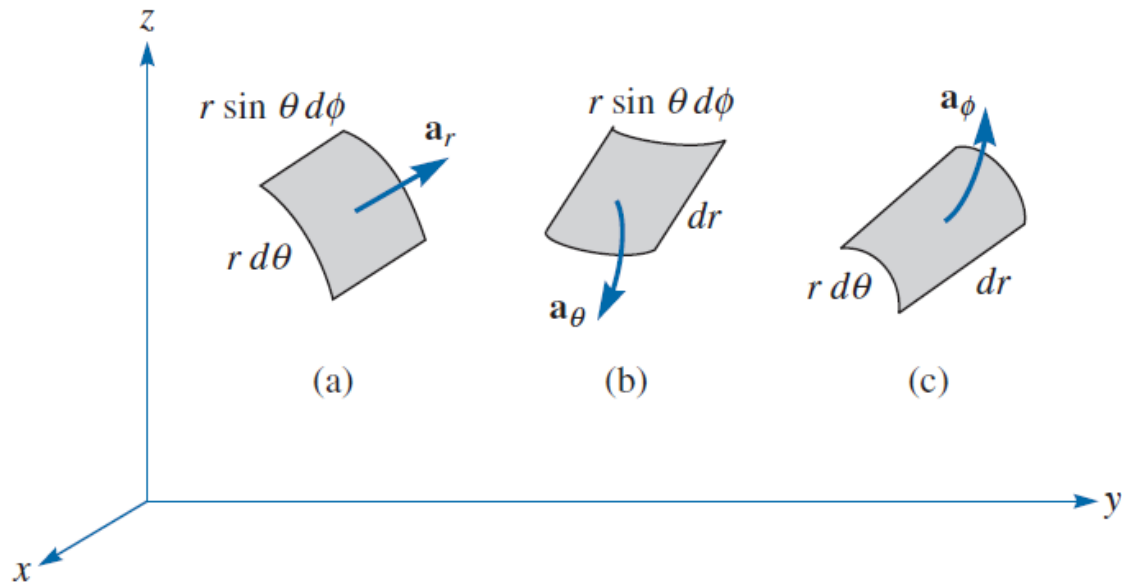
$$dv = r^2 \sin \theta dr d\theta d\phi$$



Vector Calculus – Spherical Coordinate Systems

- Differential normal surface area,

$$d\mathbf{S} = \mathbf{n}dS$$

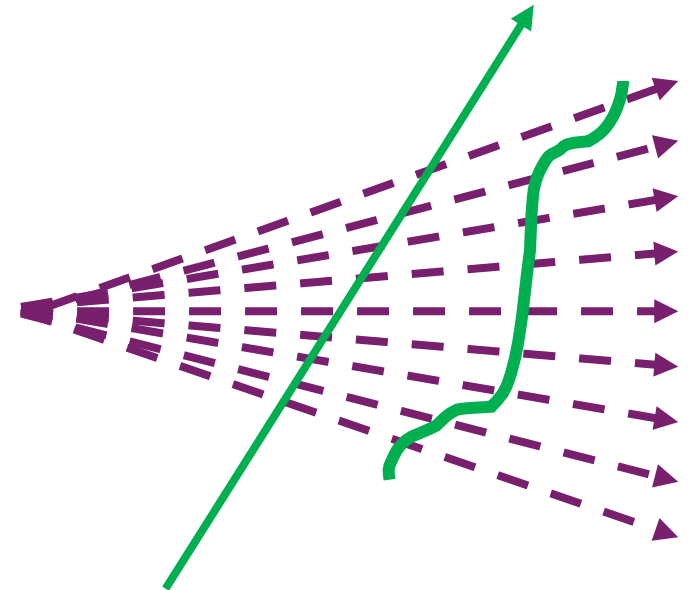
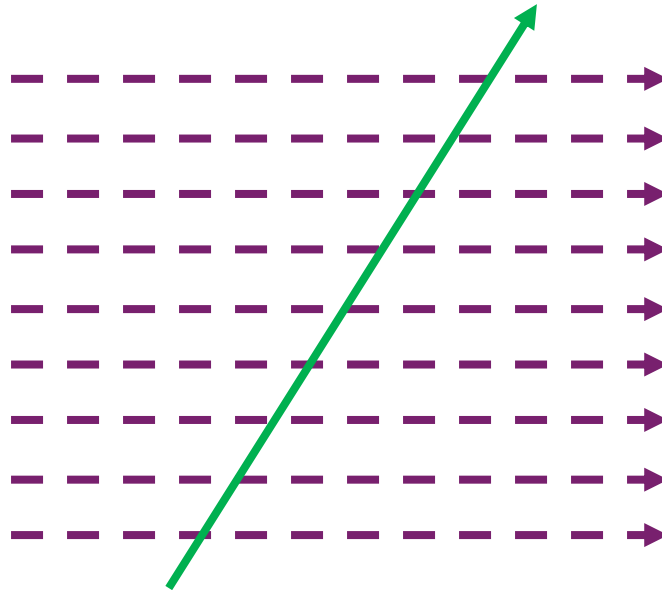
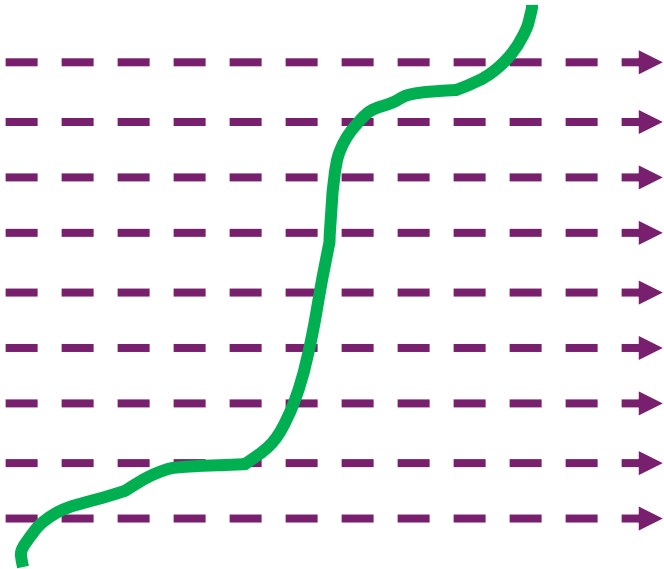


Vector Calculus

- ✓ Differential Length, Area, and Volume
- **Line, Surface, and Volume Integrals**

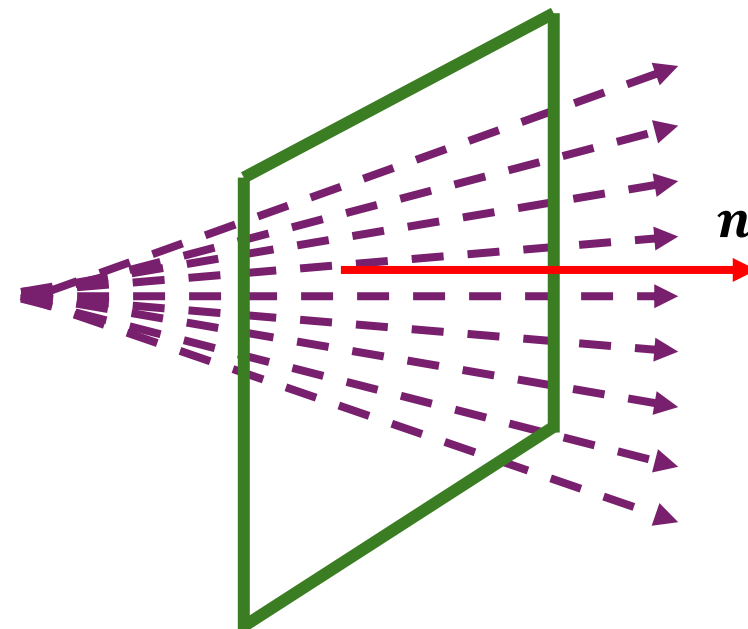
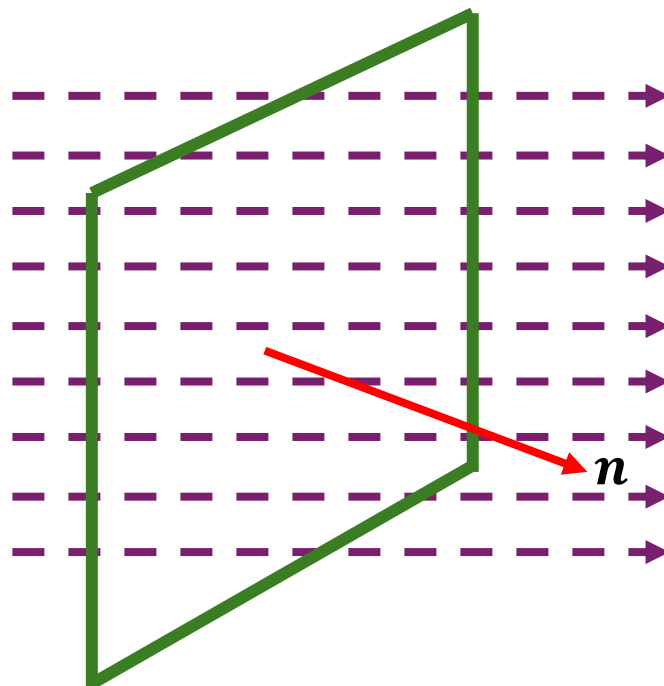
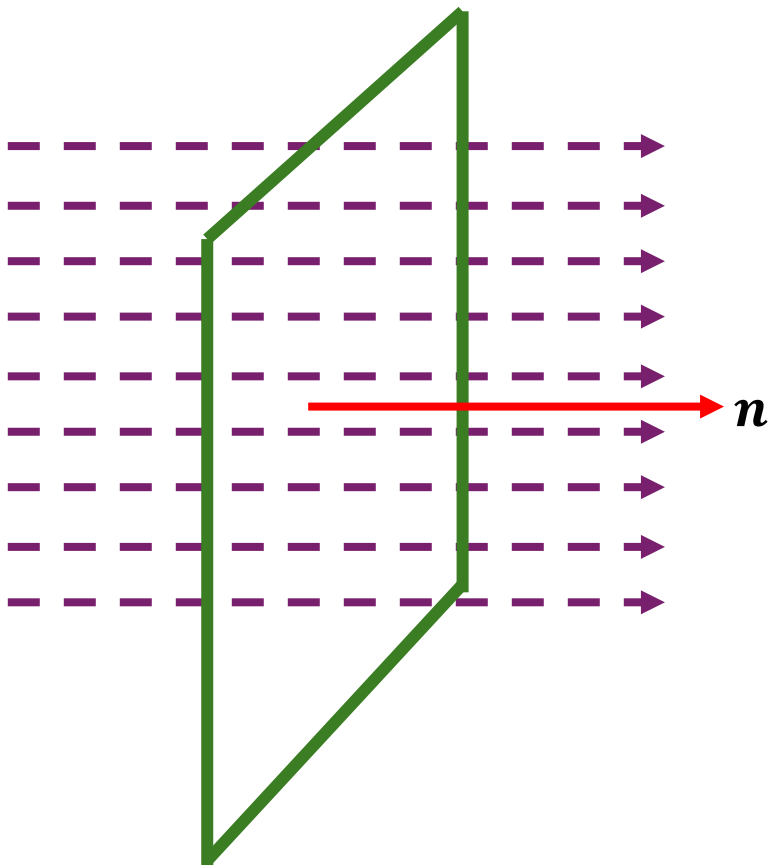
Vector Calculus – Line Integral

$$\oint_L \mathbf{A} \cdot d\mathbf{l}$$



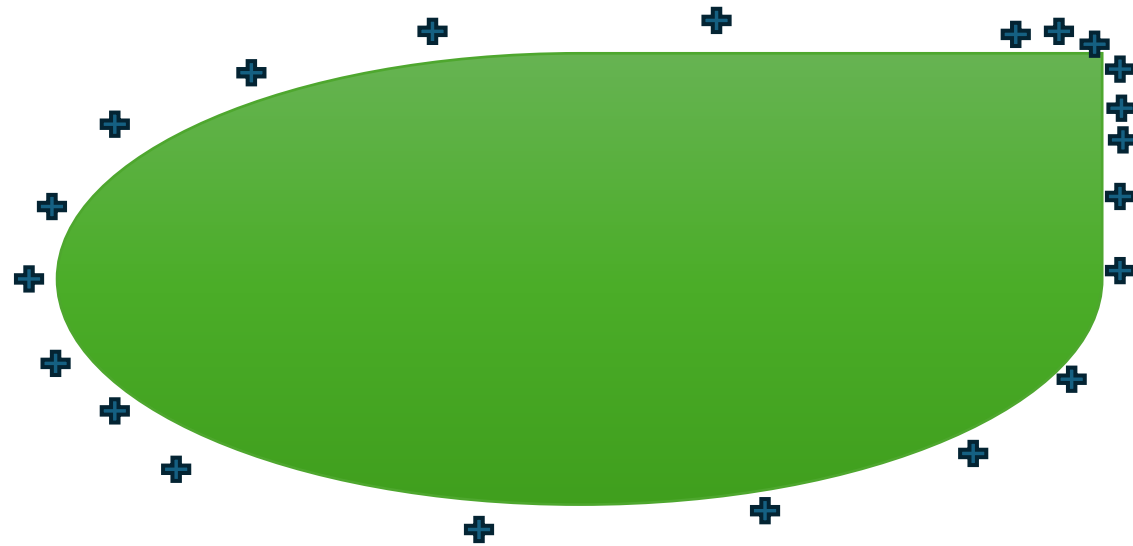
Vector Calculus – Surface Integral

$$\Psi = \int_S \mathbf{A} \cdot d\mathbf{S}$$



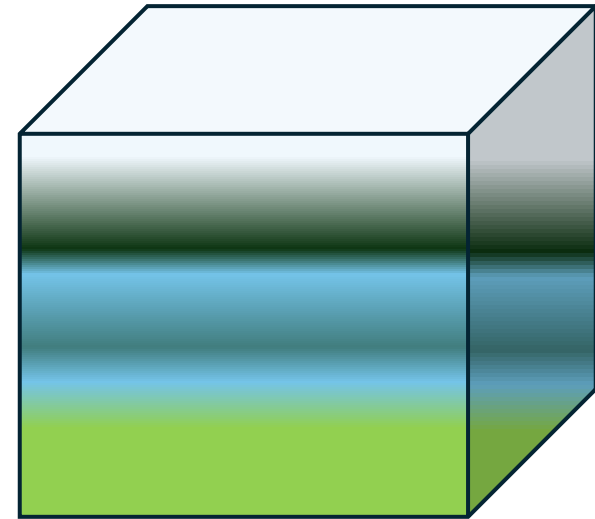
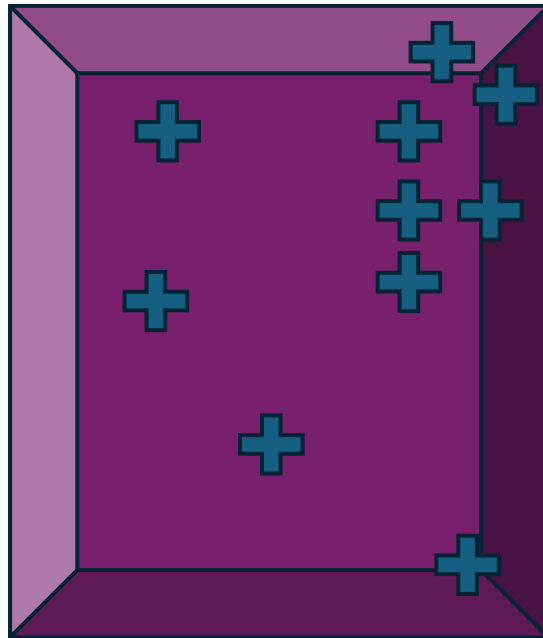
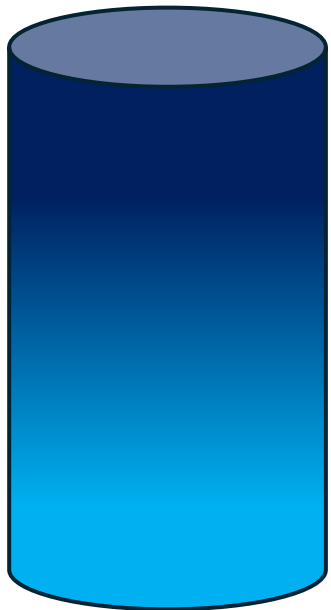
Vector Calculus – Surface Integral

$$\Psi = \oint_S \mathbf{A} \cdot d\mathbf{S}$$



Vector Calculus – Volume Integral

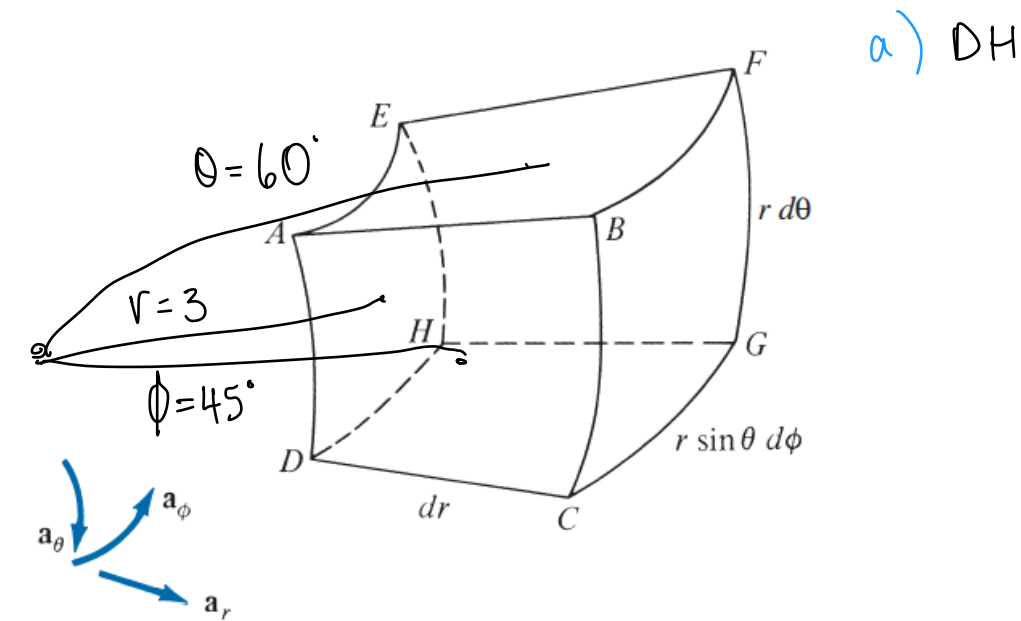
$$\int_v \rho_v dv$$



PRACTICE EXERCISE 3.1

Refer to Figure 3.8; disregard the differential lengths and imagine that the object is part of a spherical shell. It may be described as $3 \leq r \leq 5$, $60^\circ \leq \theta \leq 90^\circ$, $45^\circ \leq \phi \leq 60^\circ$ where surface $r = 3$ is the same as $AEHD$, surface $\theta = 60^\circ$ is $AEFB$, and surface $\phi = 45^\circ$ is $ABCD$. Calculate

- (a) The arc length DH
- (b) The arc length FG
- (c) The surface area $AEHD$
- (d) The surface area $ABDC$
- (e) The volume of the object

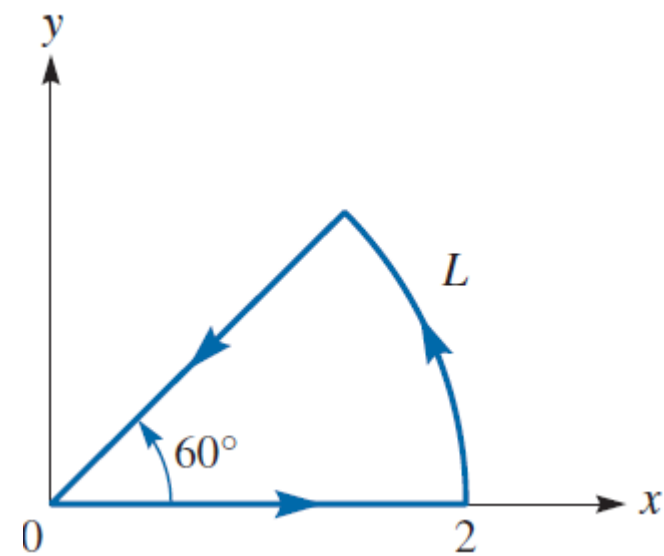


PRACTICE EXERCISE 3.2

Calculate the circulation of

$$\mathbf{A} = \rho \cos \phi \mathbf{a}_\rho + z \sin \phi \mathbf{a}_z$$

around the edge L of the wedge defined by $0 \leq \rho \leq 2$, $0 \leq \phi \leq 60^\circ$, $z = 0$ and shown in Figure 3.12.



Vector Calculus

- ✓ **Differential Length, Area, and Volume**
- ✓ **Line, Surface, and Volume Integrals**

Vector Calculus

Today's plan

- Fields, Scalar and Vector
- Del Operator (3.4 – p. 69)
- Gradient of a Scalar (3.5 – p. 71)
- Divergence of a Vector (3.6 – p. 75)

Field

What a field is?

Field

- Quantities are not confined to a single point.
- They exist everywhere in space.
- At each location, the quantity maybe different.
- We need a way to describe how a quantity depends on position.

Field

- ✓ What a field is?
- A **field** is a **physical quantity** defined at every point in **space**.
scalar → lower case

$$\underbrace{\phi(x, y, z), \quad \underline{f}(x, y, z), \quad V(x, y, z)}$$

$$\mathbf{A}(x, y, z), \quad \mathbf{E}(x, y, z)$$

$$\mathbf{A}(x, y, z) = A_x(x, y, z) \hat{a}_x + A_y(x, y, z) \hat{a}_y + A_z(x, y, z) \hat{a}_z$$

Scalar Field

What is a **scalar field**?

What if the field has magnitude only and no direction?

Scalar Field

- A scalar field describes how a quantity is distributed throughout space.
- At each point, the quantity has magnitude only, with no direction.
- The field is fully described by how this value varies with position.
- ❖ *A scalar field tells us how much of something exists at every point in space.*

Scalar Field

✓ What a scalar field is?

➤ A **scalar field** is a **physical quantity** defined at every point in space, where each point is assigned a **single scalar value**.

1. Temperature $T = T(x, y, z)$

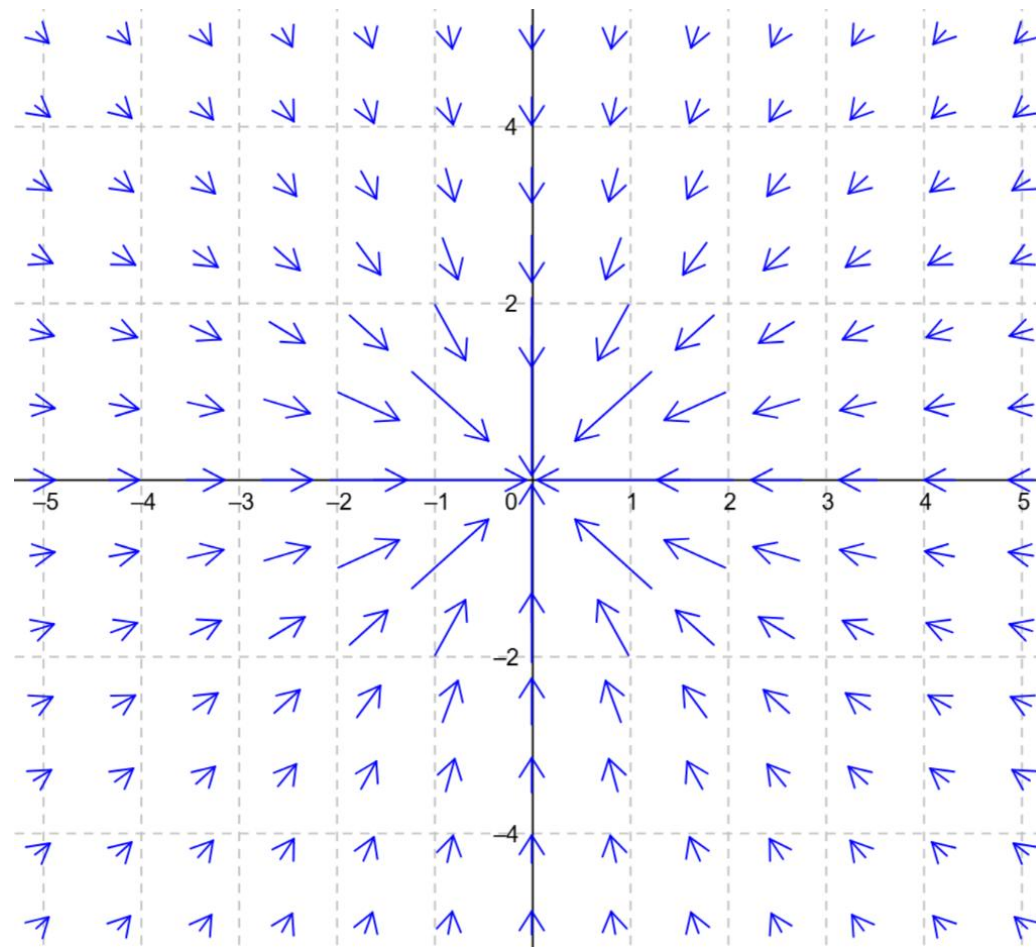
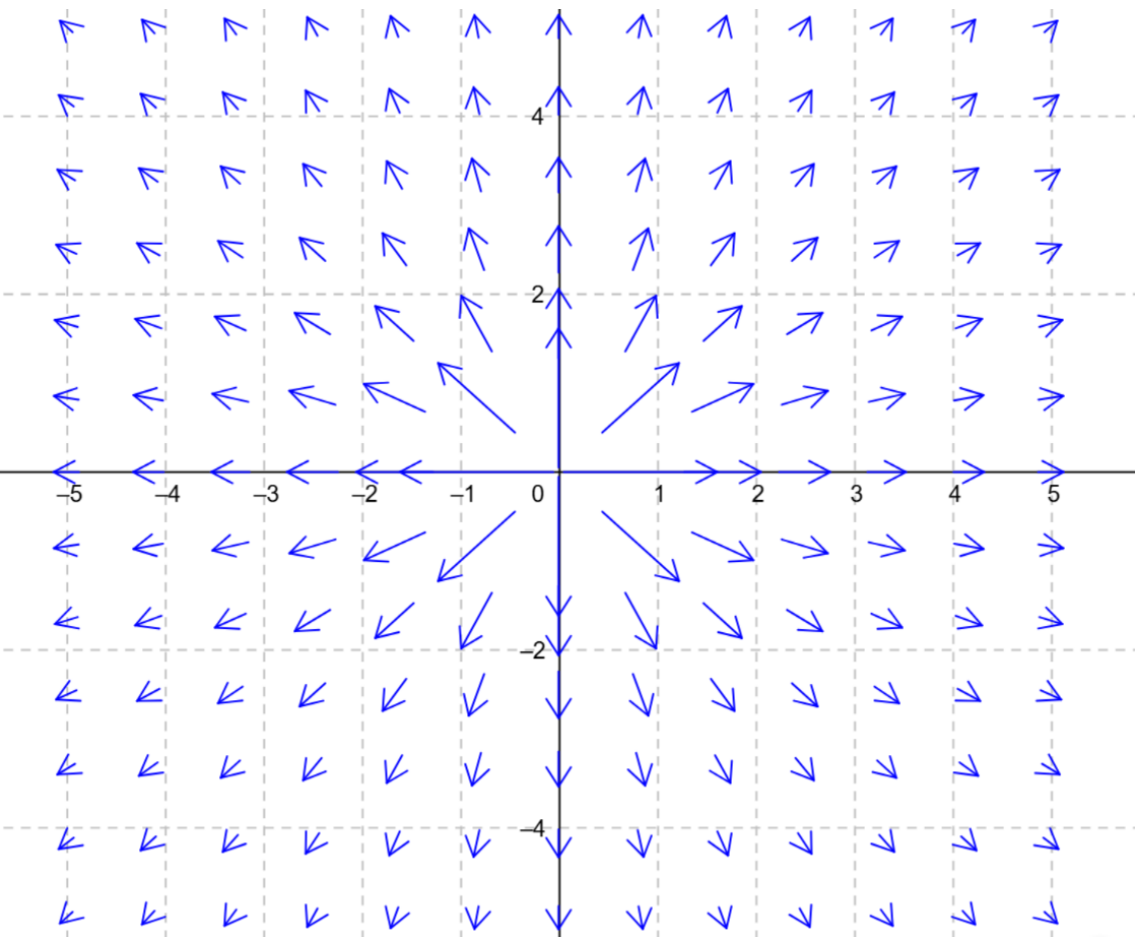
2. Electric Potential $V = V(x, y, z)$

3. Charge Density $\rho = \rho(x, y, z)$

Vector Field

What is a **vector field**?

Vector Field



Del Operator

What is the **Del Operator**?

$$\frac{\partial}{\partial x} \quad \frac{\partial}{\partial y} \quad \frac{\partial}{\partial z}$$

Del Operator

- ✓ The del operator (∇), also called Nabla, is a vector calculus operator.
- ✓ The del operator has vector-like form, but its components are differential operators, not physical vector components.
- ✓ Like other differential operators, the del operator is used to describe change.
- ✓ When applied to scalar or vector fields, it tells us how those fields vary throughout space.

Del Operator in Different Coordinate Systems

➤ Rectangular Coordinate System

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

➤ Cylindrical Coordinate System

$$\nabla = \mathbf{a}_\rho \frac{\partial}{\partial \rho} + \mathbf{a}_\phi \frac{1}{\rho} \frac{\partial}{\partial \phi} + \mathbf{a}_z \frac{\partial}{\partial z}$$

to remove ρ component from ϕ .
only want angle change, & nothing else

➤ Spherical Coordinate System

$$\nabla = \mathbf{a}_r \frac{\partial}{\partial r} + \mathbf{a}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \mathbf{a}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$$

remove unwanted components

Del Operator in Different Coordinate Systems

➤ Rectangular Coordinate System

$$\nabla = \frac{\partial}{\partial x} \mathbf{a}_x + \frac{\partial}{\partial y} \mathbf{a}_y + \frac{\partial}{\partial z} \mathbf{a}_z$$

➤ Cylindrical Coordinate System

$$\nabla = \mathbf{a}_\rho \frac{\partial}{\partial \rho} + \mathbf{a}_\phi \frac{1}{\rho} \frac{\partial}{\partial \phi} + \mathbf{a}_z \frac{\partial}{\partial z}$$

➤ Spherical Coordinate System

$$\nabla = \mathbf{a}_r \frac{\partial}{\partial r} + \mathbf{a}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \mathbf{a}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$$

Del Operator

What can the **del operator** do?

Different Operations using Del – Gradient

- For this purpose, the del operator acts **on a scalar field**.
↳ gives direction for scalar
- The result is a vector field whose magnitude represents the rate of change of the scalar field, and whose direction indicates the direction of the greatest increase of the field.
- The gradient determines the rate of change of the scalar field along a given direction.
* gradient is result of Δ operator on scalar field.
◦ gives direction of max change / ROC
↳ ie where change is the strongest

Gradient in different Coordinate Systems

➤ Rectangular Coordinate System

$$\nabla V = \frac{\partial V}{\partial x} \mathbf{a}_x + \frac{\partial V}{\partial y} \mathbf{a}_y + \frac{\partial V}{\partial z} \mathbf{a}_z$$

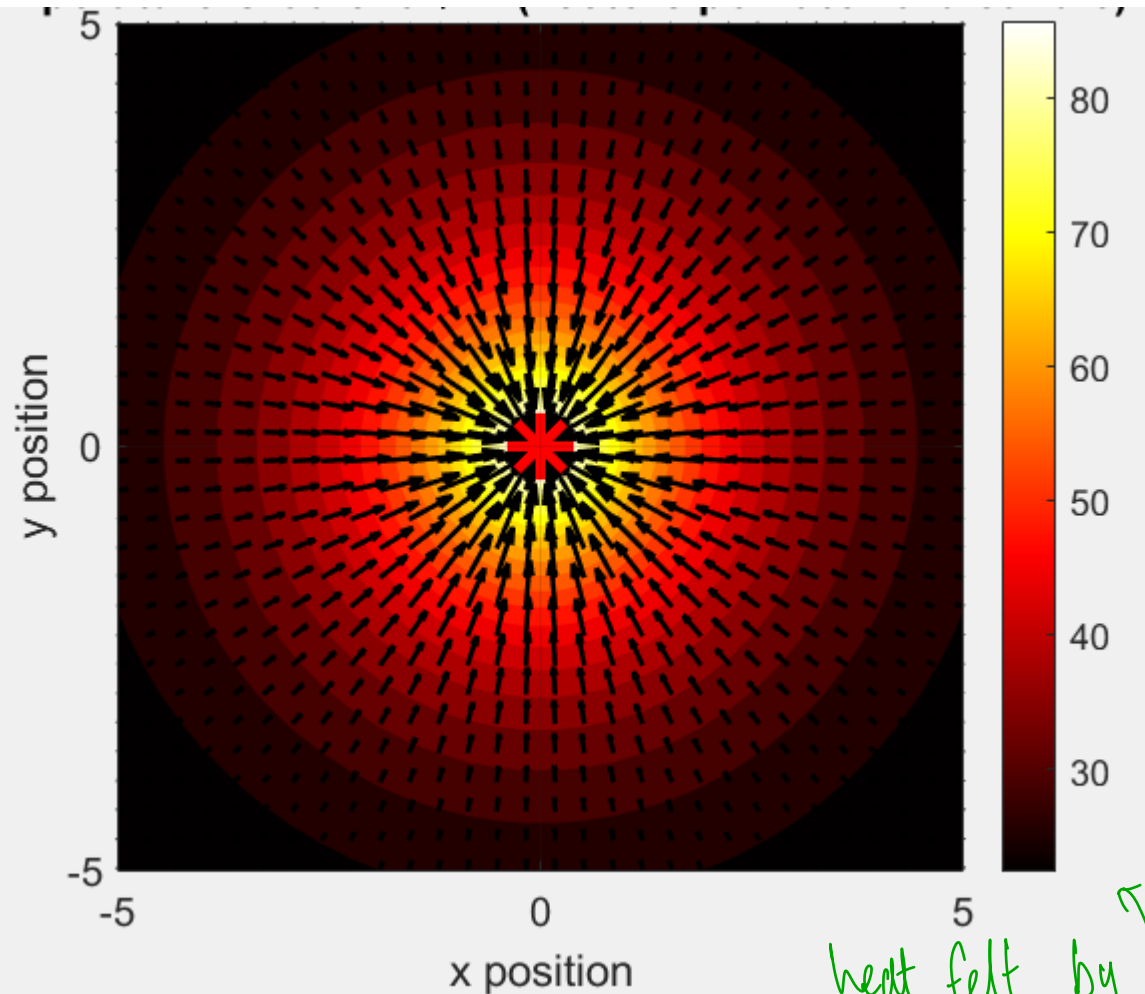
➤ Cylindrical Coordinate System

$$\nabla V = \frac{\partial V}{\partial \rho} \mathbf{a}_\rho + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi + \frac{\partial V}{\partial z} \mathbf{a}_z$$

➤ Spherical Coordinate System

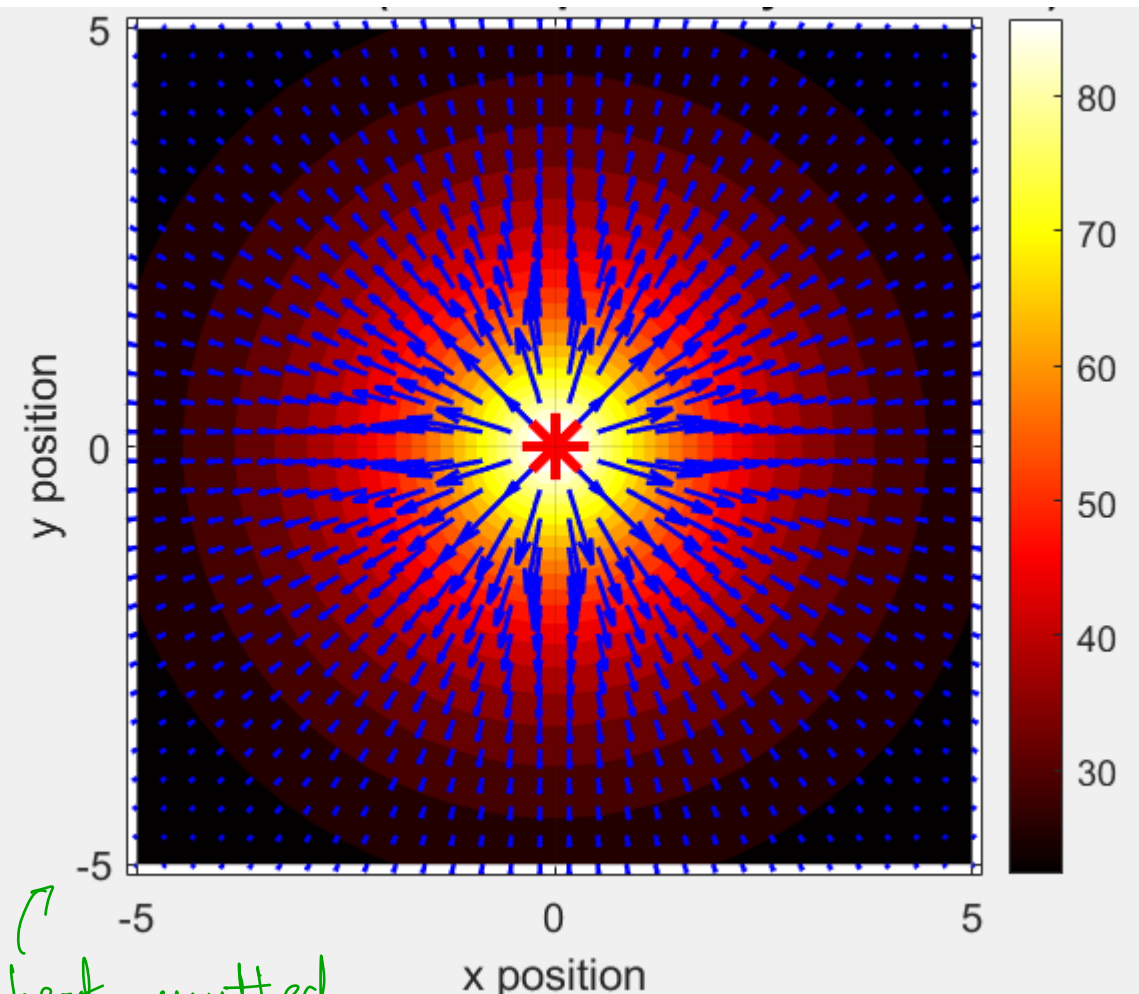
$$\nabla V = \frac{\partial V}{\partial r} \mathbf{a}_r + \frac{1}{r} \frac{\partial V}{\partial \theta} \mathbf{a}_\theta + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \mathbf{a}_\phi$$

Gradient of Temperature – Single source



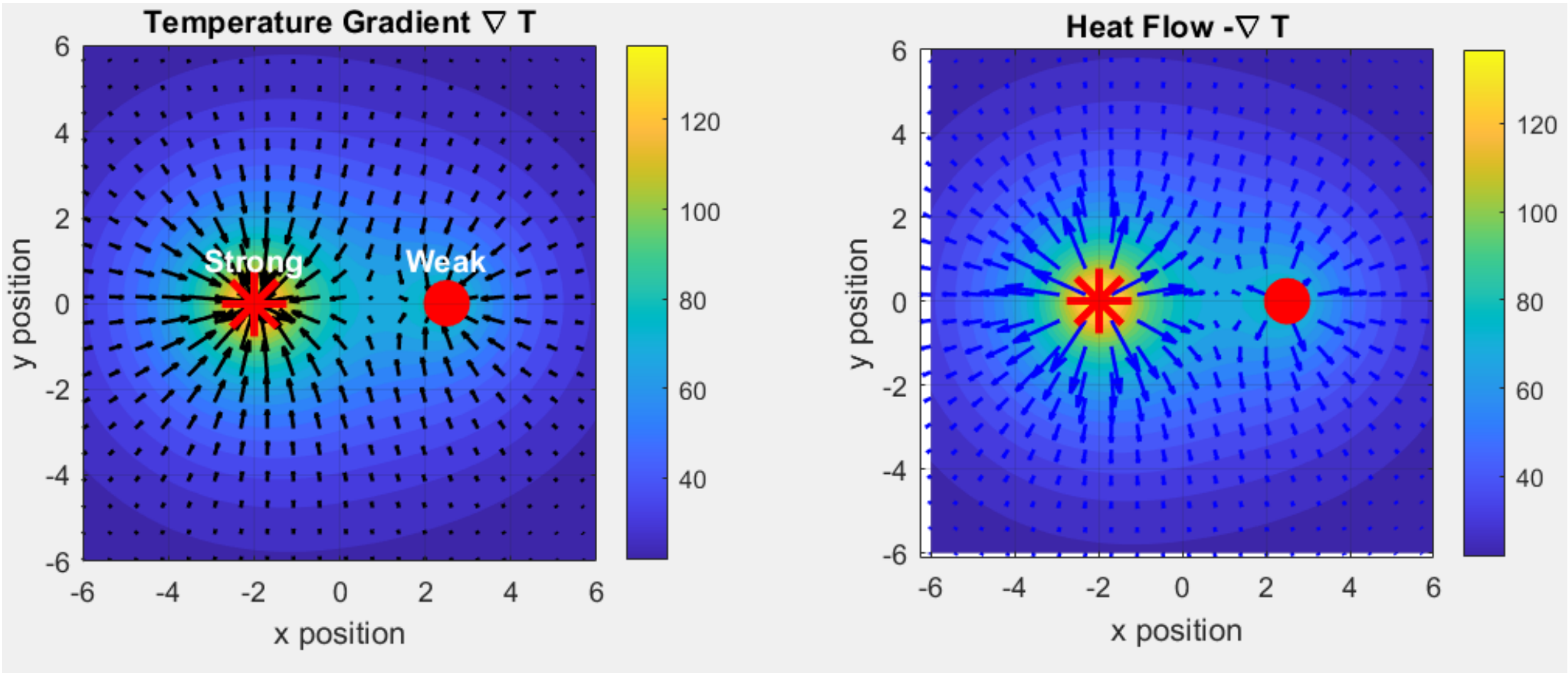
heat felt by
hand around
candle

↳ cylindrical coord system



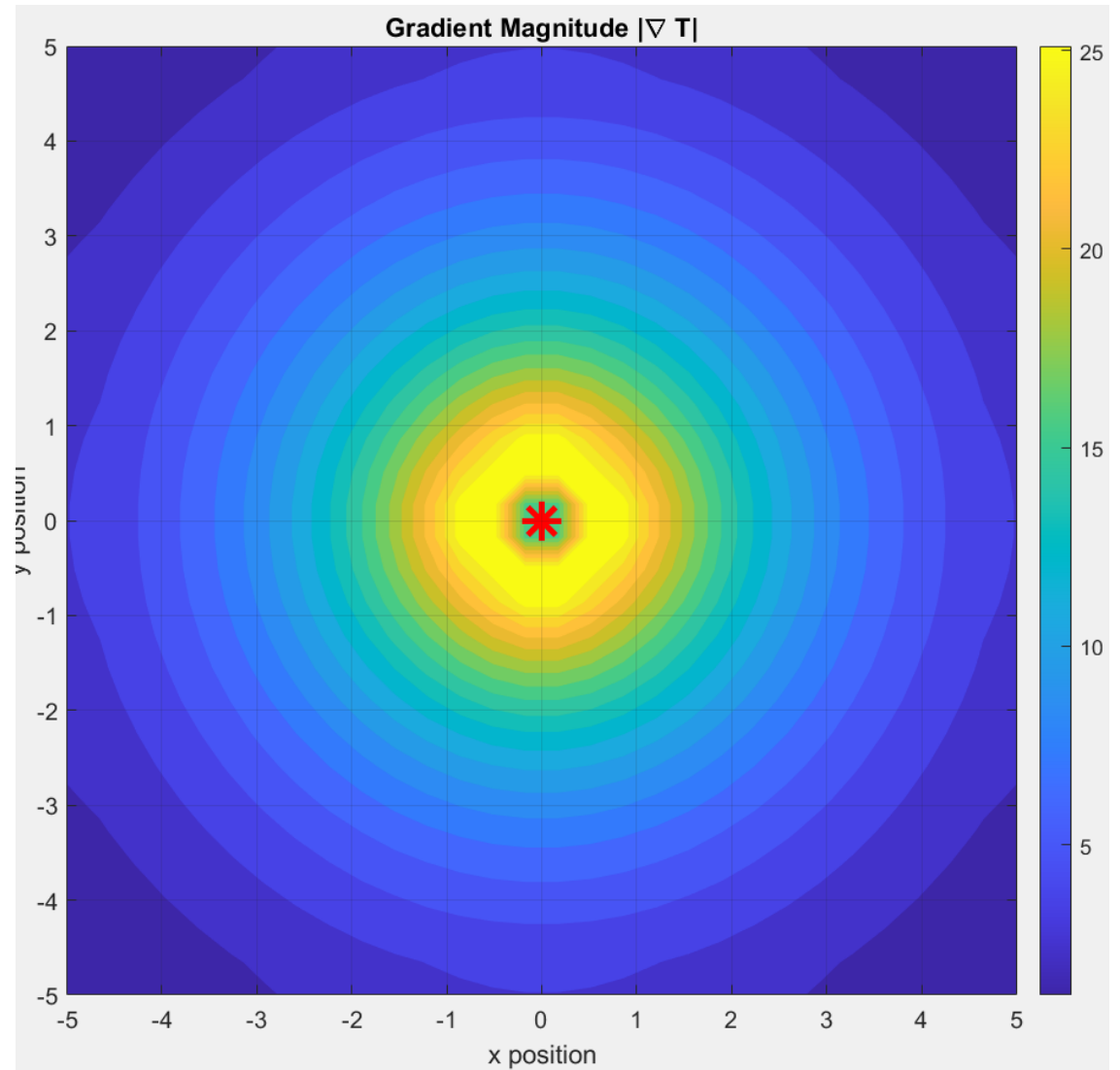
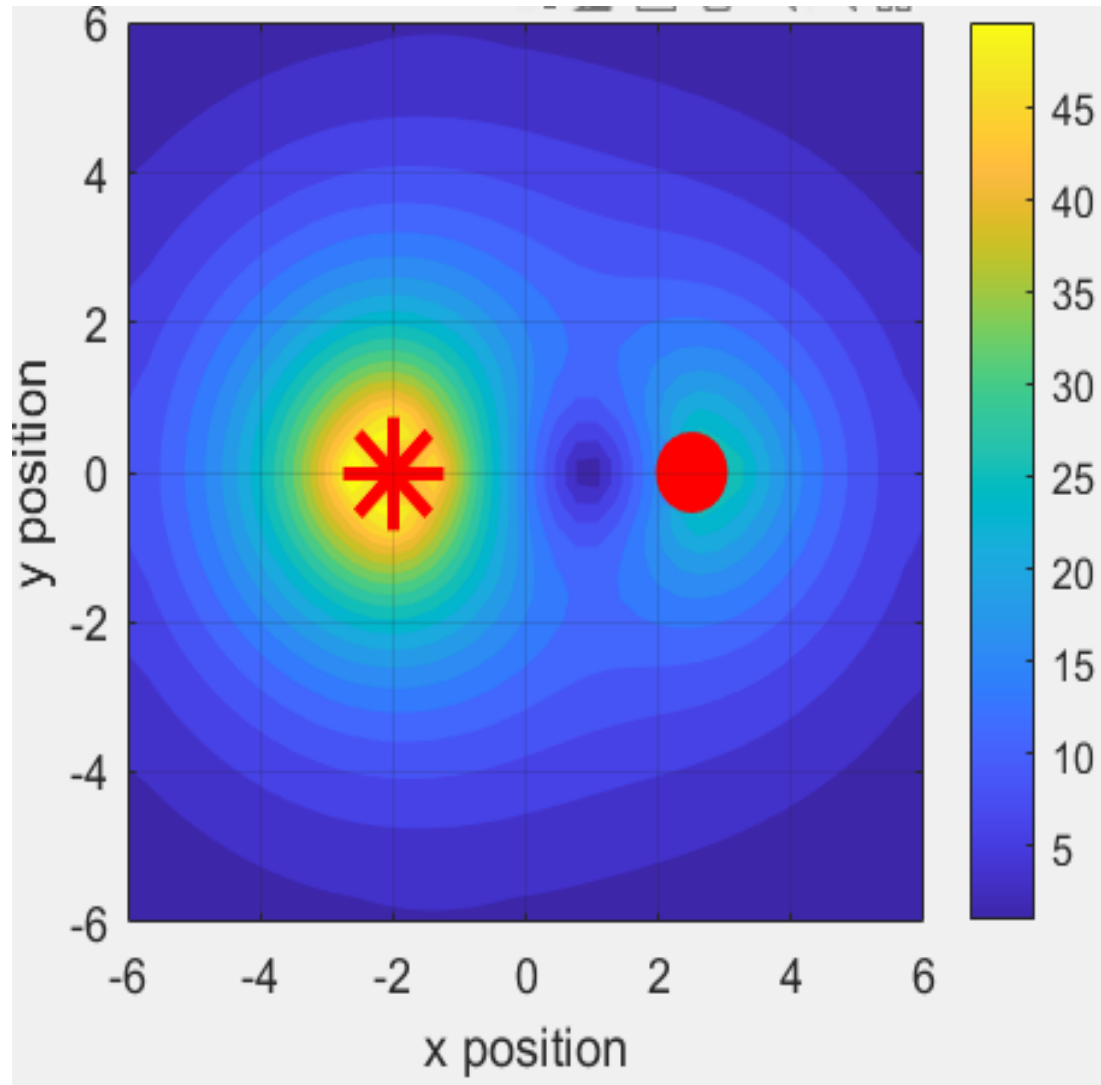
heat emitted
by candle

Gradient of Temperature – Two sources



* points to candle w/ stronger force

Gradient of Temperature – Magnitude



magnitude: $|\nabla T|$

Gradient – Example 1

PRACTICE EXERCISE 3.3

Determine the gradient of the following scalar fields:

(a) $U = x^2y + xyz$

(b) $V = \rho z \sin \phi + z^2 \cos^2 \phi + \rho^2$

(c) $f = \cos \theta \sin \phi \ln r + r^2 \phi$ ←

Ex.

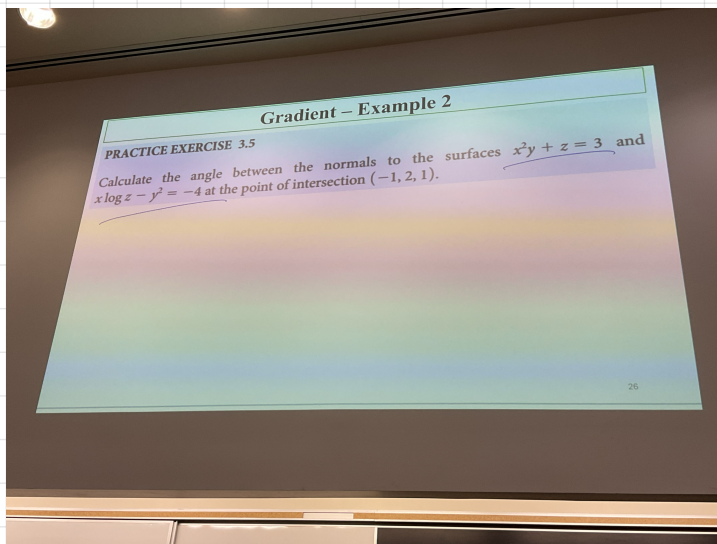
c) $f = \cos \theta \sin \phi \ln r + r^2 \phi$

$$\nabla = \vec{a}_r \frac{\partial}{\partial r} + \vec{a}_\theta \frac{1}{r} \frac{\partial}{\partial \theta} + \vec{a}_\phi \frac{1}{r \sin \theta} \frac{\partial}{\partial \phi}$$

$$\nabla f : \vec{a}_r \frac{\partial f}{\partial r} = \vec{a}_r \cos \theta \sin \phi \frac{1}{r} + 2r \phi$$

$$\nabla f : \frac{1}{r} \vec{a}_\theta \frac{\partial f}{\partial \theta} = \frac{1}{r} \vec{a}_\theta \sin \theta \sin \phi \ln r$$

$$\nabla f : \frac{1}{r \sin \theta} \vec{a}_\phi \frac{\partial f}{\partial \phi} = \frac{1}{r \sin \theta} \sim \text{~~~~~}$$



Ex 2.

$$F_1 = x^2y + z - 3$$

$$F_2 = x \log z - y^2 + 4$$

$$\vec{n}_1 = \nabla F_1$$

$$\vec{n}_2 = \nabla F_2$$

Different Operations using Del – Divergence

- When the del operator takes the divergence of a vector field, it results in a scalar field that tells us how much the field spreads out or converges at each point in space.

$\text{divergence: } \nabla \cdot \vec{A}$

$\leftarrow$ dot product
 $\leftarrow$ answer = scalar
- The divergence is positive where field lines spread out and negative where they converge.
- In physical fields, divergence is often related to the presence of sources or sinks.
- Divergence is a local quantity. It depends only on the field behavior in an infinitesimal region around a point.

Divergence in different Coordinate Systems

➤ Rectangular Coordinate System

$$\nabla \cdot \mathbf{A} = \frac{\partial A_x}{\partial x} + \frac{\partial A_y}{\partial y} + \frac{\partial A_z}{\partial z}$$

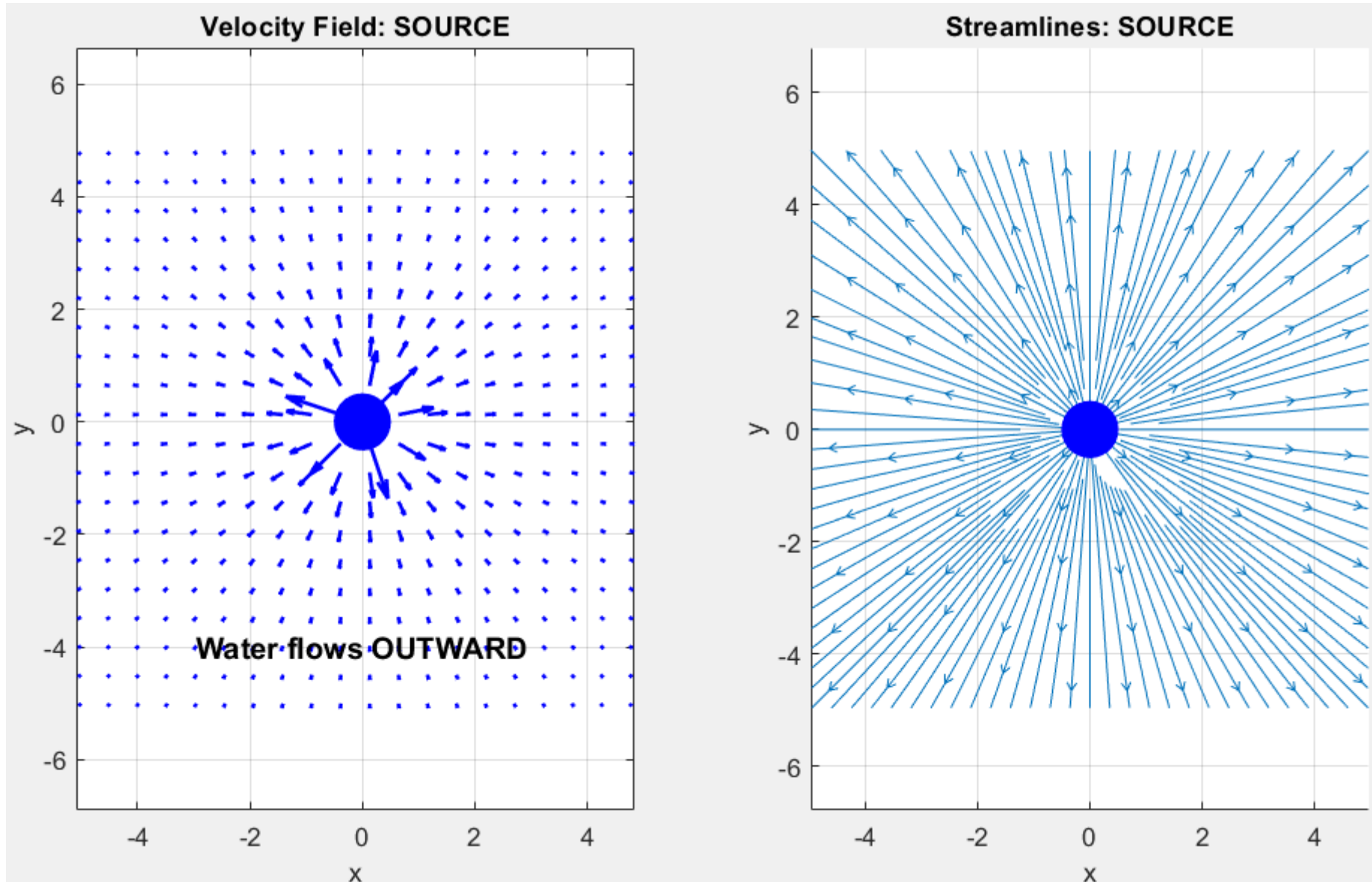
➤ Cylindrical Coordinate System

$$\nabla \cdot \mathbf{A} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho A_\rho) + \frac{1}{\rho} \frac{\partial A_\phi}{\partial \phi} + \frac{\partial A_z}{\partial z}$$

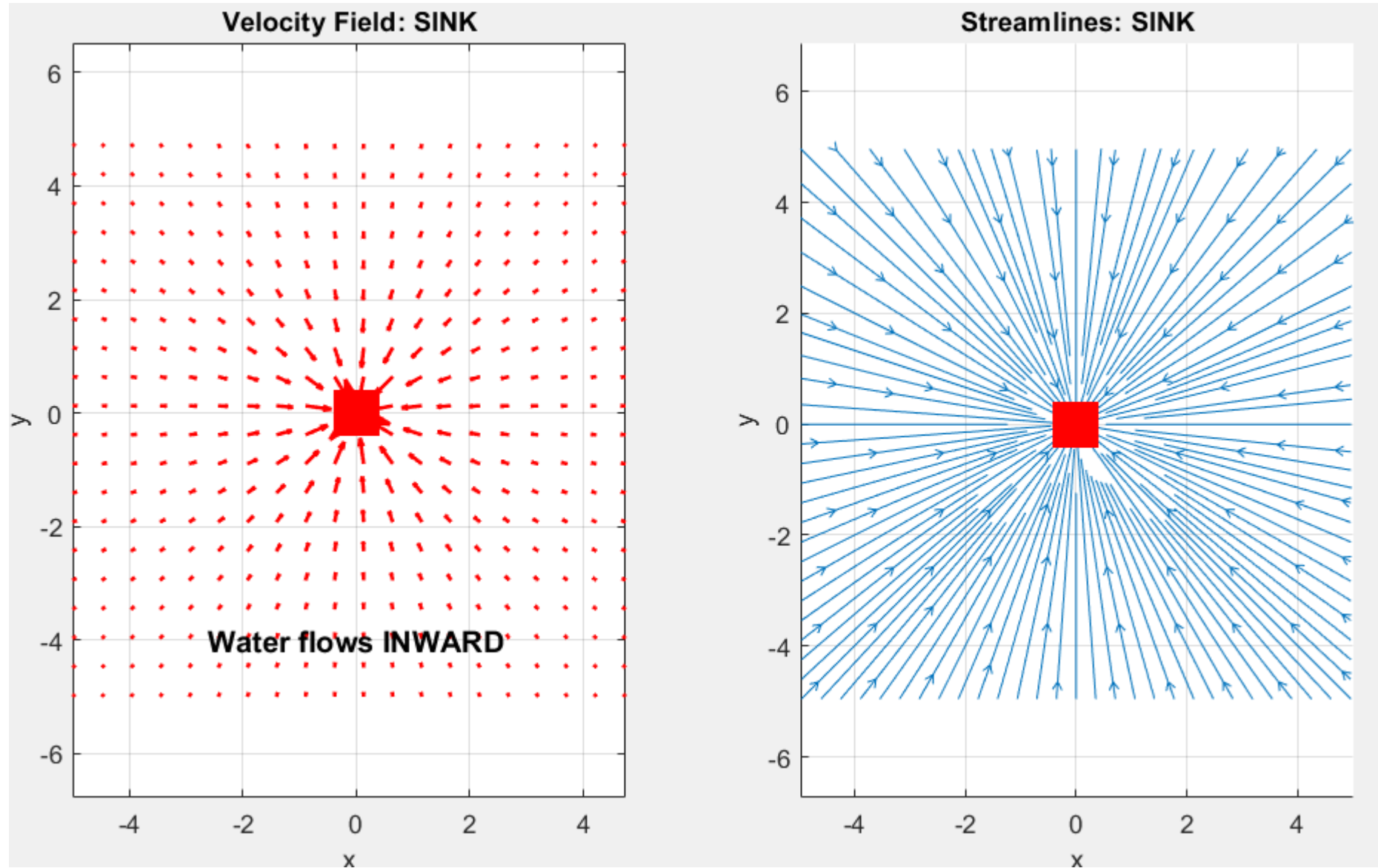
➤ Spherical Coordinate System

$$\nabla \cdot \mathbf{A} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 A_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (A_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial A_\phi}{\partial \phi}$$

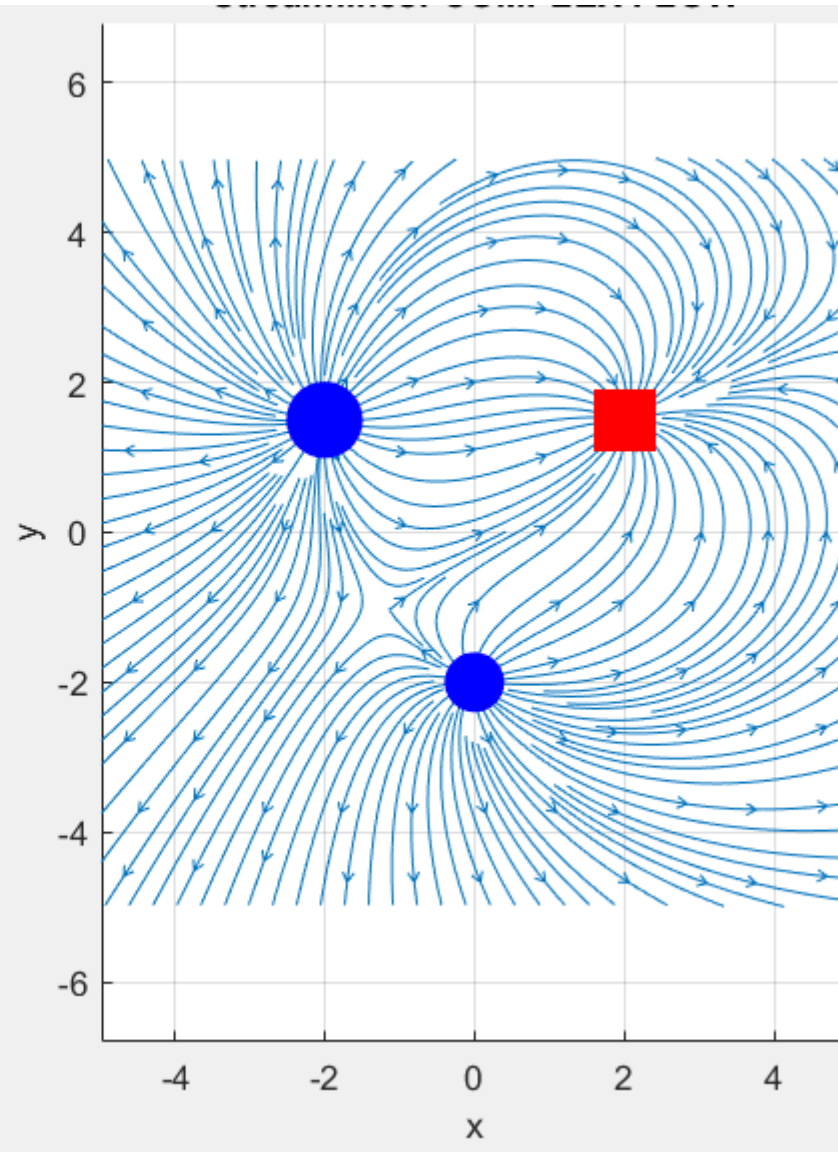
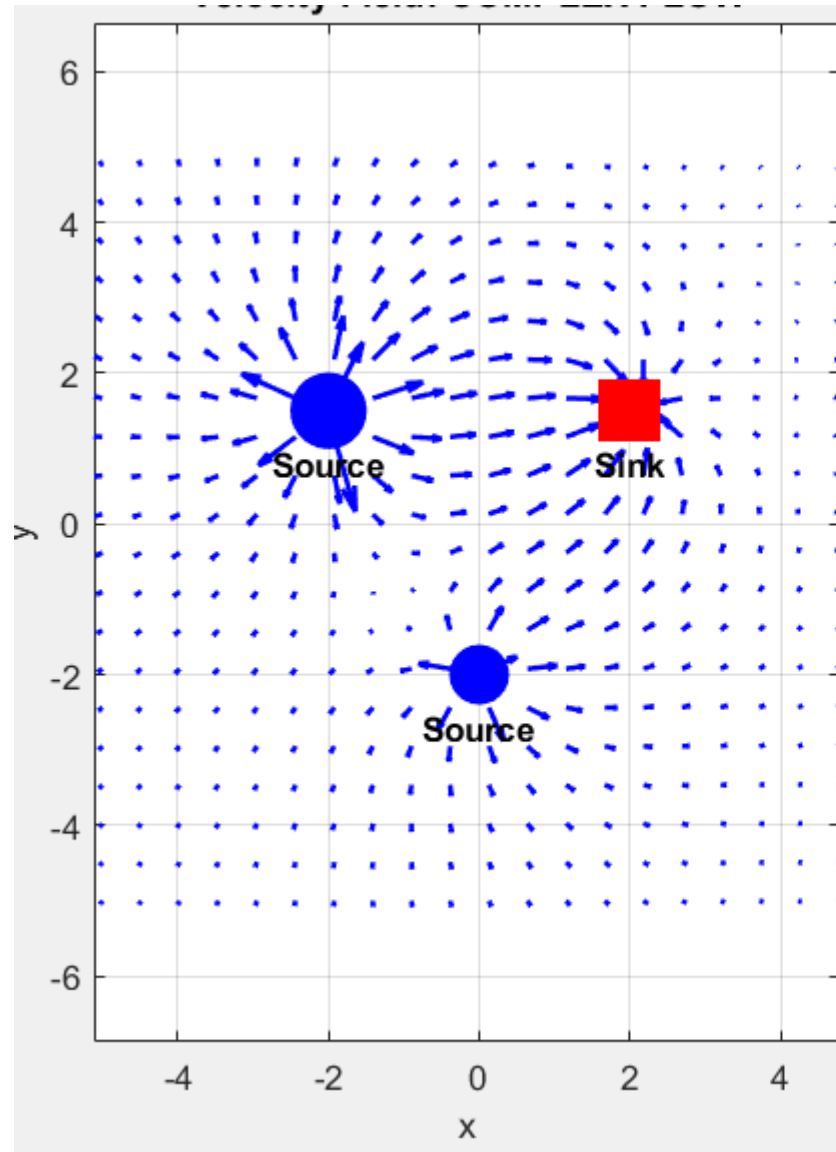
Divergence of a Water Spring



Divergence of a Water Drain



Divergence of two Sources and a sink



source = $+\vec{a}$
sink = $-\vec{a}$

Vector Calculus

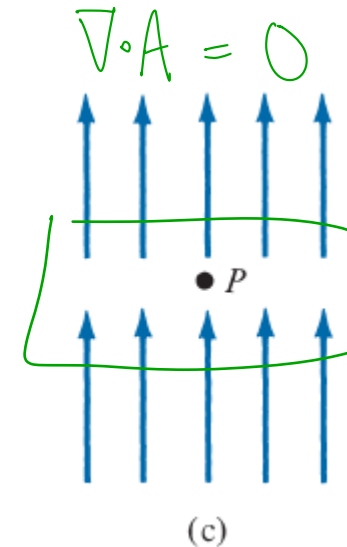
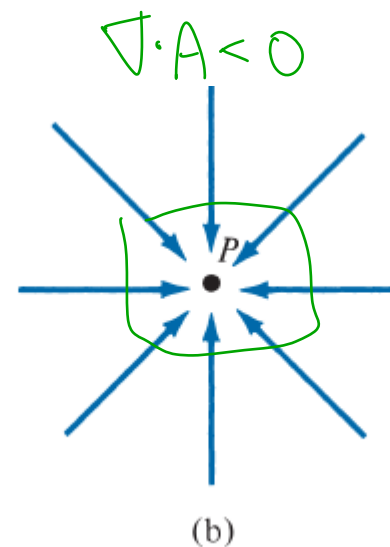
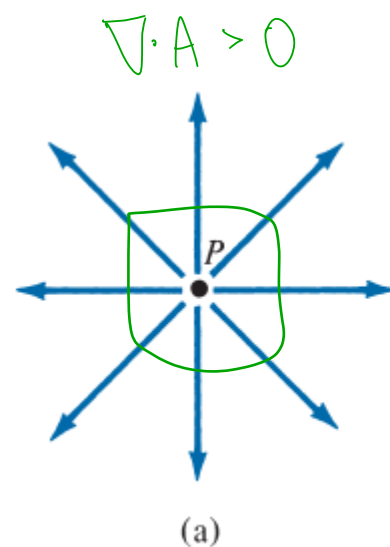
Today's plan

- Divergence Theorem (3.6 – p. 78)
- Curl of a Vector and Stokes's Theorem (3.7 – p. 82)
- CLASSIFICATION OF VECTOR FIELDS (3.9 – p. 92)

Divergence Theorem

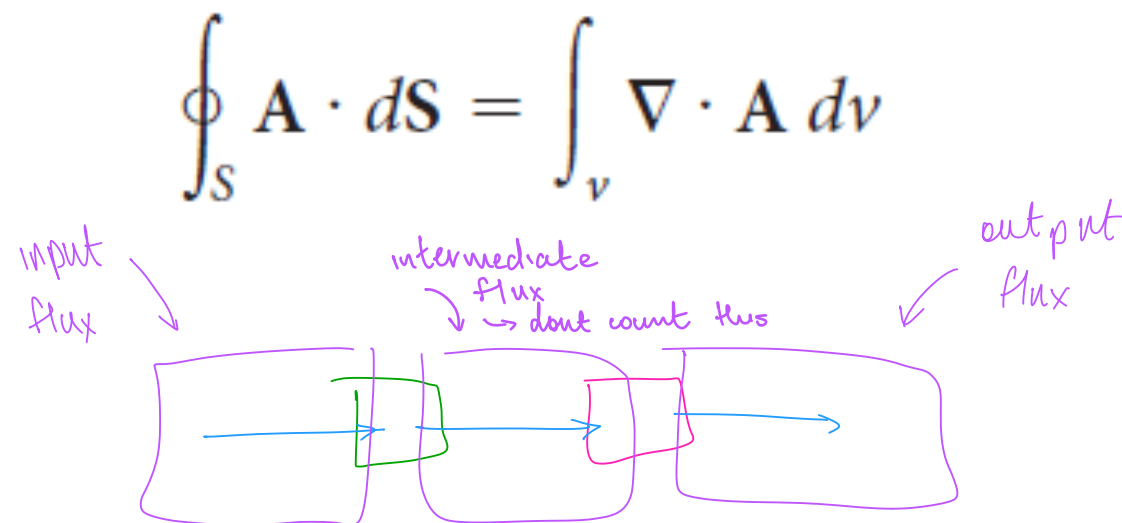
- The divergence of an arbitrary vector field ($\mathbf{A}$) at point P is the net outward flux of $\mathbf{A}$ through a small closed surface surrounding P , divided by the volume enclosed, as that volume shrinks to zero.

$$\text{div } \mathbf{A} = \nabla \cdot \mathbf{A} = \lim_{\Delta v \rightarrow 0} \frac{\oint_S \mathbf{A} \cdot d\mathbf{S}}{\Delta v}$$



Divergence Theorem

- The divergence theorem states that the total outward flux of a vector field $\mathbf{A}$ through a closed surface S is equal to the sum of the divergences of $\mathbf{A}$ throughout the volume enclosed by that surface.
- ❖ *What flows out through the boundary of a region is exactly equal to what is generated inside the region.*



Divergence

- One physical meaning, two mathematical expressions; **Divergence**
- **Point Form**, what is happening **at a single point**

$$\nabla \cdot \mathbf{A}$$

- Use this when
 1. You want the divergence at **a point**
 2. You are given $A(x, y, z)$ explicitly
 3. You are analyzing sources and sinks locally
- Applications
 1. Charge density in electromagnetics
 2. Fluid flow, compression or expansion at a point
 3. Heat flow or diffusion problems

Divergence

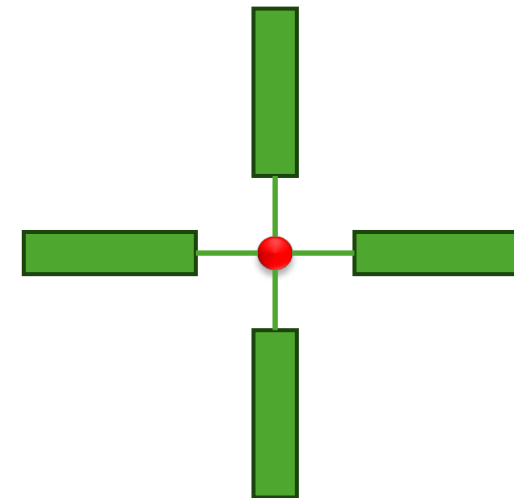
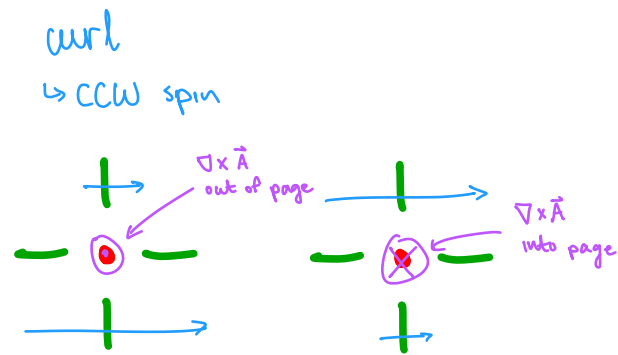
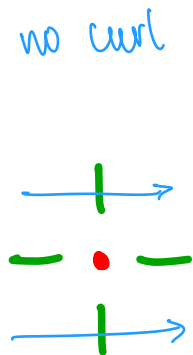
- One physical meaning, two mathematical expression; **Divergence**
- Integral or flux form, what is happening over a finite region

$$\oint_S \mathbf{A} \cdot d\mathbf{S} = \int_V (\nabla \cdot \mathbf{A}) dV$$

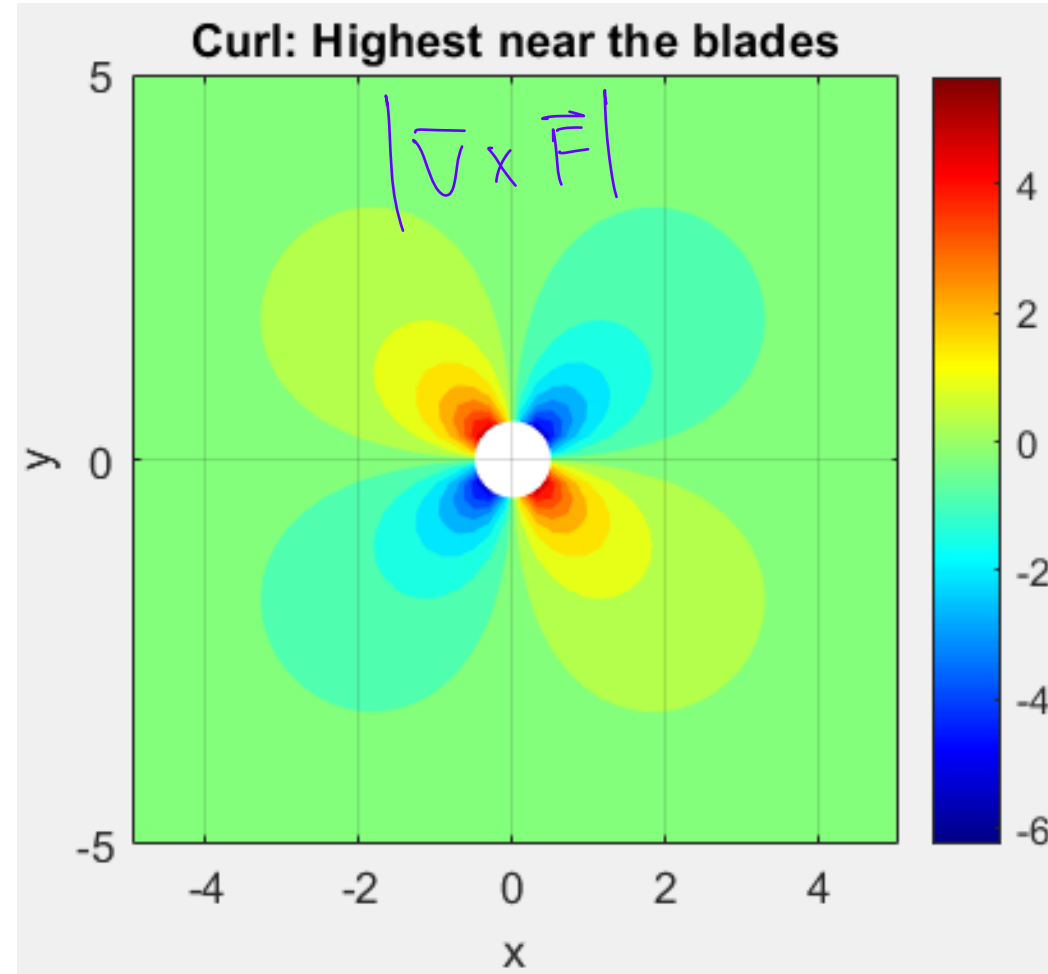
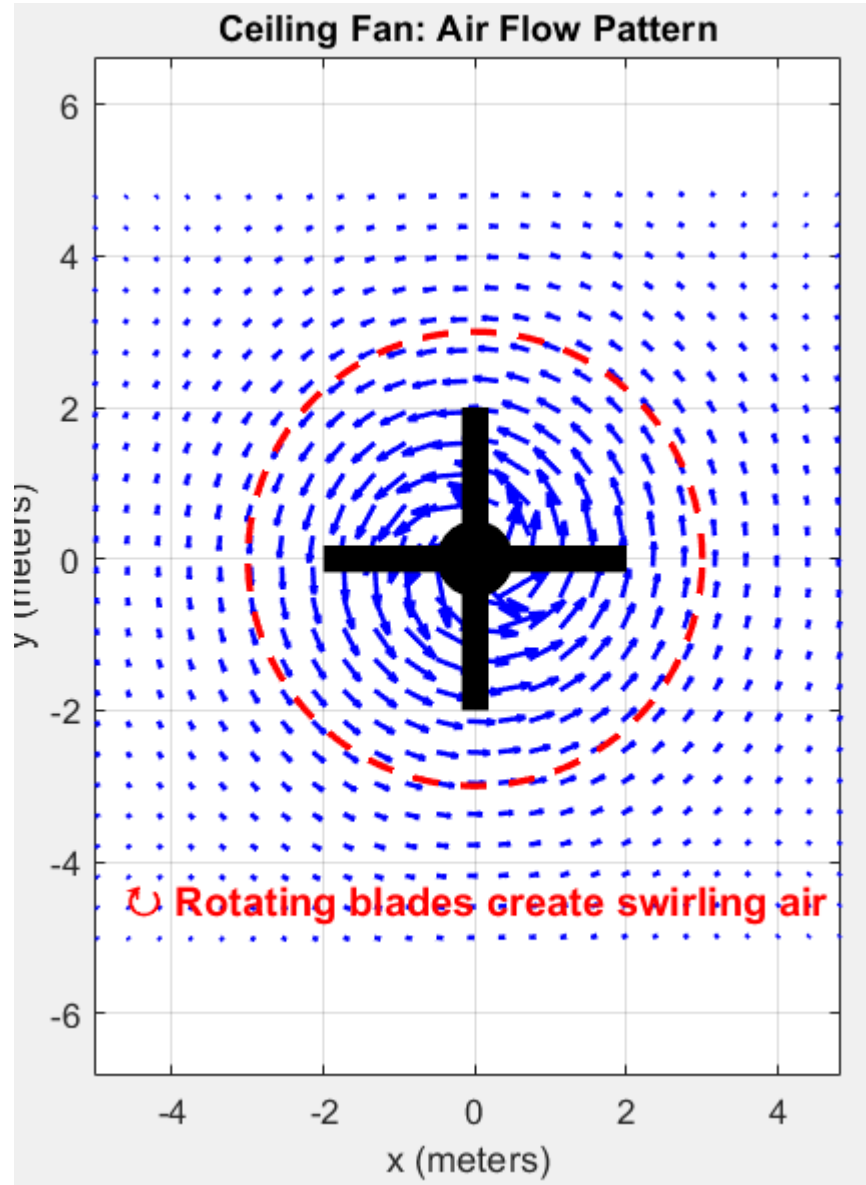
- Use this when
 1. You want the total flux through a surface
 2. The surface has symmetry
 3. Direct surface integration is hard, but volume behavior is simple
- Applications
 1. Gauss's law in electromagnetics
 2. Total charge enclosed
 3. Verifying conservation laws

Curl

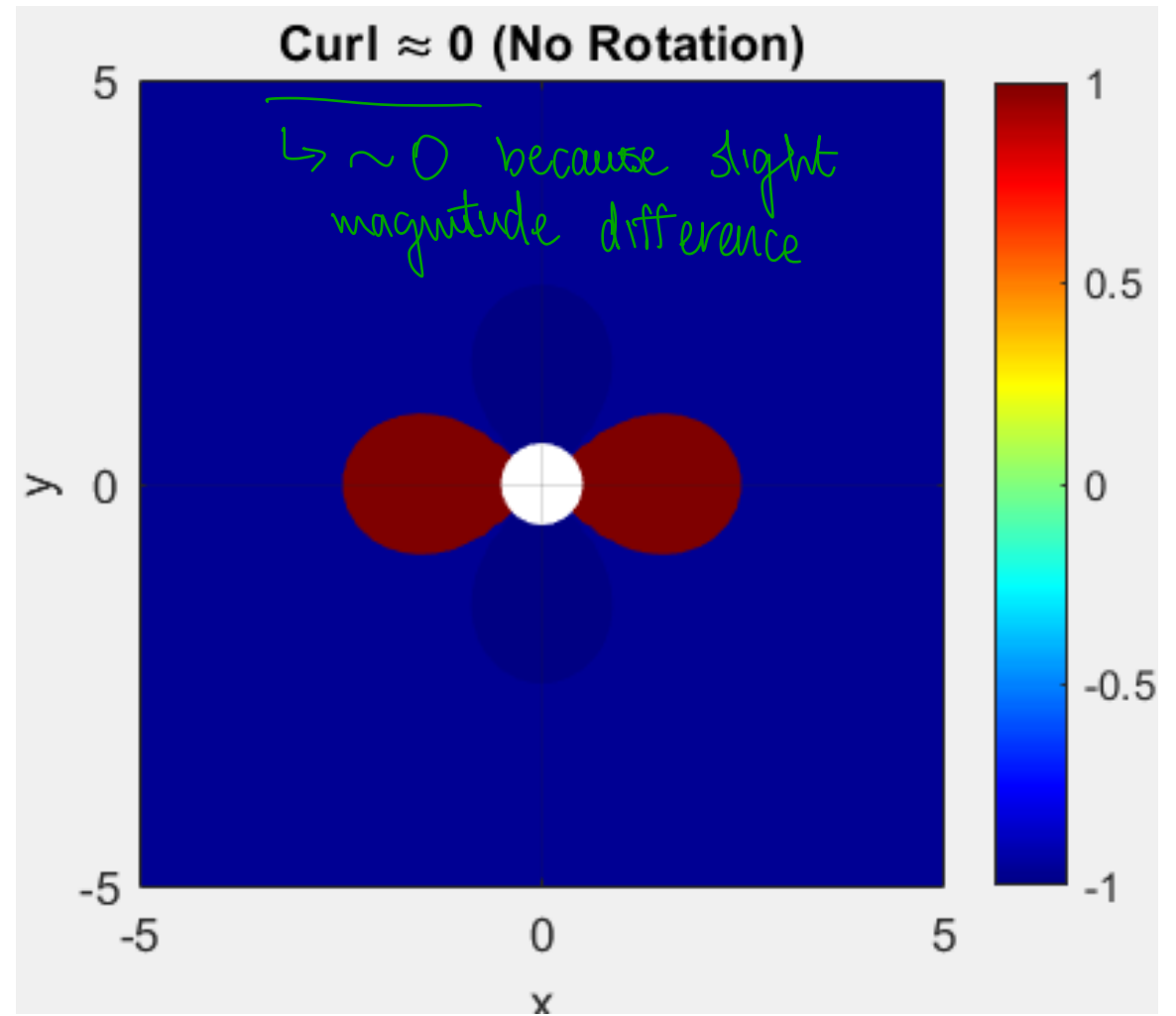
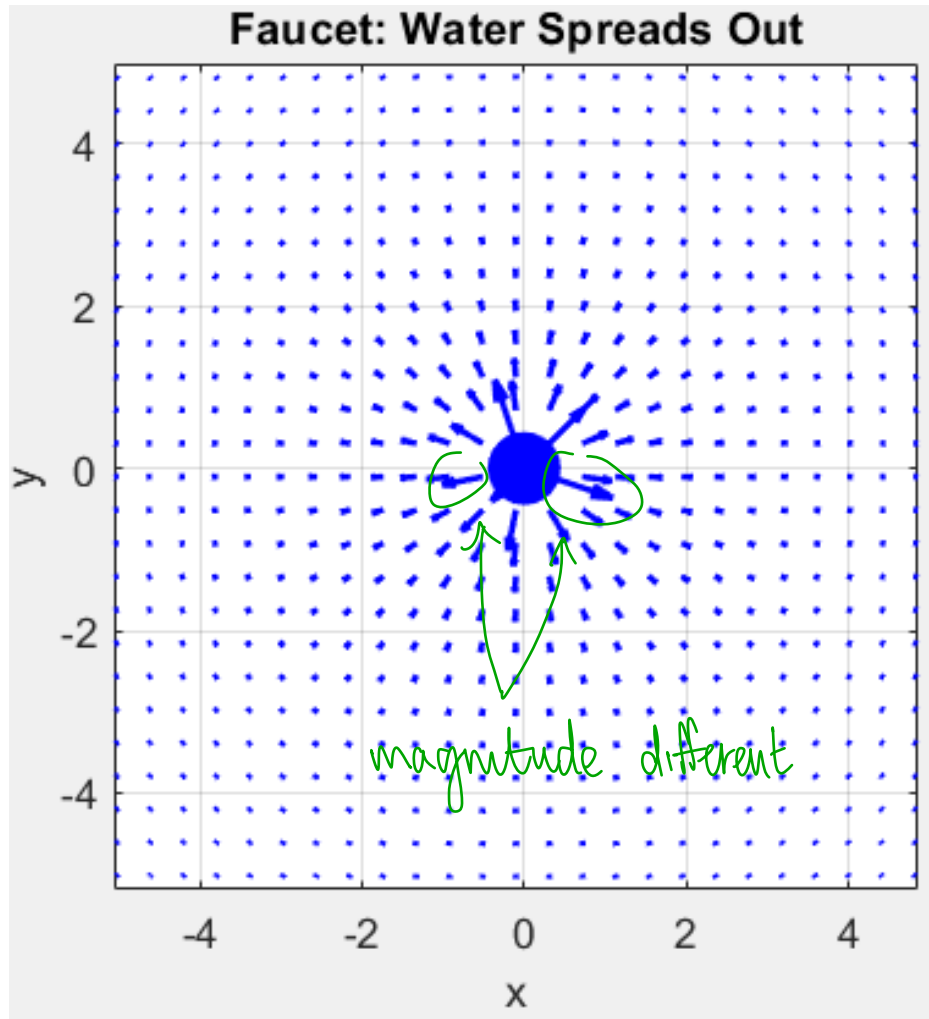
- If I put a tiny paddle wheel at this point, will it spin?
- Direction of curl = axis of rotation
- **Zero curl** does NOT mean **zero field**
- Curl of a vector field at a point measures how much and in which direction the field wants to rotate about that point.



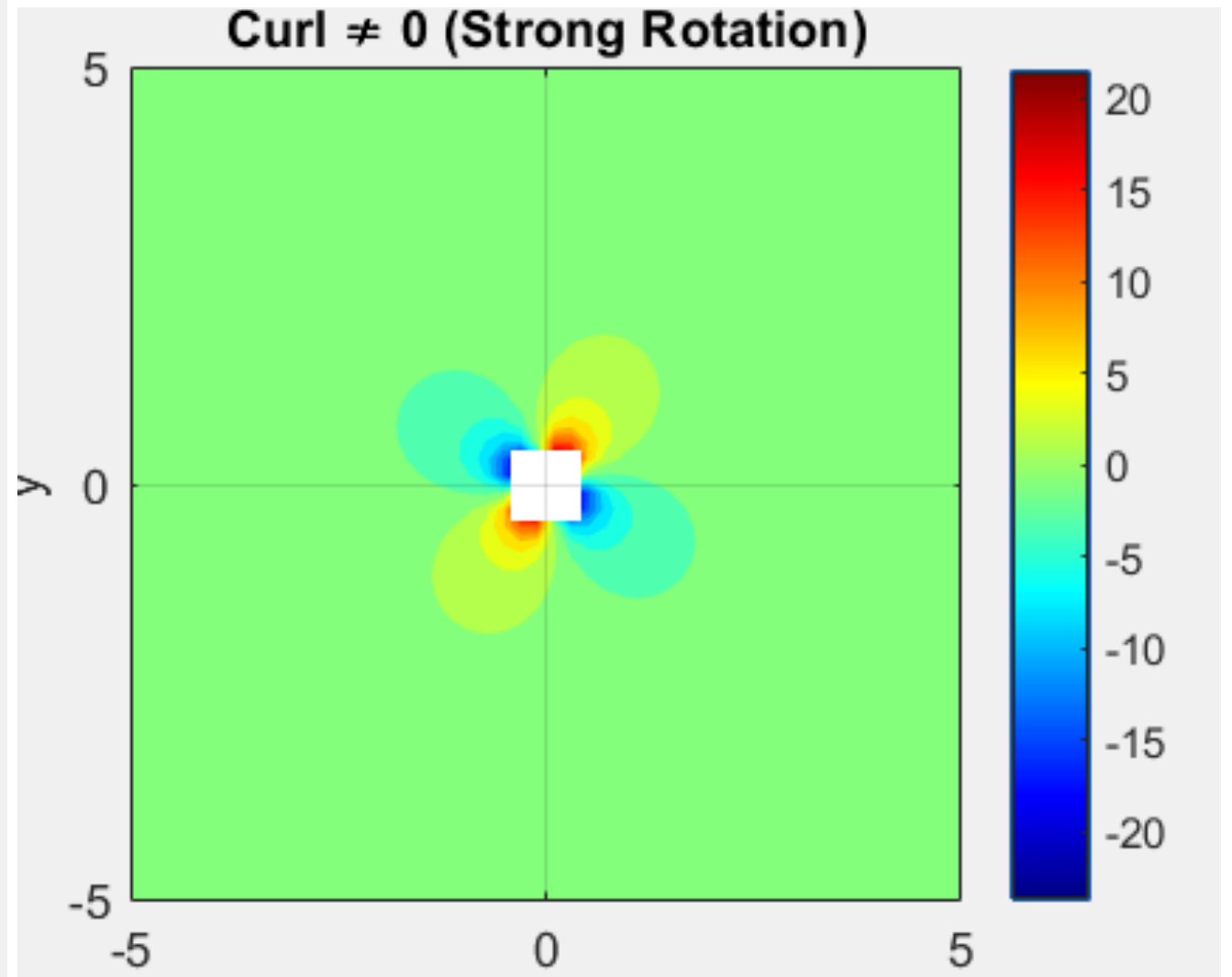
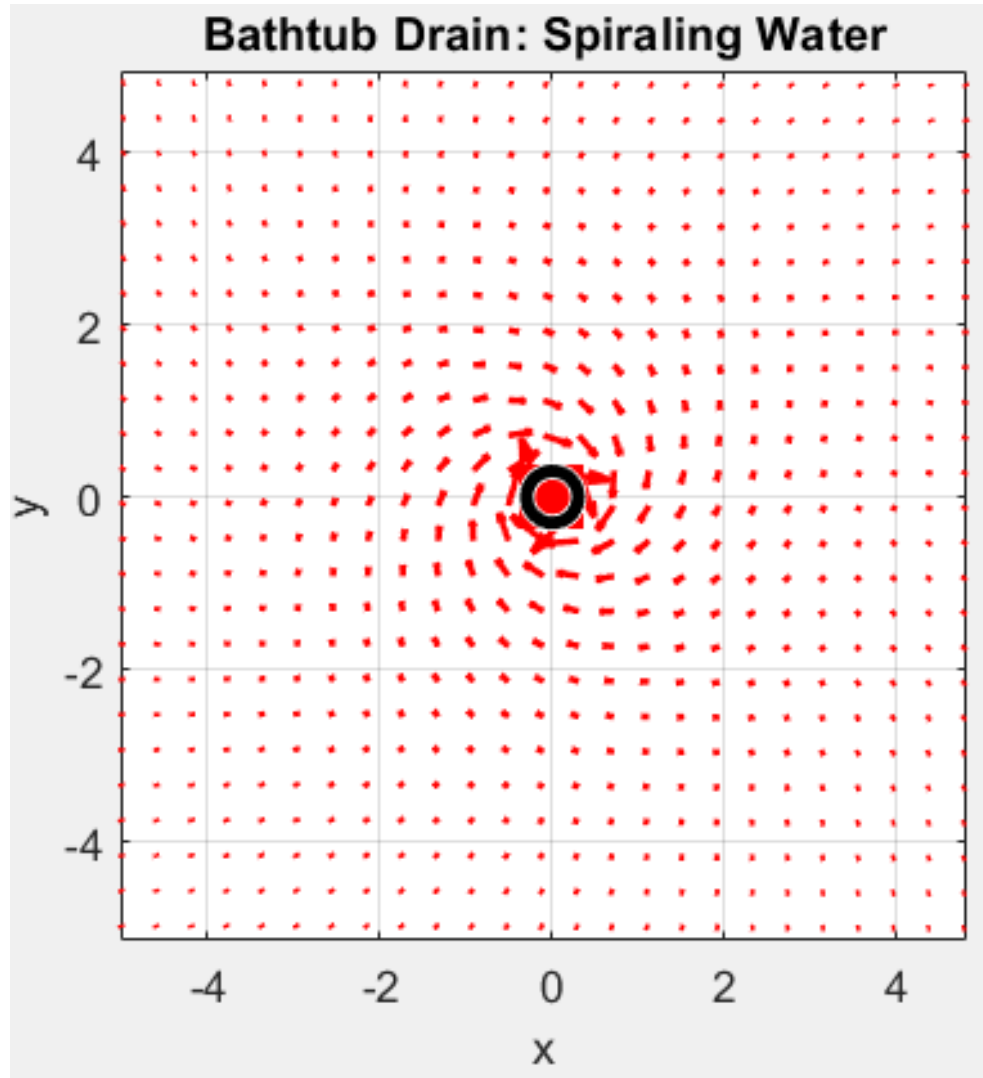
Curl



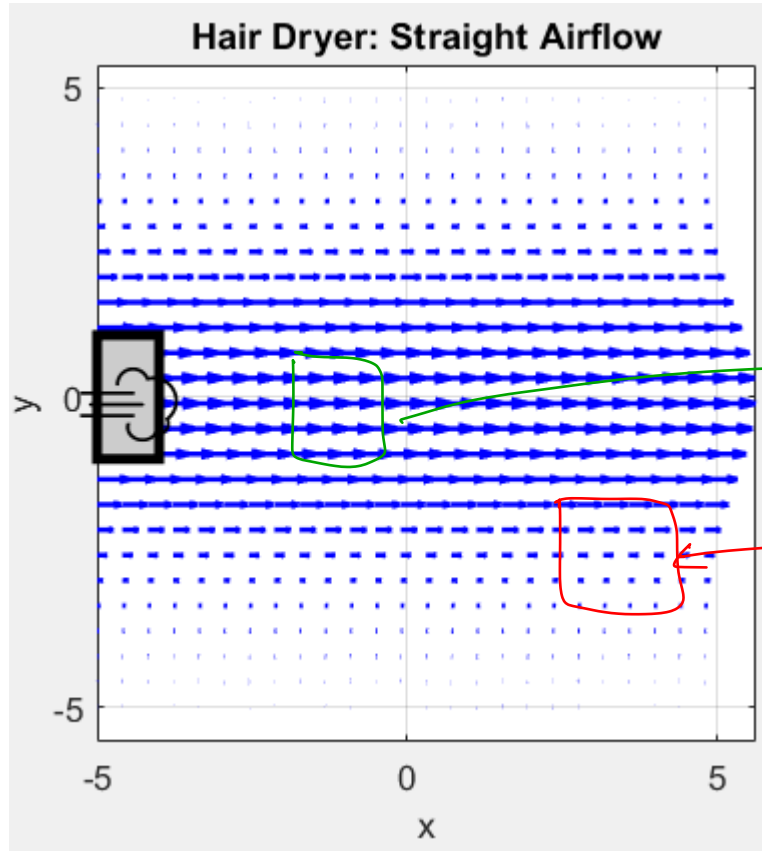
Curl



Curl

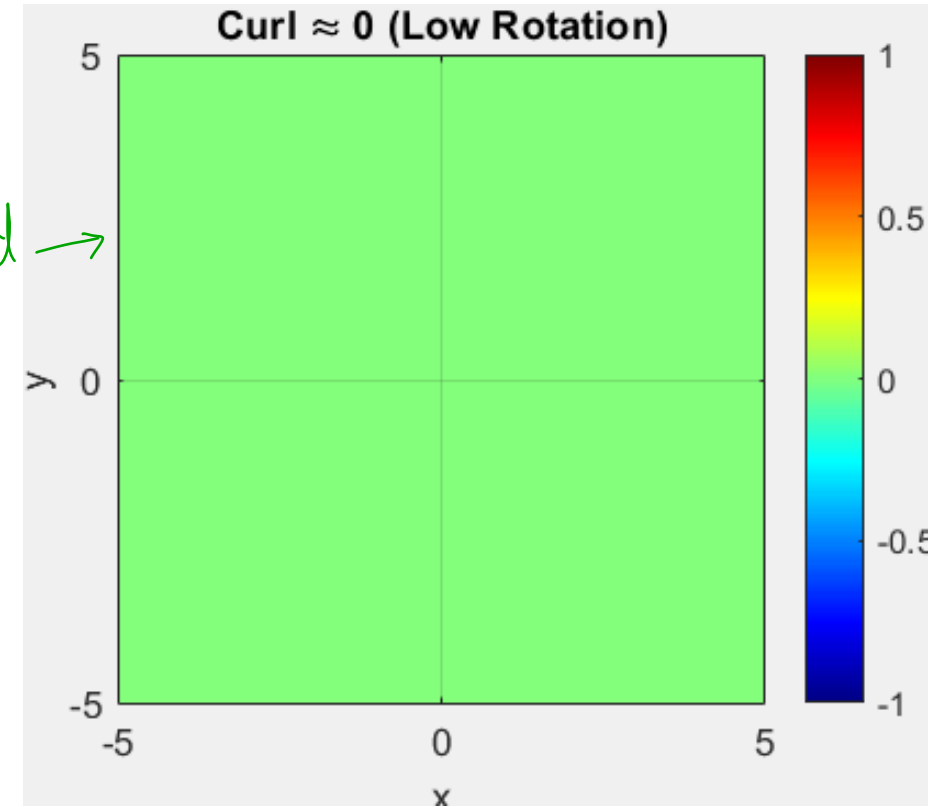


Curl

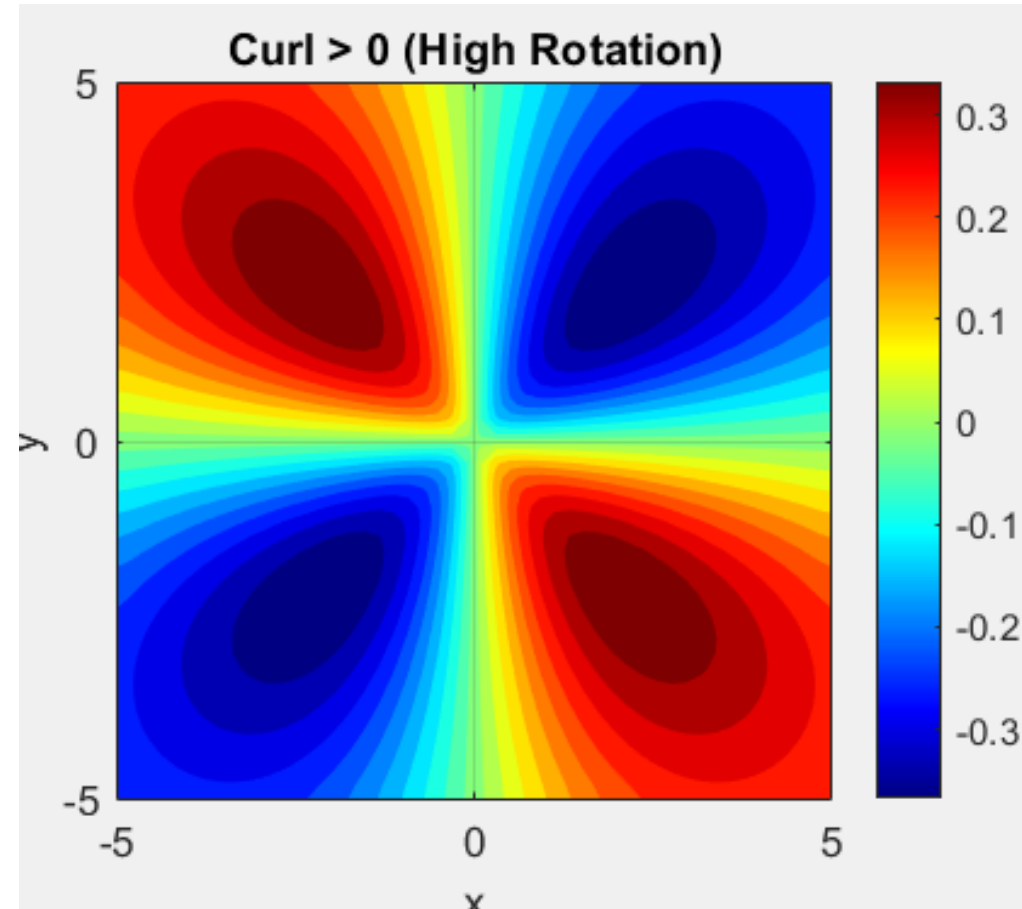
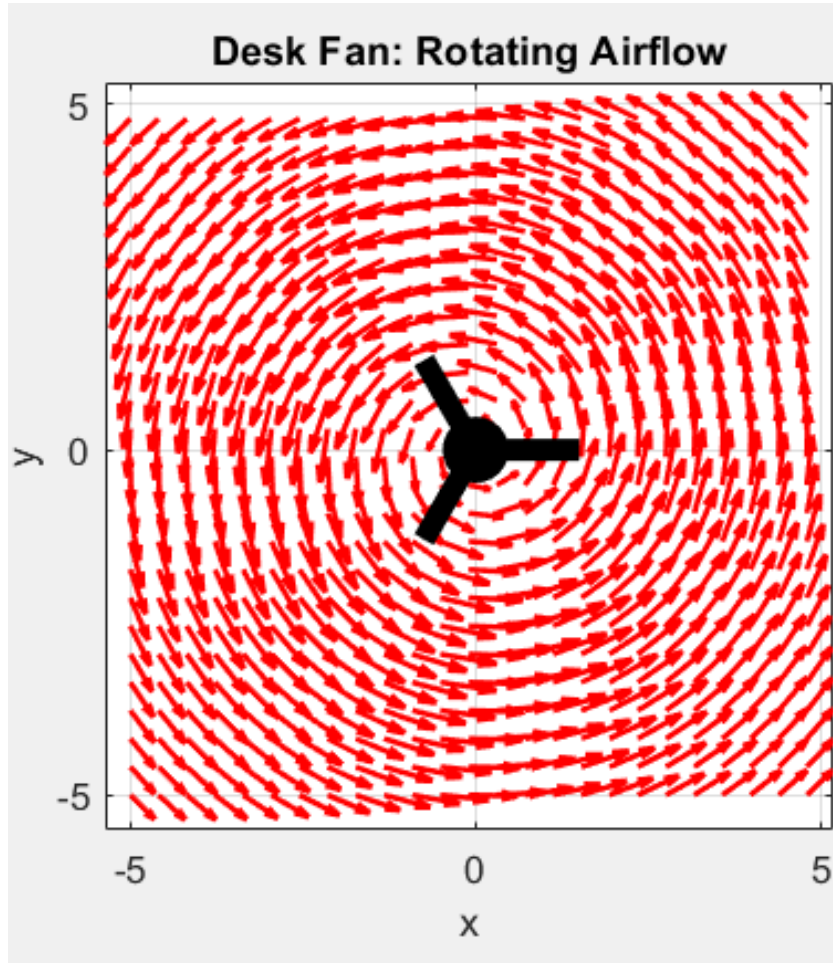


uniform represented →

← not uniform



Curl



Curl

➤ Rectangular Coordinate System

$$\nabla \times \mathbf{A} = \begin{vmatrix} \mathbf{a}_x & \mathbf{a}_y & \mathbf{a}_z \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ A_x & A_y & A_z \end{vmatrix}$$

➤ Cylindrical Coordinate System

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

➤ Spherical Coordinate System

$$\nabla \times \mathbf{A} = \frac{1}{r^2 \sin \theta} \begin{vmatrix} \mathbf{a}_r & r \mathbf{a}_\theta & r \sin \theta \mathbf{a}_\phi \\ \frac{\partial}{\partial r} & \frac{\partial}{\partial \theta} & \frac{\partial}{\partial \phi} \\ A_r & r A_\theta & r \sin \theta A_\phi \end{vmatrix}$$

Curl – Cylindrical Coordinate System

➤ Cylindrical Coordinate System $\mathbf{a}_\rho = \cos \phi \mathbf{a}_x + \sin \phi \mathbf{a}_y$ $\mathbf{a}_\phi = -\sin \phi \mathbf{a}_x + \cos \phi \mathbf{a}_y$

$$\nabla \times \mathbf{A} = \frac{1}{\rho} \begin{vmatrix} \mathbf{a}_\rho & \rho \mathbf{a}_\phi & \mathbf{a}_z \\ \frac{\partial}{\partial \rho} & \frac{\partial}{\partial \phi} & \frac{\partial}{\partial z} \\ A_\rho & \rho A_\phi & A_z \end{vmatrix}$$

$$\vec{A} = \vec{A}_\rho \vec{a}_\rho + \vec{A}_\phi \vec{a}_\phi + \vec{A}_z \vec{a}_z$$

$$\vec{\nabla} = \vec{a}_\rho \frac{\partial}{\partial \rho} + \frac{1}{\rho} \vec{a}_\phi \frac{\partial}{\partial \phi} + \vec{a}_z \frac{\partial}{\partial z}$$

$$\nabla \times \mathbf{A} = \left(\vec{a}_\rho \frac{\partial}{\partial \rho} + \frac{1}{\rho} \vec{a}_\phi \frac{\partial}{\partial \phi} + \vec{a}_z \frac{\partial}{\partial z} \right) \times \vec{A}$$

$$\vec{a}_\rho \times \vec{a}_\rho = 0$$

$$\vec{a}_\phi \times \vec{a}_\rho = -\vec{a}_z$$

$$\vec{a}_z \times \vec{a}_\rho = \vec{a}_\phi$$

first term:

Curl

- ✓ **Curl tells us how much the field wants to spin, and which way it spins.**

$$\text{curl } \mathbf{A} = \nabla \times \mathbf{A} = \left(\lim_{\Delta S \rightarrow 0} \frac{\oint_L \mathbf{A} \cdot d\mathbf{l}}{\Delta S} \right)_{\max} \mathbf{a}_n$$

- The curl of a vector field tells us how much the field tends to rotate around a point, and the direction of the curl shows the axis of that rotation.
- The curl of $\mathbf{A}$ measures how strongly the field circulates around a point. Its direction shows the axis about which the field rotates, and its size tells how intense that rotation is.
- The curl of $\mathbf{A}$ is a vector that represents the local rotational behavior of the field. Its magnitude equals the circulation per unit area as the area becomes very small, and its direction is perpendicular to the surface where this circulation is greatest.

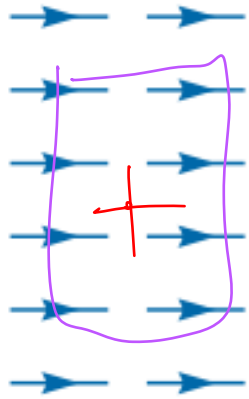
Stokes's theorem

- ✓ **Stokes's theorem:** circulation of a vector field $\mathbf{A}$ around a closed path C is equal to the surface integral of the curl of $\mathbf{A}$ over any surface S bounded by C .
- The direction of C and the surface normal $d\mathbf{S}$ must satisfy the right-hand rule.

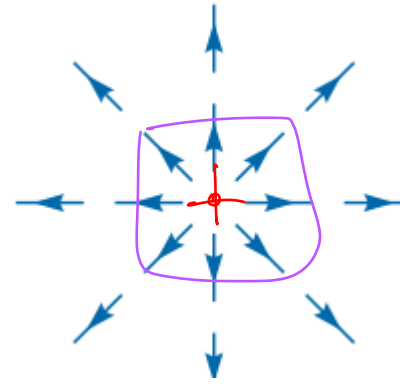
$$\oint_{\text{Line}} \mathbf{A} \cdot d\mathbf{l} = \int_{\text{Surface}} (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$



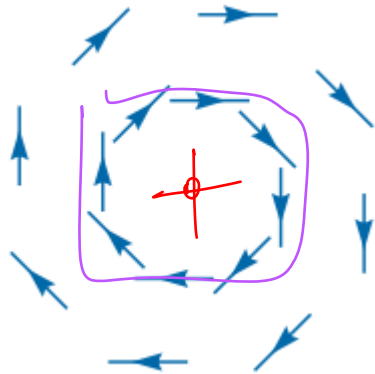
Classification of Vector Fields



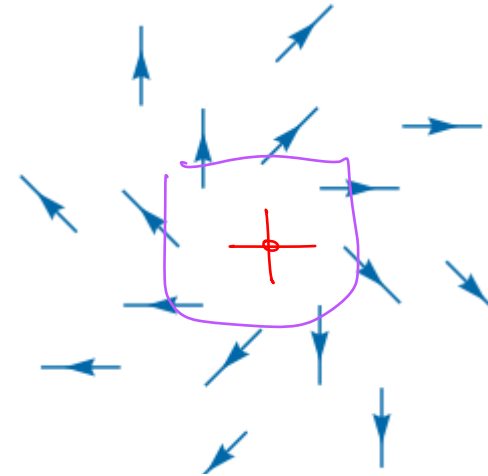
$$\nabla \cdot \mathbf{A} = 0, \nabla \times \mathbf{A} = \mathbf{0}$$



$$\nabla \cdot \mathbf{A} \neq 0, \nabla \times \mathbf{A} = \mathbf{0}$$



$$\nabla \cdot \mathbf{A} = 0, \nabla \times \mathbf{A} \neq \mathbf{0}$$



$$\nabla \cdot \mathbf{A} \neq 0, \nabla \times \mathbf{A} \neq \mathbf{0}$$

Divergence, scaling-factor form

Curl, scaling-factor form

Example 1

PRACTICE EXERCISE 3.6

Determine the divergence of the following vector fields and evaluate them at the specified points.

(a) $\mathbf{A} = yz\mathbf{a}_x + 4xy\mathbf{a}_y + y\mathbf{a}_z$ at $(1, -2, 3)$

(b) $\mathbf{B} = \rho z \sin \phi \mathbf{a}_\rho + 3\rho z^2 \cos \phi \mathbf{a}_\phi$ at $(5, \pi/2, 1)$

(c) $\mathbf{C} = 2r \cos \theta \cos \phi \mathbf{a}_r + r^{1/2} \mathbf{a}_\phi$ at $(1, \pi/6, \pi/3)$

PRACTICE EXERCISE 3.7

Determine the flux of $\mathbf{D} = \rho^2 \cos^2 \phi \mathbf{a}_\rho + z \sin \phi \mathbf{a}_\phi$ over the closed surface of the cylinder $0 \leq z \leq 1$, $\rho = 4$. Verify the divergence theorem for this case.

Vector Calculus

Today's plan

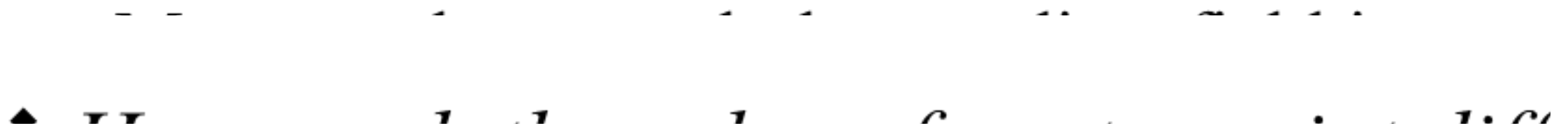
- Laplacian (3.8 – p. 90)
- Divergence and Curl with scaling factors
- Chapter Review

Laplacian

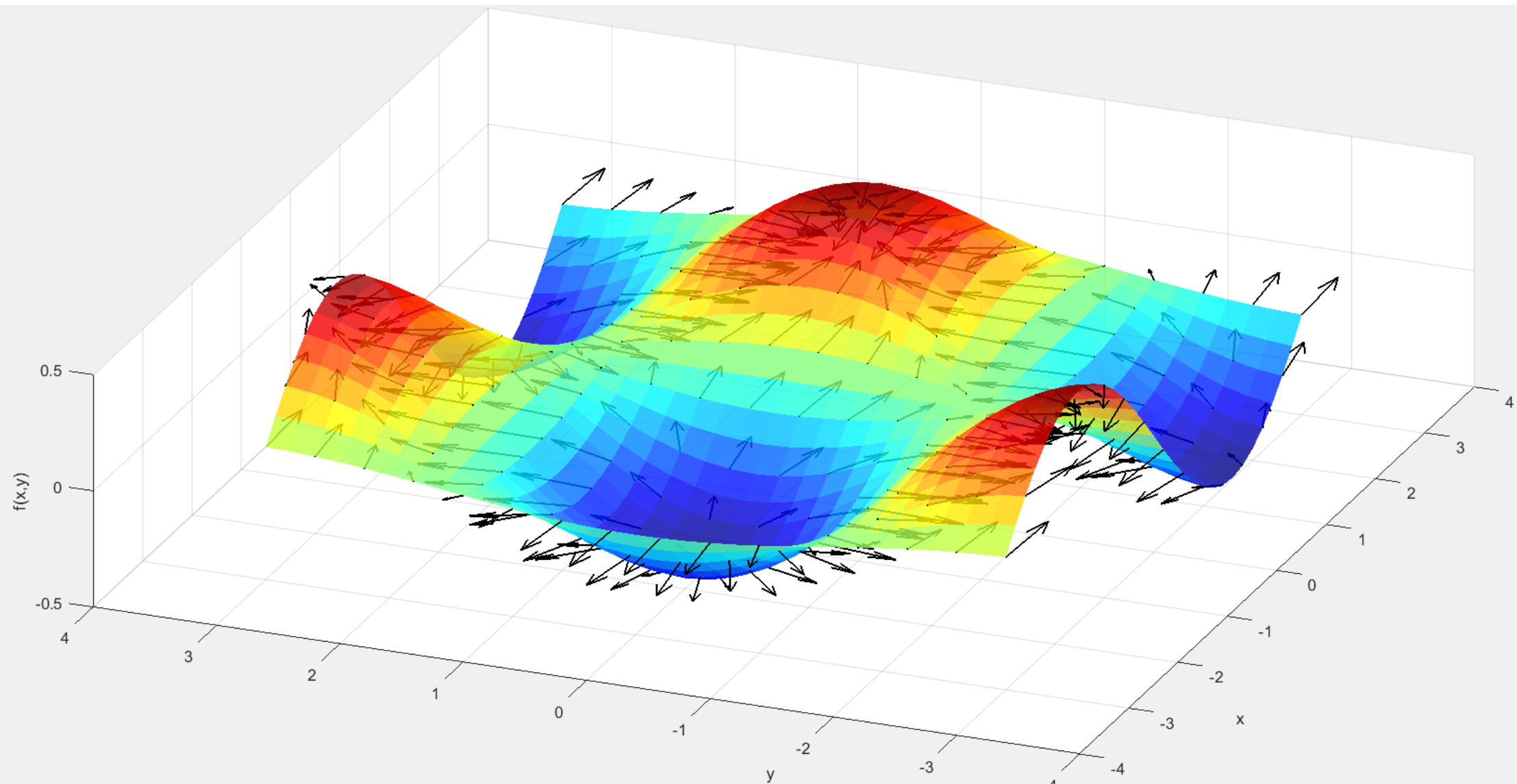
✓ Laplacian of a scalar field is the divergence of

Take gradient of a scalar field (φ) $\nabla\varphi$

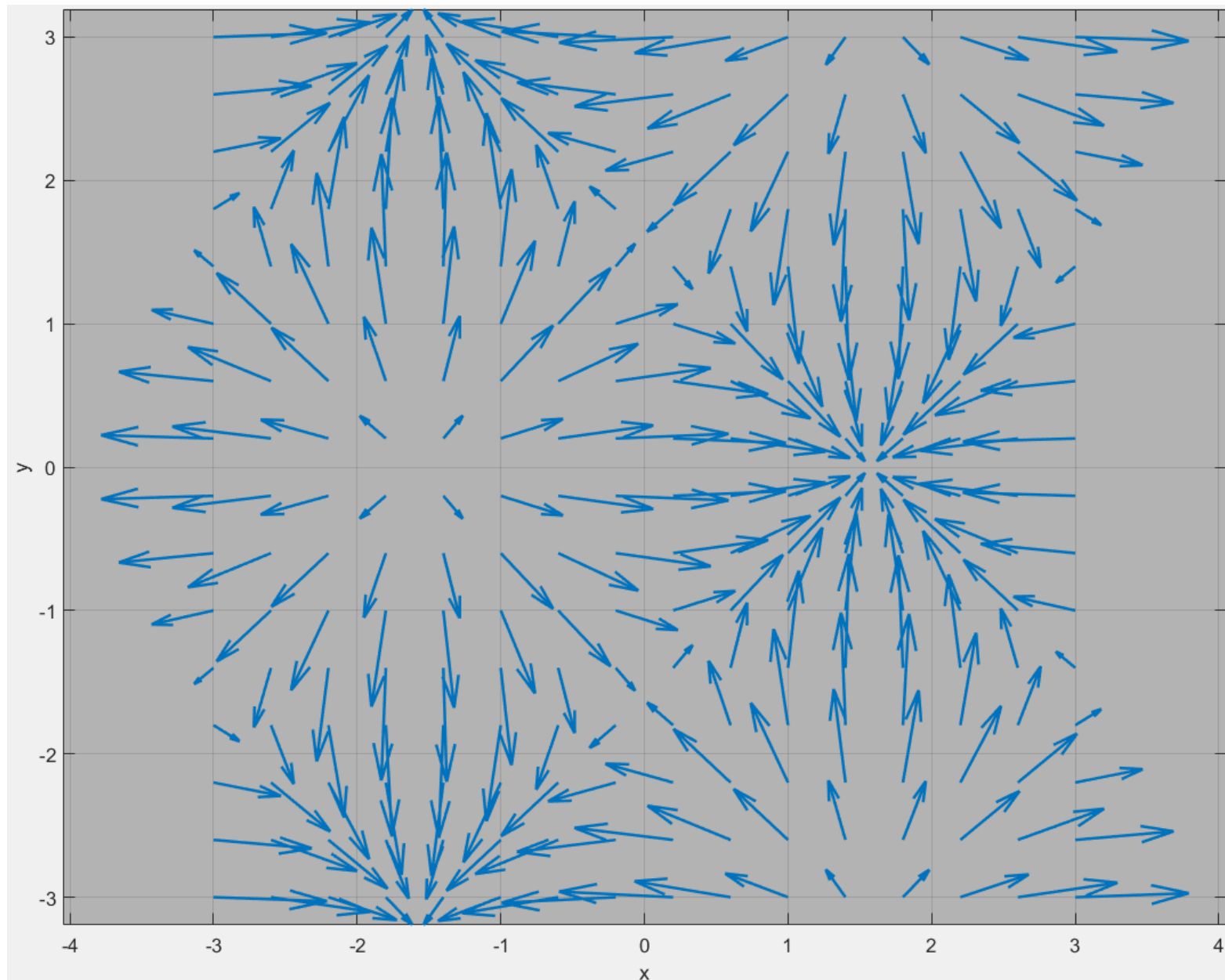
➤ What does the divergence of a vector field tell

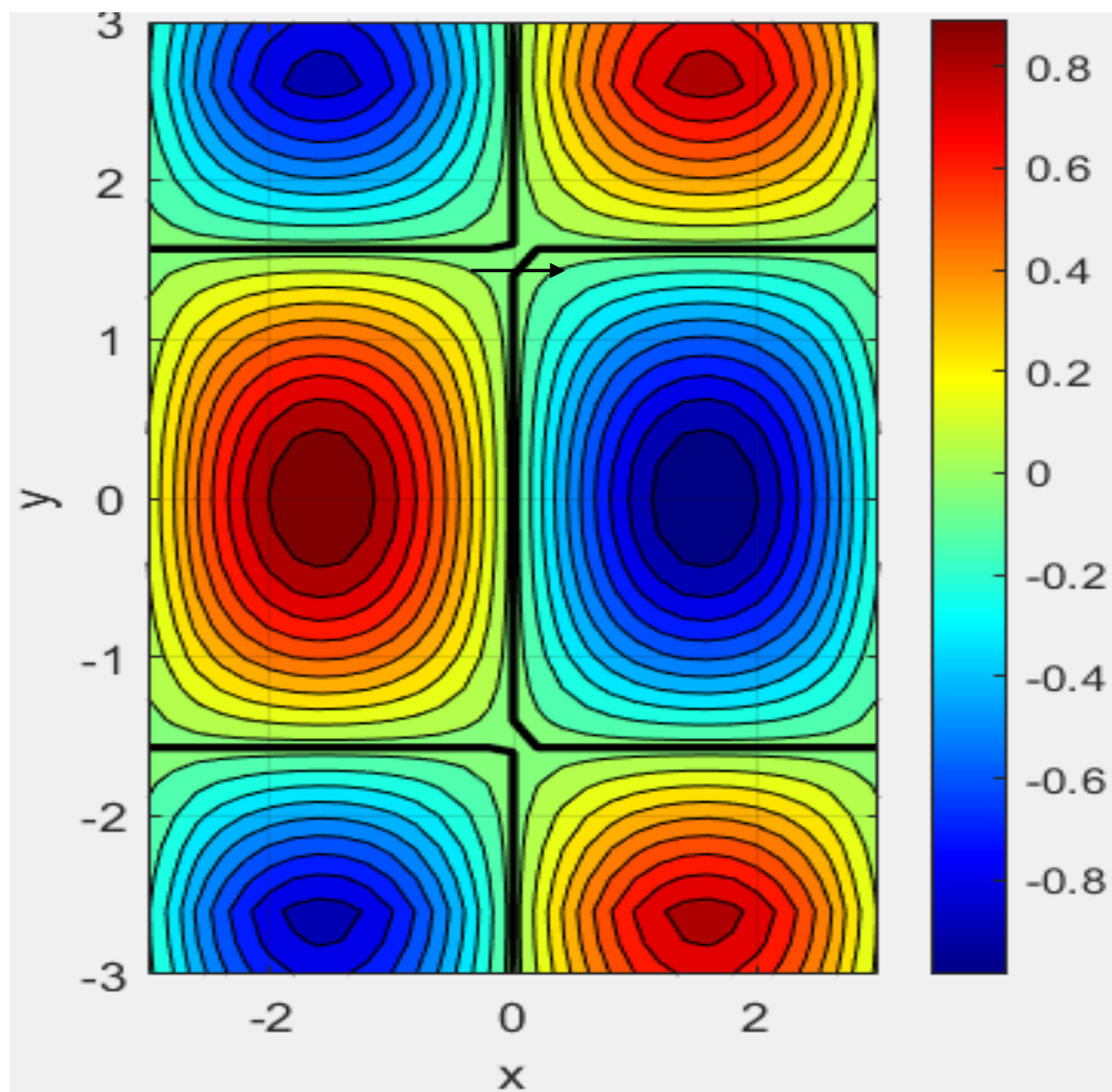


Scalar field $f(x,y)$



0.1 0.1 0.1 0.1 0.1

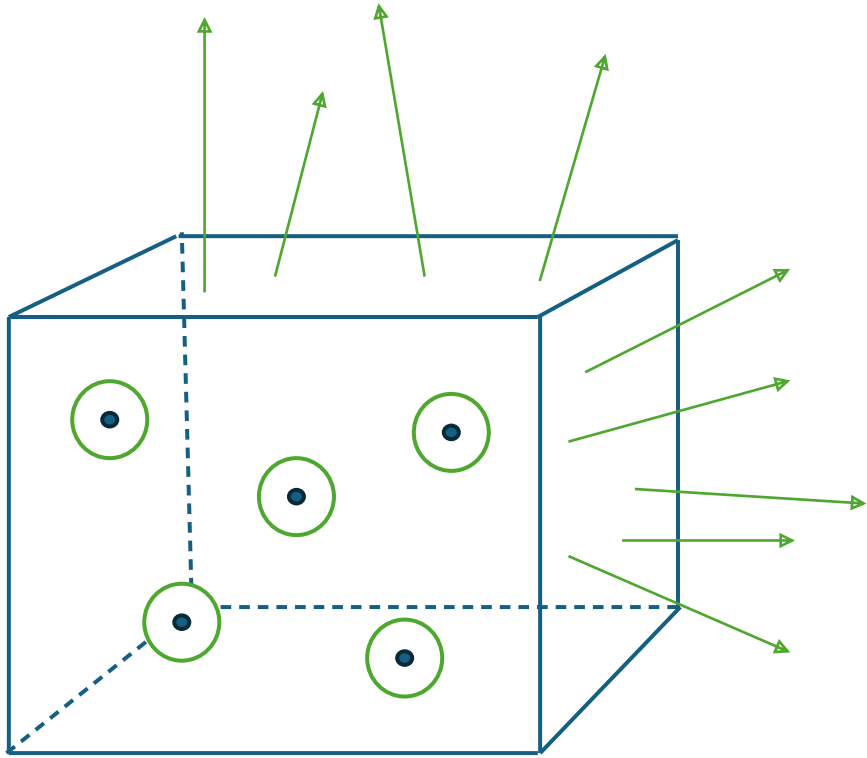




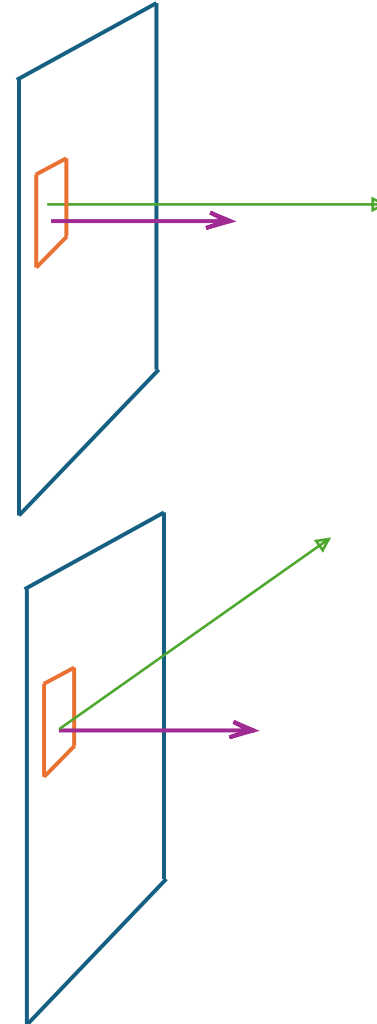
Divergence, scaling-factor form

Curl, scaling-factor form

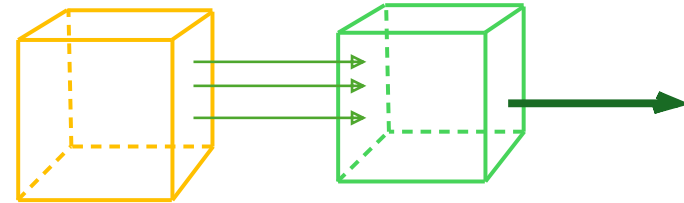
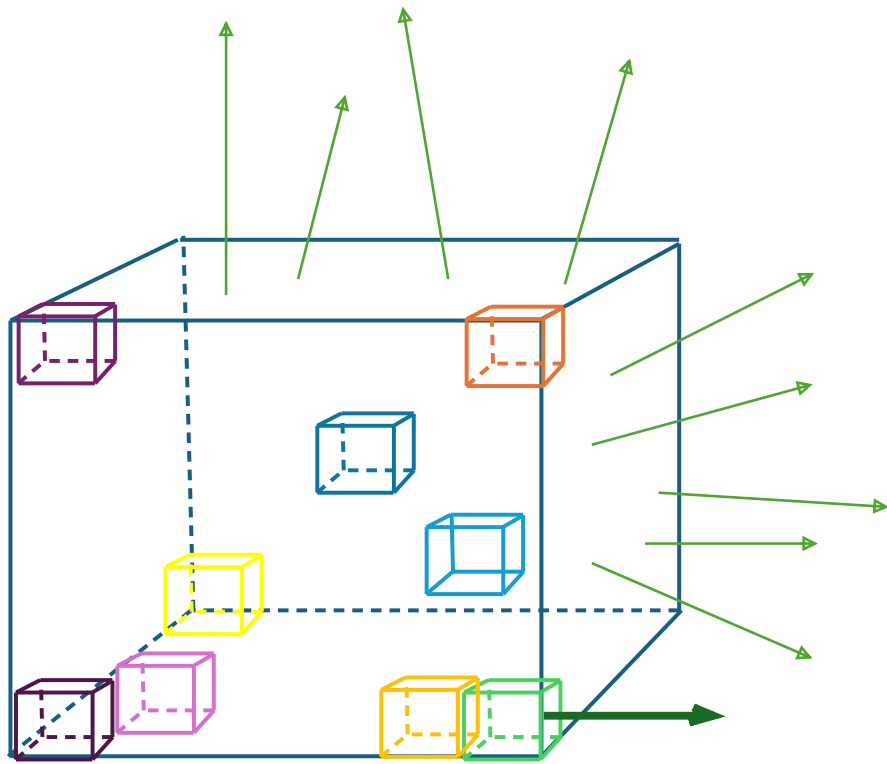
Divergence and Divergence Theorem



$$\oint_S \mathbf{A} \cdot d\mathbf{S}$$

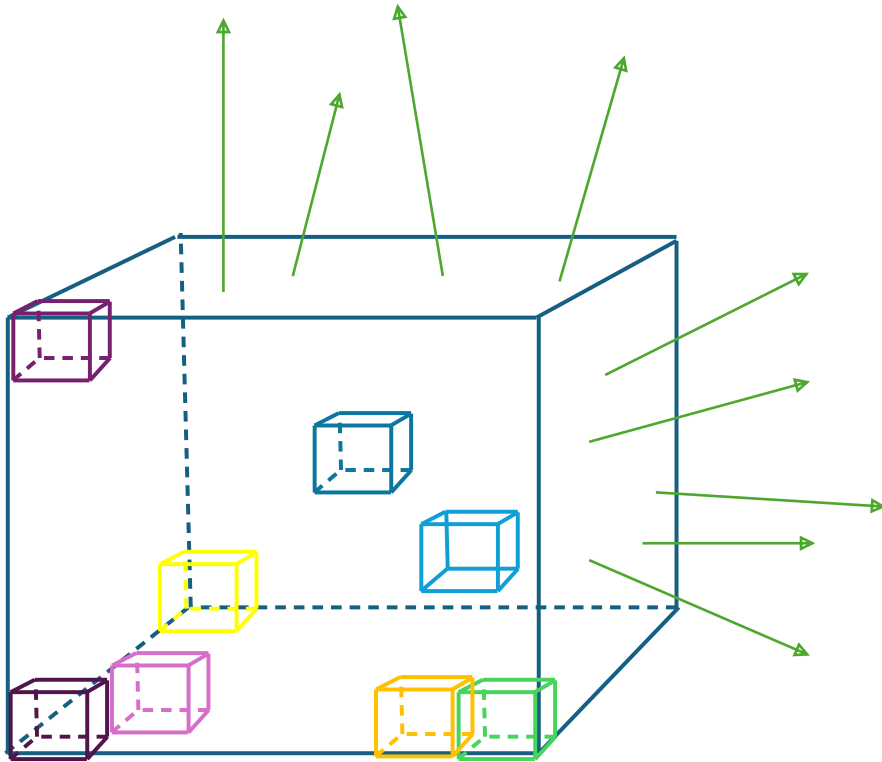


Divergence and Divergence Theorem

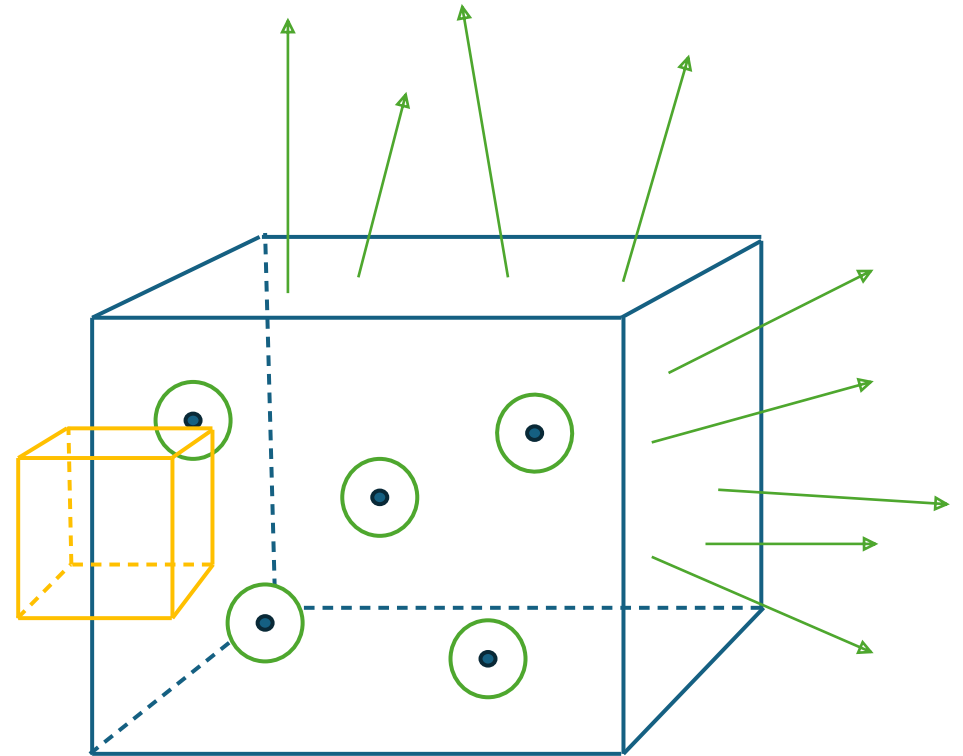


$$\int_v \nabla \cdot \mathbf{A} \, dv$$

Divergence and Divergence Theorem

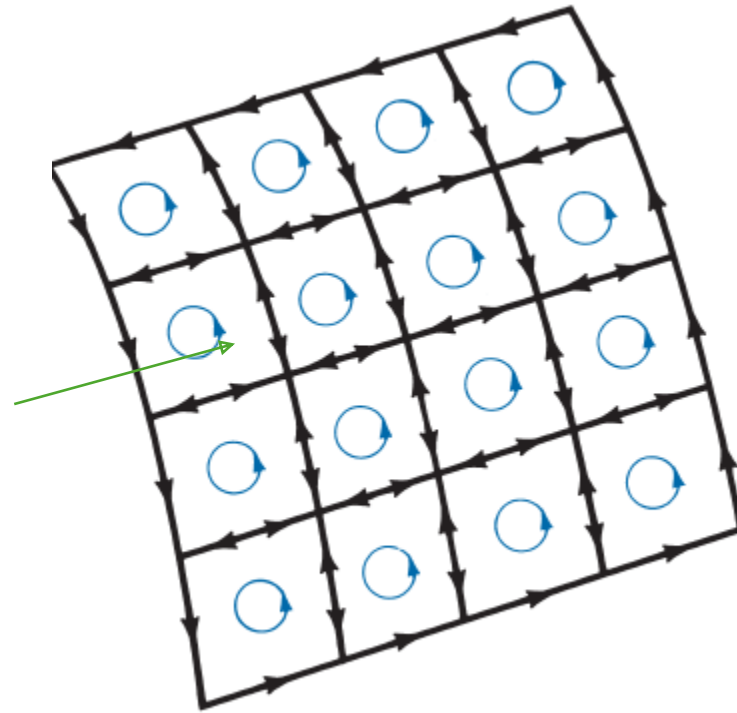


$$\int_V \nabla \cdot \mathbf{A} \, dv$$



$$\oint_S \mathbf{A} \cdot d\mathbf{S}$$

Stokes's Theorem



$$\oint_L \mathbf{A} \cdot d\mathbf{l} = \int_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}$$

Example 1

PRACTICE EXERCISE 3.6

Determine the divergence of the following vector fields and evaluate them at the specified points.

- (a) $\mathbf{A} = yz\mathbf{a}_x + 4xy\mathbf{a}_y + y\mathbf{a}_z$ at $(1, -2, 3)$
- (b) $\mathbf{B} = \rho z \sin \phi \mathbf{a}_\rho + 3\rho z^2 \cos \phi \mathbf{a}_\phi$ at $(5, \pi/2, 1)$
- (c) $\mathbf{C} = 2r \cos \theta \cos \phi \mathbf{a}_r + r^{1/2}\mathbf{a}_\phi$ at $(1, \pi/6, \pi/3)$

PRACTICE EXERCISE 3.7

Determine the flux of $\mathbf{D} = \rho^2 \cos^2 \phi \mathbf{a}_\rho + z \sin \phi \mathbf{a}_\phi$ over the closed surface of the cylinder $0 \leq z \leq 1$, $\rho = 4$. Verify the divergence theorem for this case.

PRACTICE EXERCISE 3.2

Calculate the circulation of

$$\mathbf{A} = \rho \cos \phi \mathbf{a}_\rho + z \sin \phi \mathbf{a}_z$$

around the edge L of the wedge defined by $0 \leq \rho \leq 2$, $0 \leq \phi \leq 60^\circ$, $z = 0$ and shown in Figure 3.12.

PRACTICE EXERCISE 3.9

Use Stokes's theorem to confirm your result in Practice Exercise 3.2.

Electrostatics

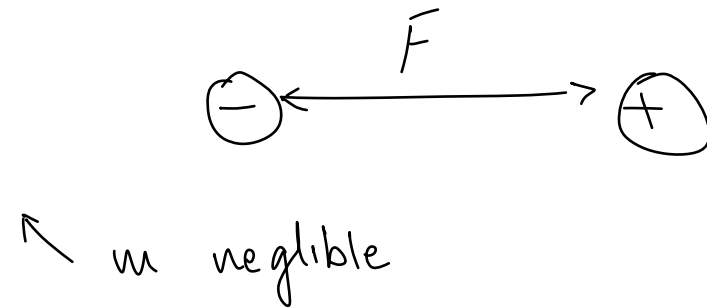
Today's plan

- Coulomb's Law and Field Intensity (4.2 – p. 112)
- Electric Fields Due to Continuous Charge Distribution (4.3 – p. 119)

Coulomb's Law

- There is a force between electrical charges
- This force is called the electric force
- When considering two point charges, Q_1 & Q_2 the force between them, F , has the following properties
 1. The force is along the line joining the charges
 2. The force is proportional to the product of the charges, $Q_1 Q_2$
 3. The force is inversely proportional to the square of the distance between them, R
- The mathematical definition of this force is,

$$F = k \frac{Q_1 Q_2}{R^2}$$



Coulomb's Law

➤ Details about the formula:

The charges can be like or unlike.
The unit of electrical charge is C

$$1\ C = \frac{1.6 \times 10^{+19}}{n_e} e^-$$

$$F = k \frac{Q_1 Q_2}{R^2}$$

The force between the charges, the
unit for force is N .

R is the distance between the
charges, and its unit is m .

$$\frac{1}{k} = \frac{Q_1 Q_2}{F R^2}$$

k is the proportionality constant, and its
unit is $[k] \neq \frac{C^2}{N \cdot m^2}$.

$$k = \frac{N \cdot m^2}{C^2}$$

Coulomb's Law

➤ Details about k :

ϵ_0 - free space permittivity ← lowest permittivity possible
 ϵ_d - any medium permittivity

$$k = \frac{1}{4\pi\epsilon_0}$$

Permittivity of Free Space and is a constant. Its value differs from one medium to another. The unit for this constant is F/m .

The value for k is,
 $9 \times 10^9 m/F$.

The value of the permittivity is,
 $\epsilon_0 = 8.854 \times 10^{-12} \cong \frac{10^{-9}}{36\pi} F/m$

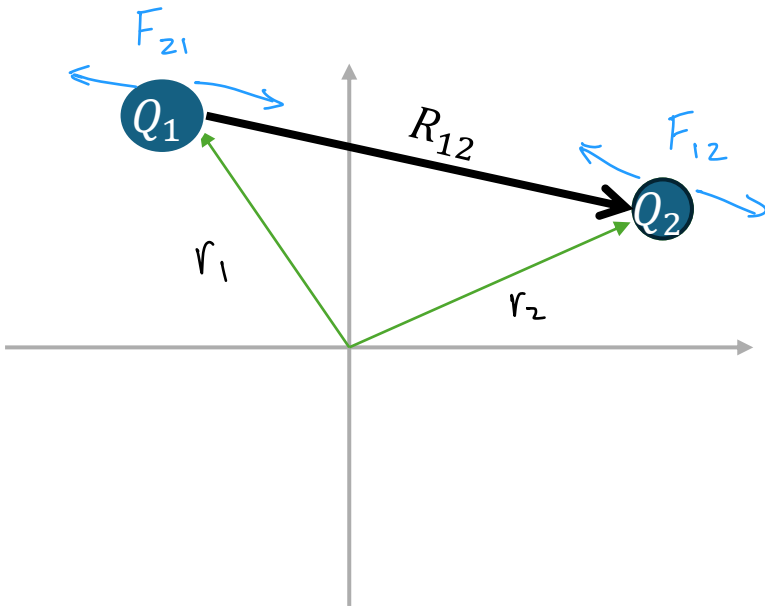
unit = energy stored in
defined space

Farads for capacitance

Coulomb's Law

➤ Electrical Force Equation as a Vector

$$\vec{F}_{21} = -\vec{F}_{12}$$

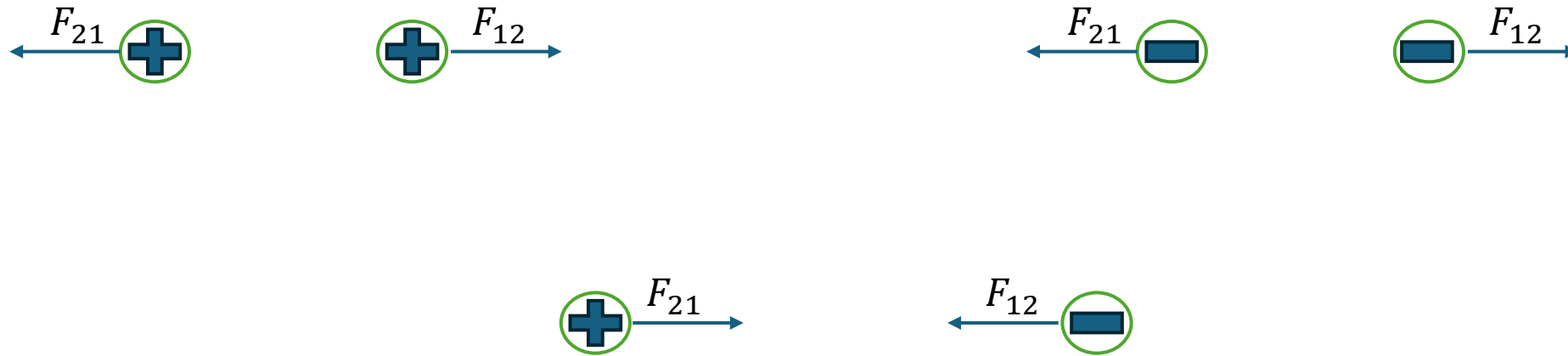


$$\mathbf{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \mathbf{a}_{R_{12}}$$

$$\mathbf{a}_{R_{12}} = \frac{\vec{R}_{12}}{|\vec{R}_{12}|}$$

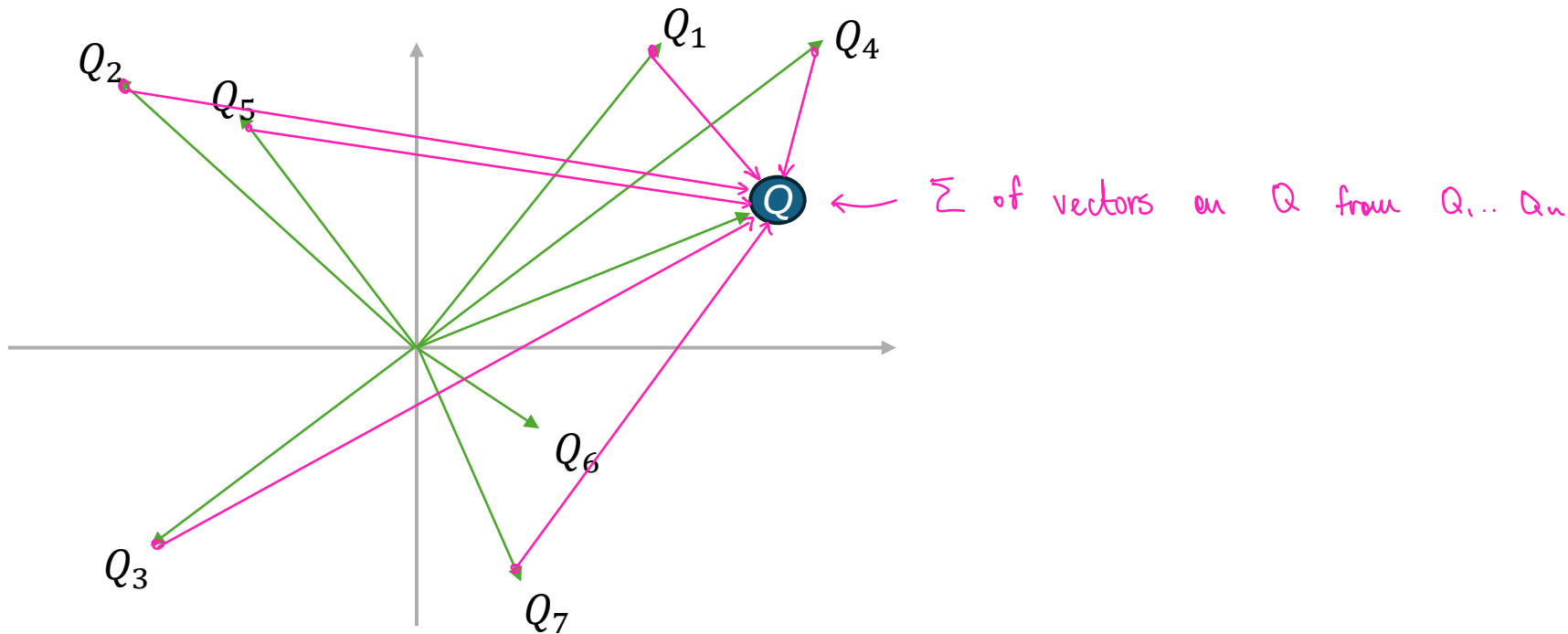
Coulomb's Law

- Electrical force due to like and unlike charges



Coulomb's Law

- Electrical force due to more than one charge (superposition)



$$\mathbf{F} = \frac{Q}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3} \rightarrow F = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} a_{R12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \cdot \frac{\vec{R}_{12}}{R} = \frac{Q_1}{4\pi\epsilon_0} \cdot Q_2$$

Coulomb's Law

➤ Why does the force have this mathematical form?

$$F = \frac{k Q_1 Q_2}{R^2} = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1 Q_2}{R^2}$$

$V = \frac{4}{3}\pi R^3$
 $SA = V'$
 $= 4\pi R^2$

The diagram illustrates the derivation of the R^2 term in Coulomb's Law. It shows the formula for the force $F = \frac{k Q_1 Q_2}{R^2}$ and its equivalent form $F = \frac{1}{4\pi\epsilon_0} \cdot \frac{Q_1 Q_2}{R^2}$. Below these, the volume of a sphere is given as $V = \frac{4}{3}\pi R^3$, and its surface area is derived as $SA = V' = 4\pi R^2$. Two blue curved arrows originate from the 4π in the denominator of the force equation and point to the 4π in the surface area equation, indicating that the surface area of a sphere is the geometric basis for the R^2 term in the force law.

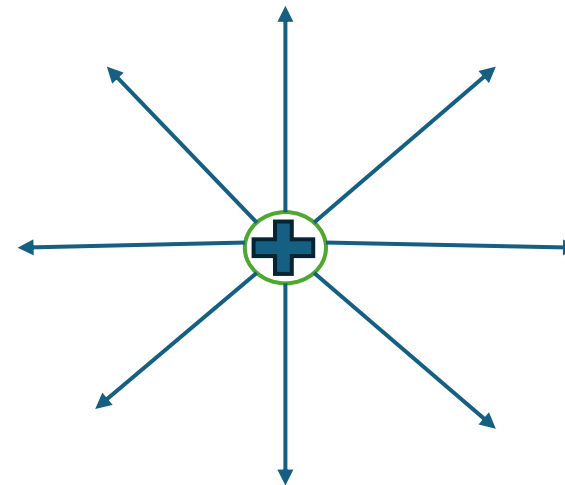
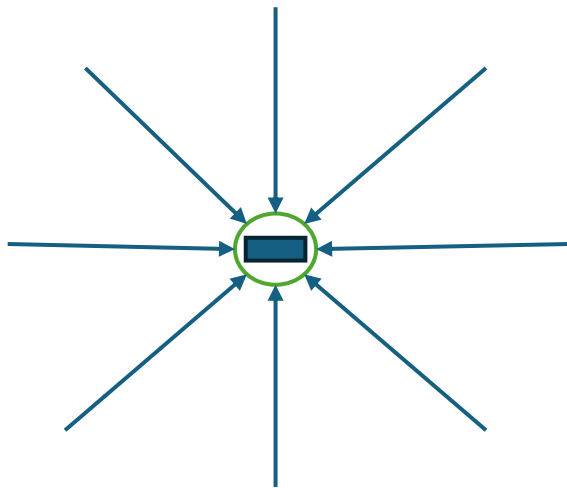
Coulomb's Law

➤ Electric field due to a charge

+ test charge → ideally want $Q = 0$ to find accurate F & Q
→ also don't want test charge to affect surrounding charges
→ should always be +ve

$$E = \lim_{Q \rightarrow 0} \frac{F}{Q}$$

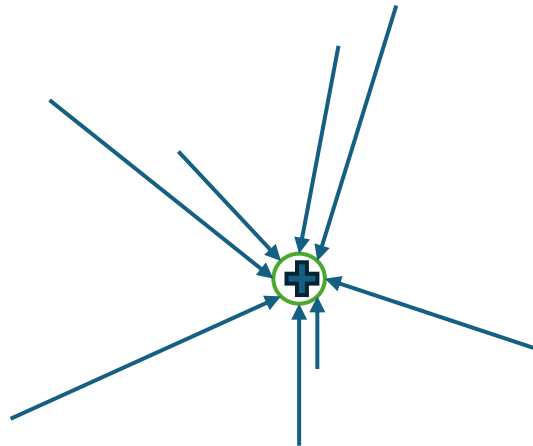
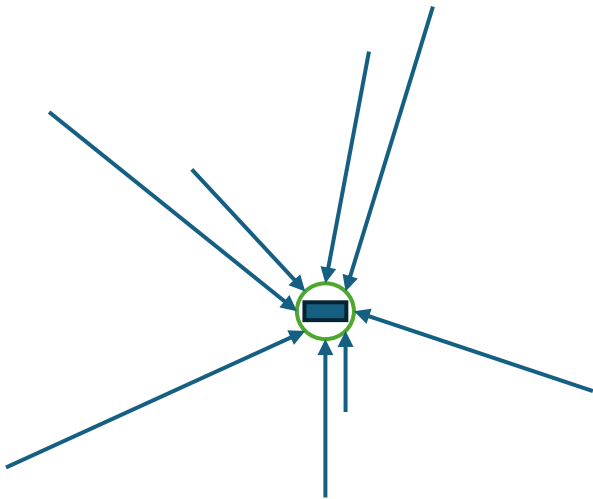
$$E = \frac{F}{Q}$$



Coulomb's Law

- Electric field due to multiple charges (superposition)

$$\mathbf{E} = \frac{1}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3}$$



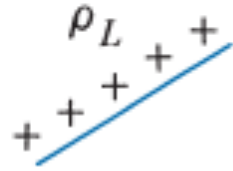
Coulomb's Law

➤ Electrical charge density = ρ ← general density formula



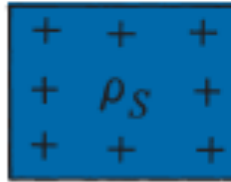
Point
charge

0D



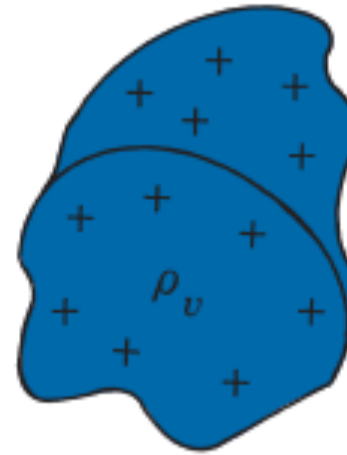
Line
charge

1D



Surface
charge

2D



Volume
charge

3D

Coulomb's Law

Coulomb's Law

Coulomb's Law

Coulomb's Law

Electrostatics

Today's plan

- Electric Fields Due to Continuous Charge Distribution (4.3 – p. 119)

Electrostatics

- Electric field of a single point charge and multiple point charges

Q

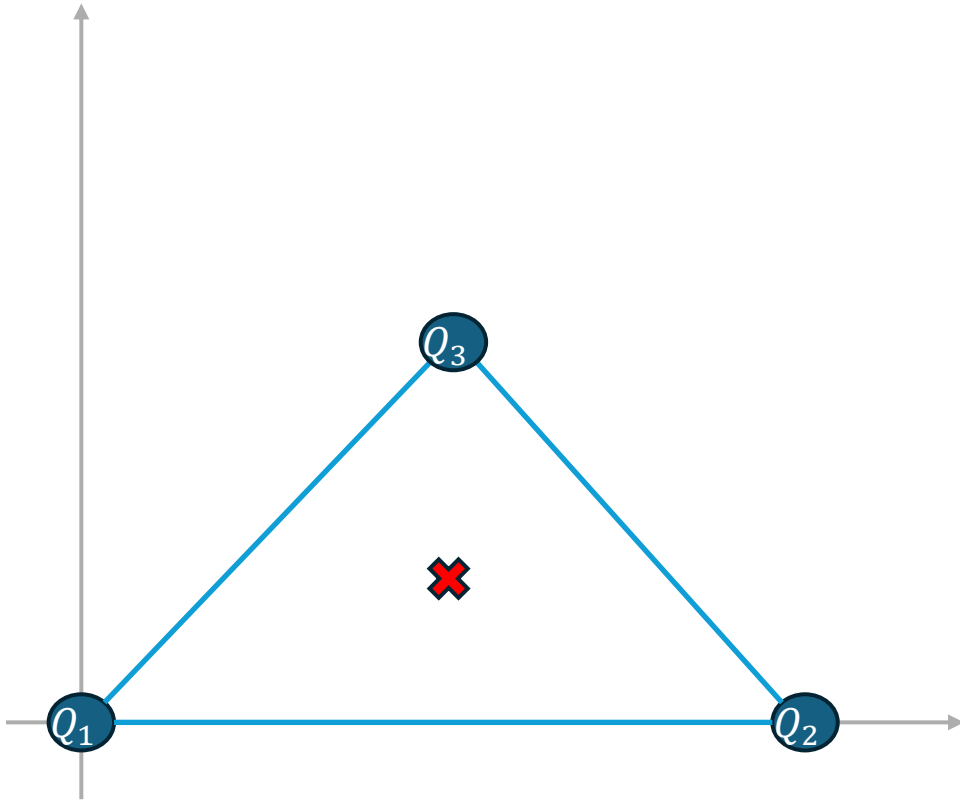
$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R = \frac{Q(\mathbf{r} - \mathbf{r}')}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}'|^3} = \frac{Q}{4\pi\epsilon_0 r^2} \mathbf{a}_r$$

$Q_1 \quad Q_2 \quad \dots \quad Q_N$

$$\begin{aligned} \mathbf{E} &= \mathbf{E}_1 + \mathbf{E}_2 + \mathbf{E}_3 + \dots + \mathbf{E}_N \\ &= \frac{Q_1(\mathbf{r} - \mathbf{r}_1)}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_1|^3} + \frac{Q_2(\mathbf{r} - \mathbf{r}_2)}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_2|^3} + \dots + \frac{Q_N(\mathbf{r} - \mathbf{r}_N)}{4\pi\epsilon_0 |\mathbf{r} - \mathbf{r}_N|^3} \\ &= \frac{1}{4\pi\epsilon_0} \sum_{k=1}^N \frac{Q_k(\mathbf{r} - \mathbf{r}_k)}{|\mathbf{r} - \mathbf{r}_k|^3} \end{aligned}$$

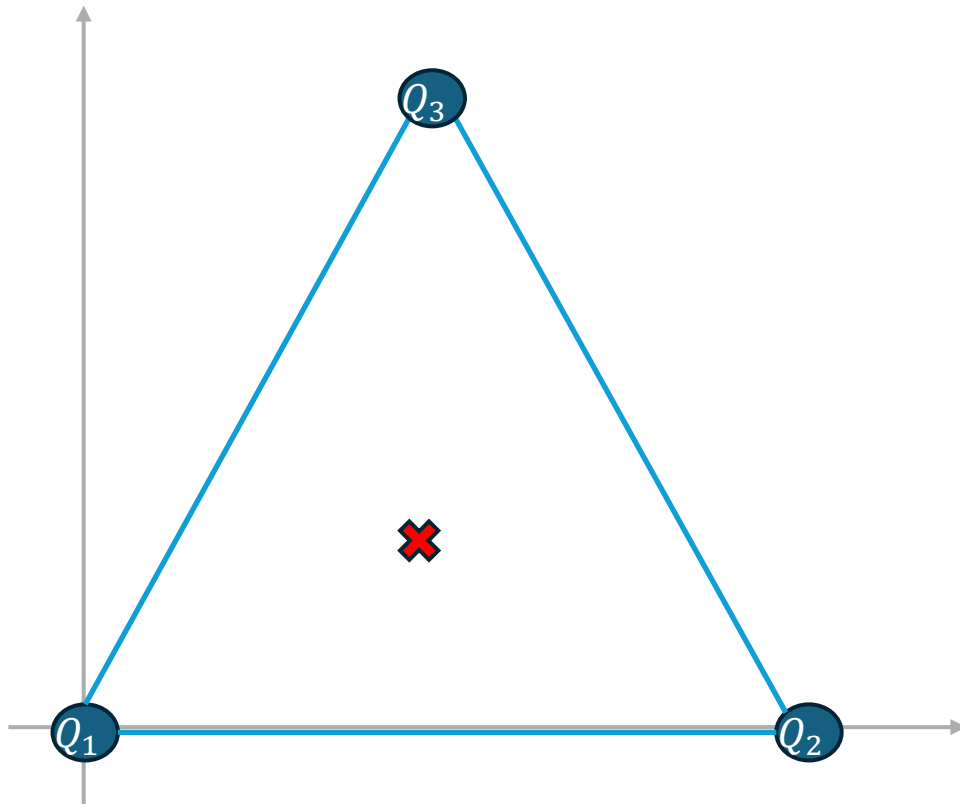
Electrostatics

- Electric field due to a few point charges



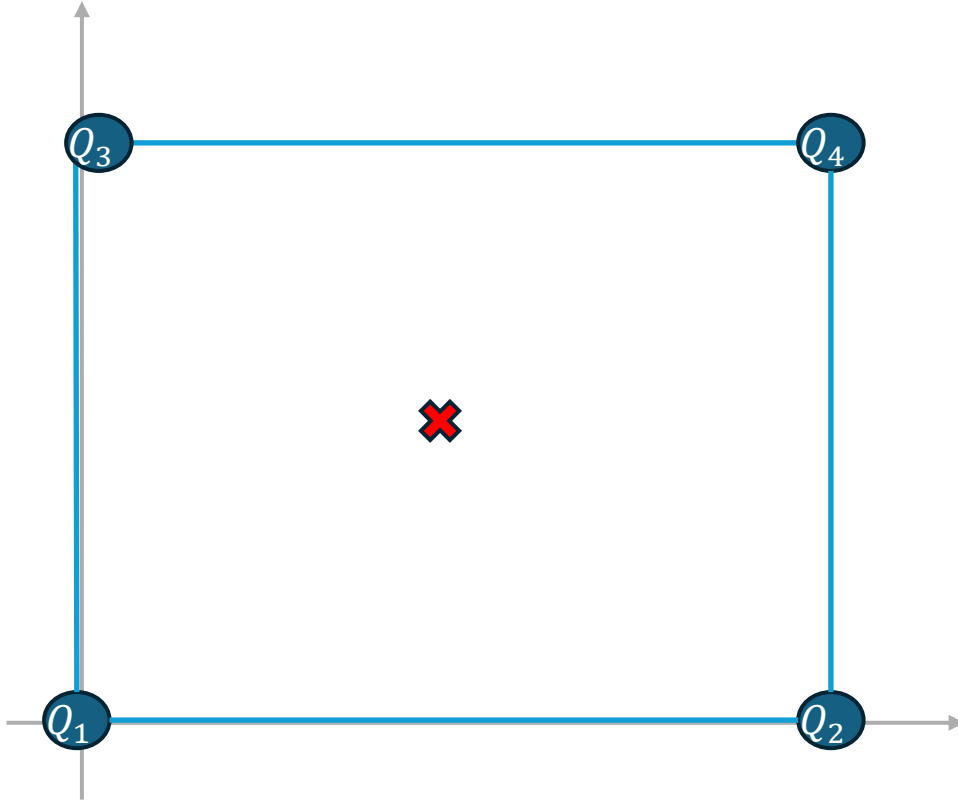
Electrostatics

- Electric field due to a few point charges



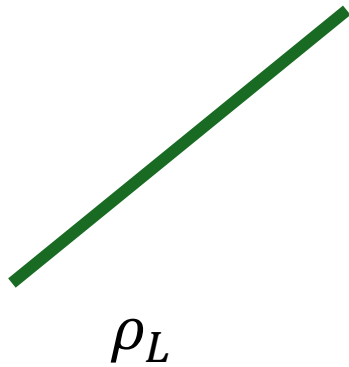
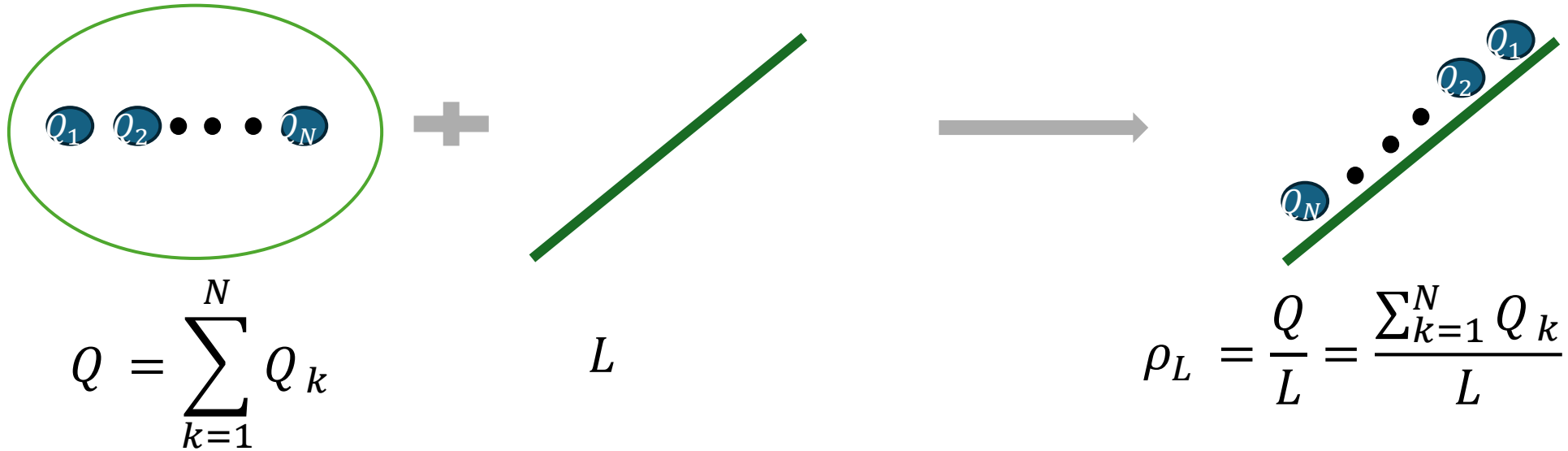
Electrostatics

- Electric field due to a few point charges



Electrostatics

- The electrical field due to a line distribution of charge

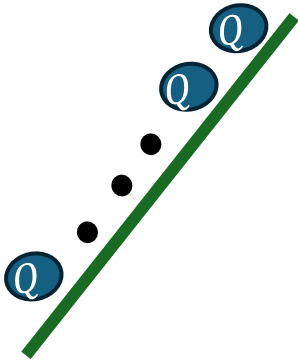


$$Q = \int_L \rho_L dl$$

Electrostatics

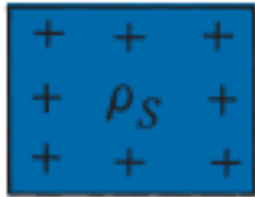
- The electrical field due to a line distribution of charge

$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R \quad \longrightarrow \quad \mathbf{E} = \int_L \frac{\rho_L dl}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$



Electrostatics

- The electrical field due to a surface and a volume distribution of charge



Surface
charge

$$dQ = \rho_S dS \rightarrow Q = \int_S \rho_S dS$$

$$\mathbf{E} = \int_S \frac{\rho_S dS}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$



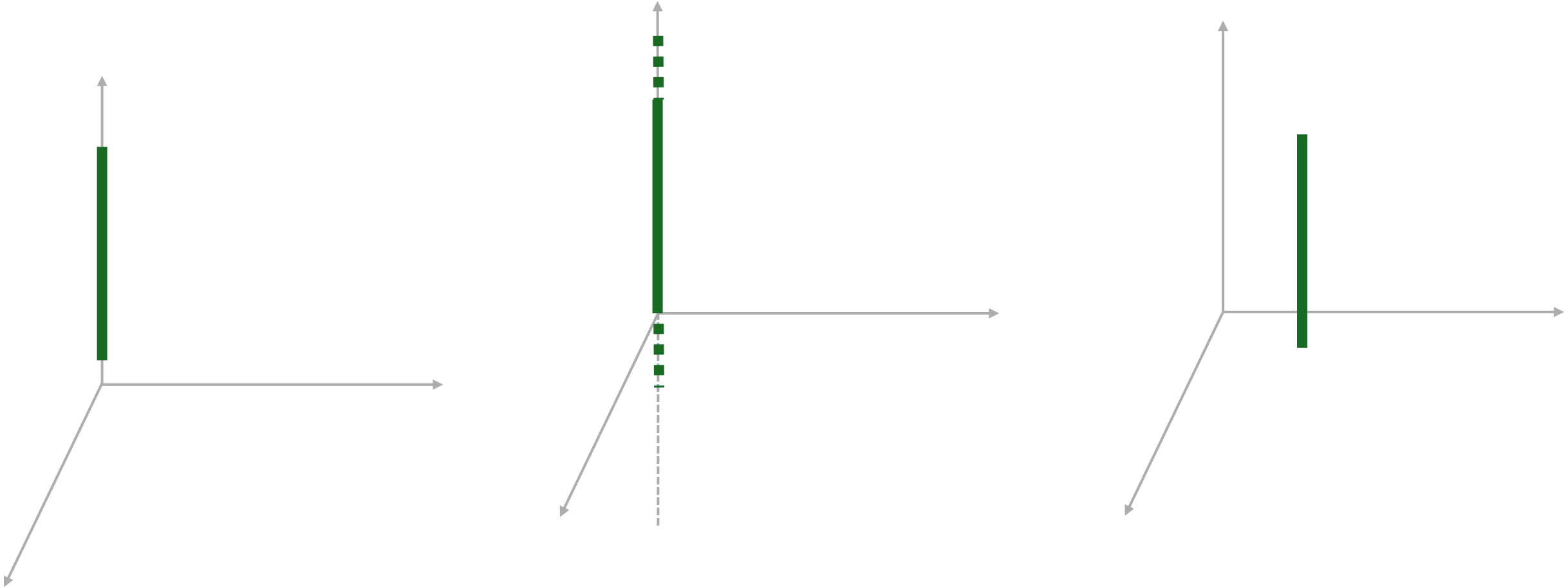
Volume
charge

$$dQ = \rho_v dv \rightarrow Q = \int_v \rho_v dv$$

$$\mathbf{E} = \int_v \frac{\rho_v dv}{4\pi\epsilon_0 R^2} \mathbf{a}_R$$

Electrostatics

- A line charge



Electrostatics

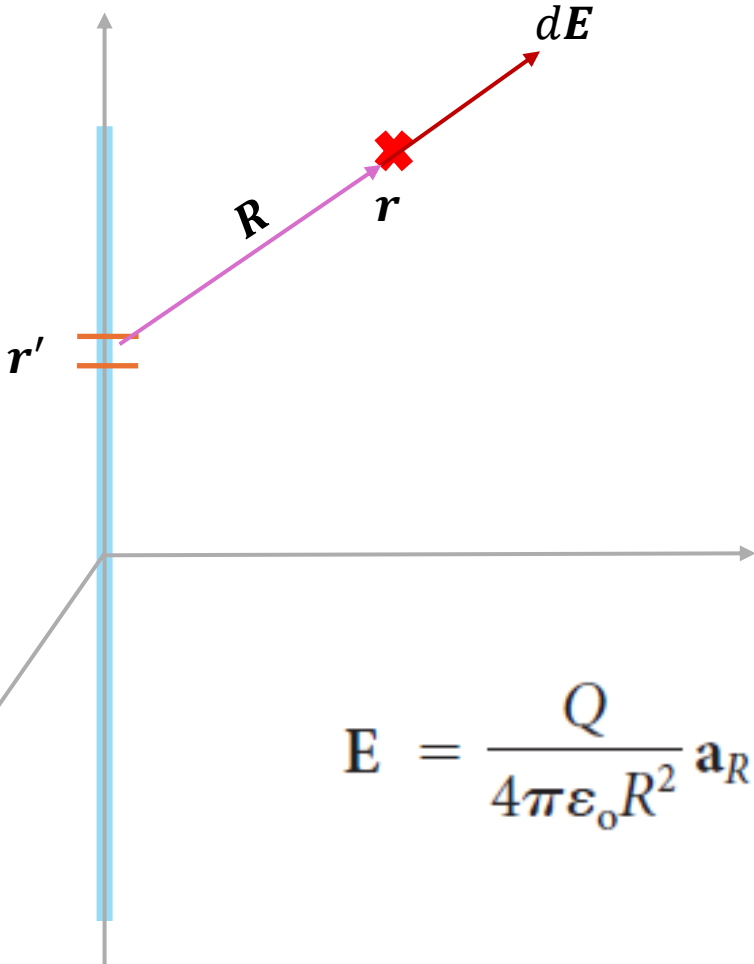
- A line charge, symmetric about xy-plane

$$\mathbf{r} = x\vec{a}_x + y\vec{a}_y + z\vec{a}_z$$

$$\mathbf{r}' = 0\vec{a}_x + 0\vec{a}_y + z'\vec{a}_z$$

$$\mathbf{R} = \mathbf{r} - \mathbf{r}' = x\vec{a}_x + y\vec{a}_y + (z - z')\vec{a}_z$$

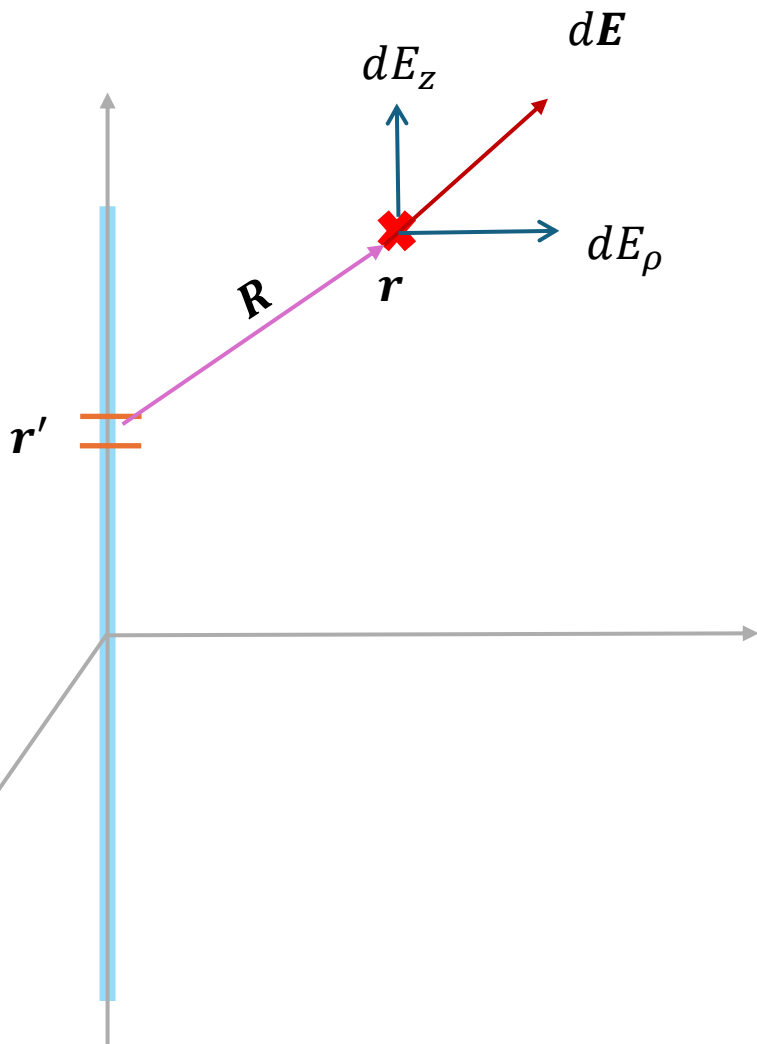
$$R = \sqrt{\rho^2 + z'^2}$$



$$\mathbf{E} = \frac{Q}{4\pi\epsilon_0 R^2} \mathbf{a}_R \xrightarrow{\lambda = \rho_L} d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dz'}{(\rho^2 + (z - z')^2)^{3/2}} (\rho\vec{a}_\rho + (z - z')\vec{a}_z)$$

Electrostatics

- A line charge, symmetric about xy-plane

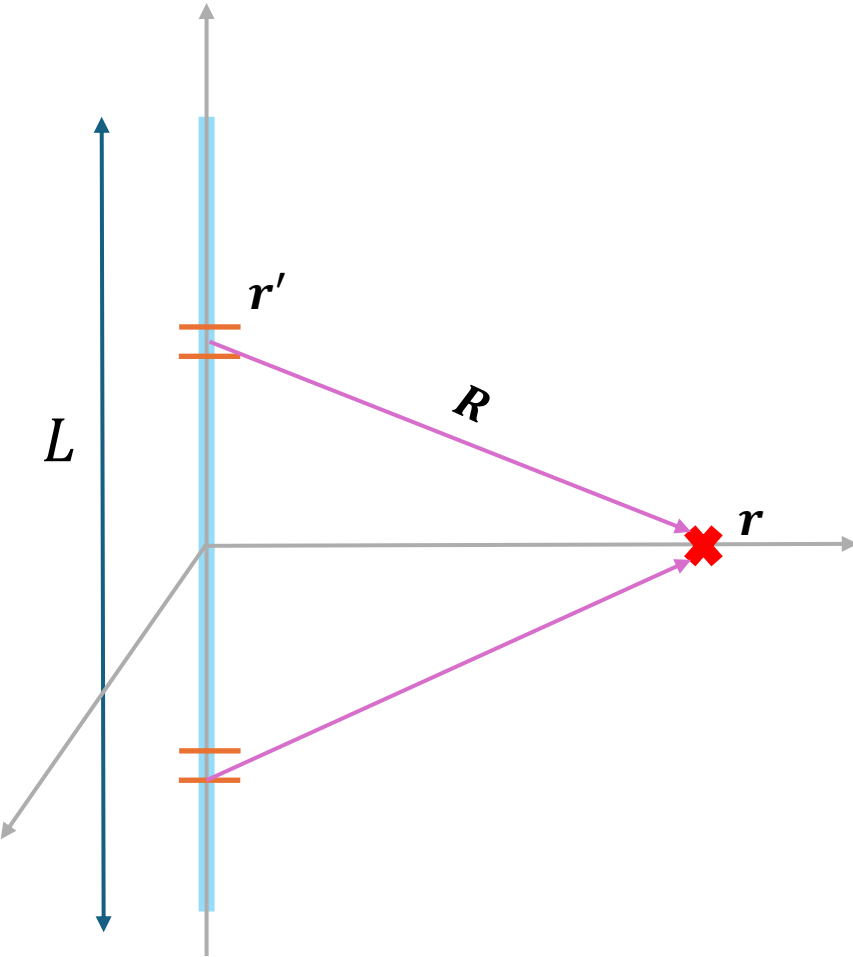


$$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{\lambda dz'}{(\rho^2 + (z - z')^2)^{3/2}} (\rho \vec{a}_\rho + (z - z') \vec{a}_z)$$

$$\begin{cases} dE_\rho = \frac{\lambda \rho}{4\pi\epsilon_0} \frac{dz'}{(\rho^2 + (z - z')^2)^{3/2}} \\ dE_z = \frac{\lambda}{4\pi\epsilon_0} \frac{(z - z') dz'}{(\rho^2 + (z - z')^2)^{3/2}} \end{cases}$$

Electrostatics

- Observation on xy-plane, symmetric finite line (limited)

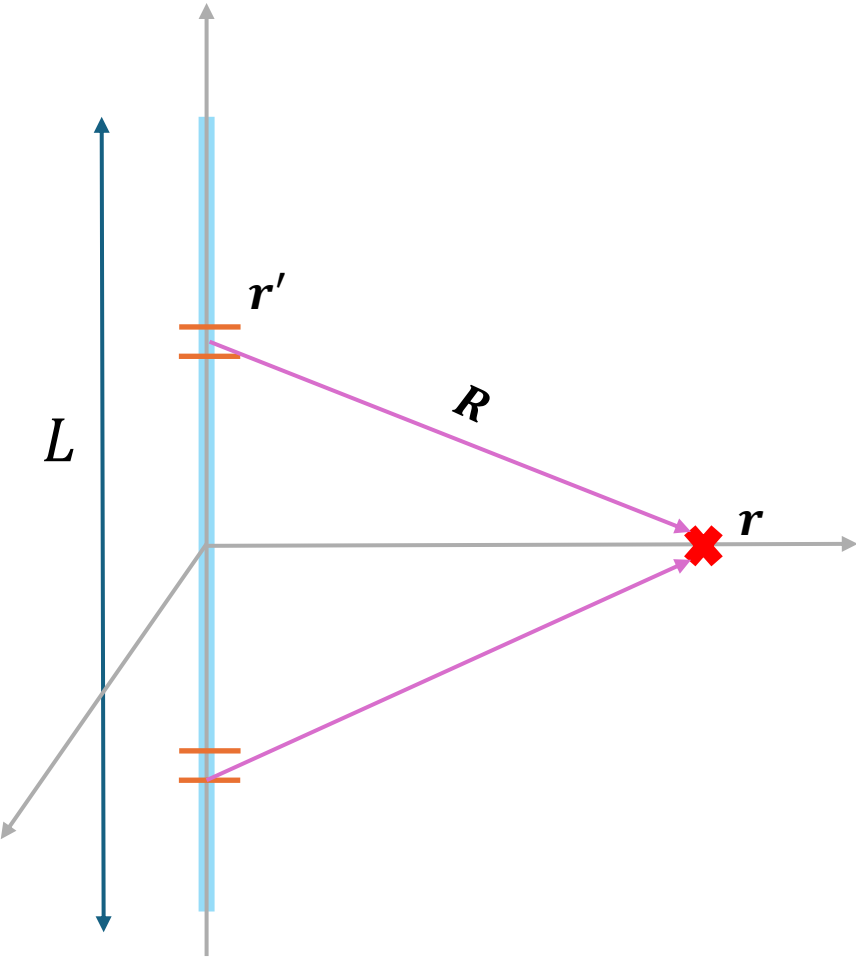


$$\begin{cases} dE_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \frac{dz'}{(\rho^2 + (z - z')^2)^{3/2}} \\ dE_z = \frac{\lambda}{4\pi\epsilon_0} \frac{(z - z')dz'}{(\rho^2 + (z - z')^2)^{3/2}} \end{cases}$$

$$dE_{\rho} = \frac{1}{4\pi\epsilon_0} \frac{\lambda \rho dz'}{(\rho^2 + z'^2)^{\frac{3}{2}}}$$

Electrostatics

- Observation on xy-plane, symmetric finite line (limited)



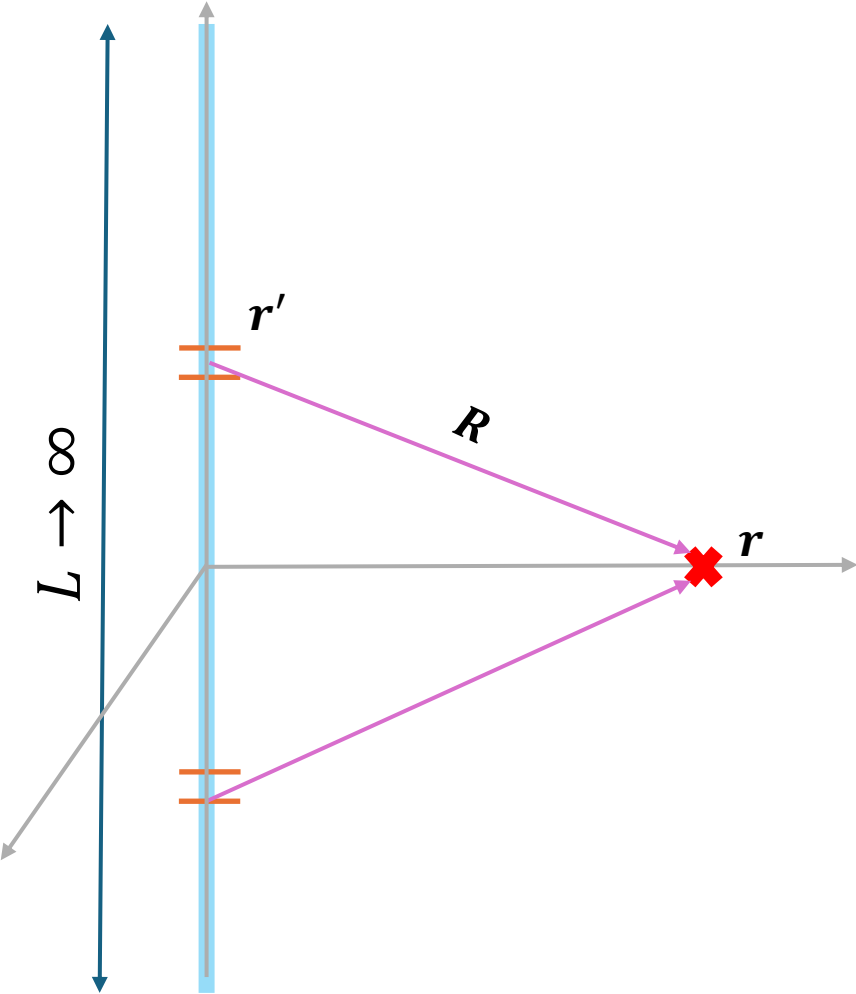
$$dE_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \int_{-L/2}^{L/2} \frac{dz'}{(\rho^2 + z'^2)^{3/2}}$$

$$\int_{-L/2}^{L/2} \frac{dz'}{(\rho^2 + z'^2)^{3/2}} = \frac{z'}{\rho^2 \sqrt{\rho^2 + z'^2}}$$

$$E_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \left[\frac{z'}{\rho^2 \sqrt{\rho^2 + z'^2}} \right]_{-L/2}^{L/2} = \frac{\lambda}{4\pi\epsilon_0 \rho} \frac{L}{\sqrt{L^2 + 4\rho^2}}$$

Electrostatics

- Observation on xy-plane, symmetric infinite line

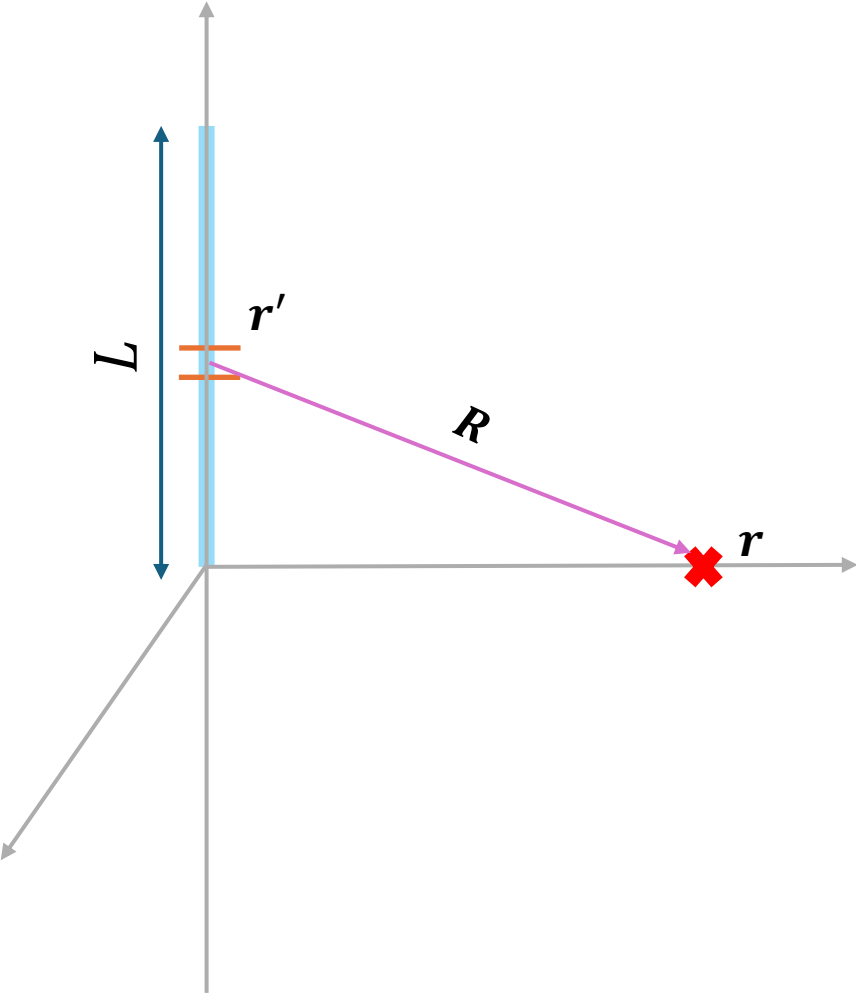


$$E_{\rho} = \frac{\lambda}{4\pi\epsilon_0\rho} \frac{L}{\sqrt{L^2 + 4\rho^2}}$$

$$E_{\rho} = \frac{\lambda}{4\pi\epsilon_0\rho} \lim_{L \rightarrow \infty} \frac{L}{\sqrt{L^2 + 4\rho^2}} = \frac{\lambda}{2\pi\epsilon_0\rho}$$

Electrostatics

- Observation on xy-plane, asymmetric finite line

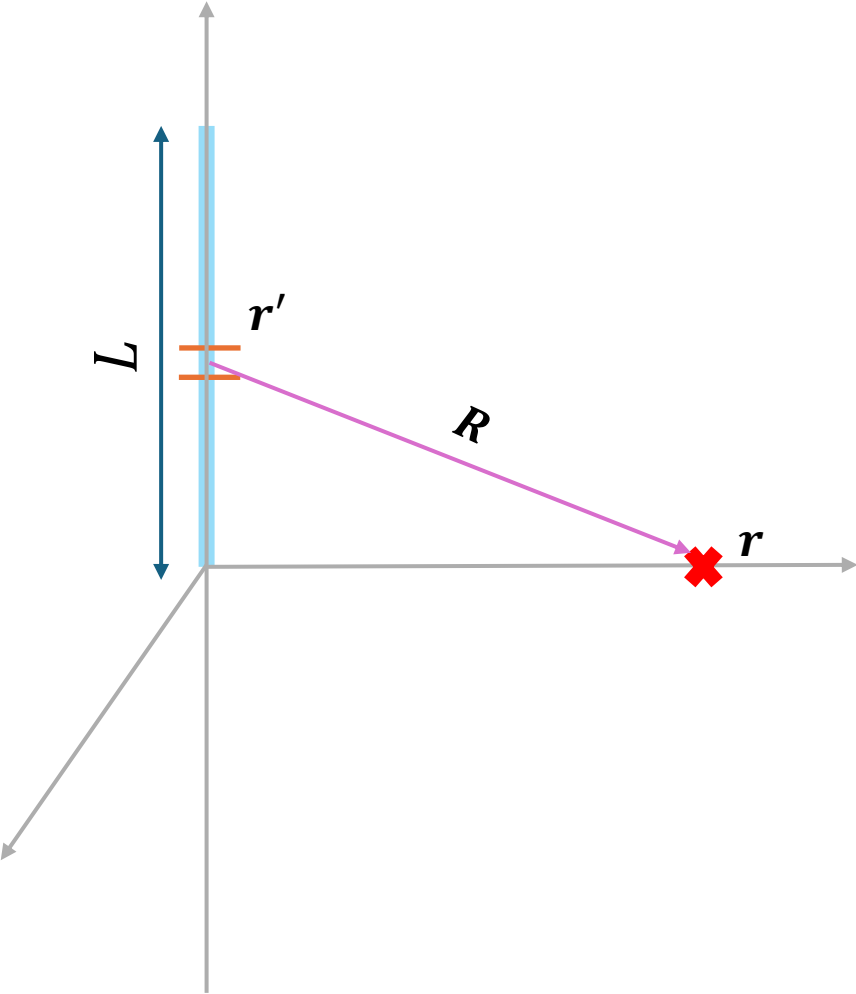


$$dE_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \frac{dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

$$dE_z = \frac{\lambda}{4\pi\epsilon_0} \frac{-z' dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

Electrostatics

- Observation on xy-plane, asymmetric finite line

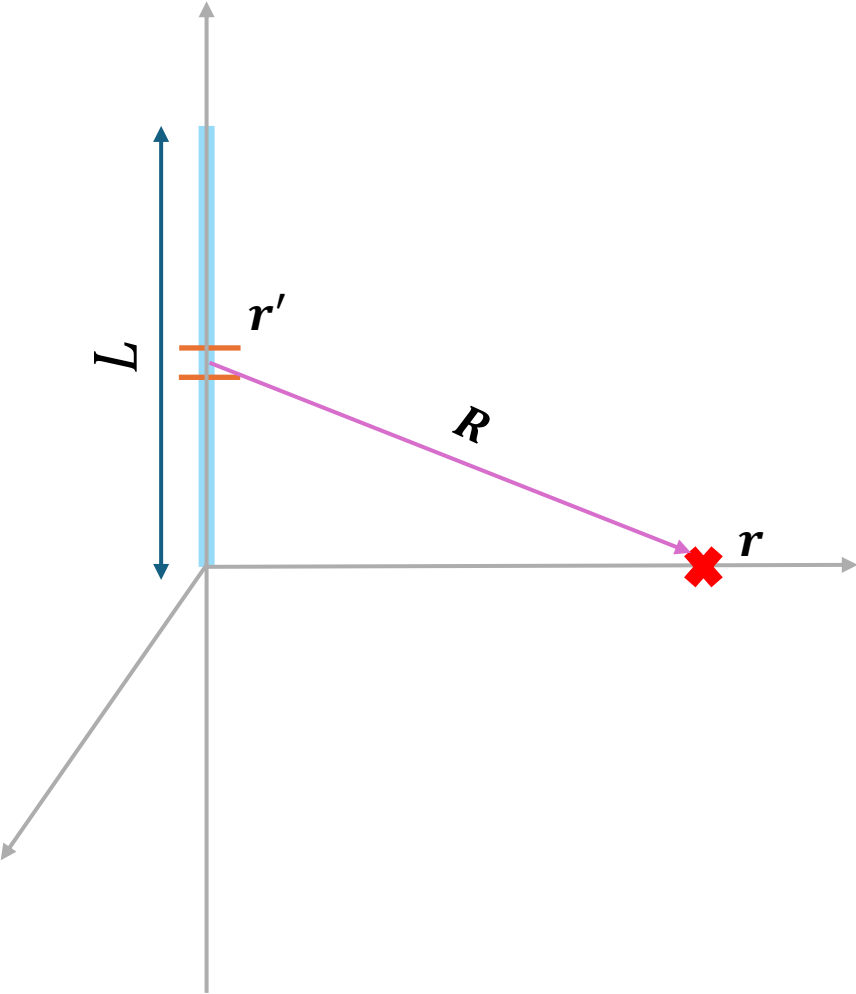


$$E_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \int_0^L \frac{dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{-z' dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

Electrostatics

- Observation on xy-plane, asymmetric finite line

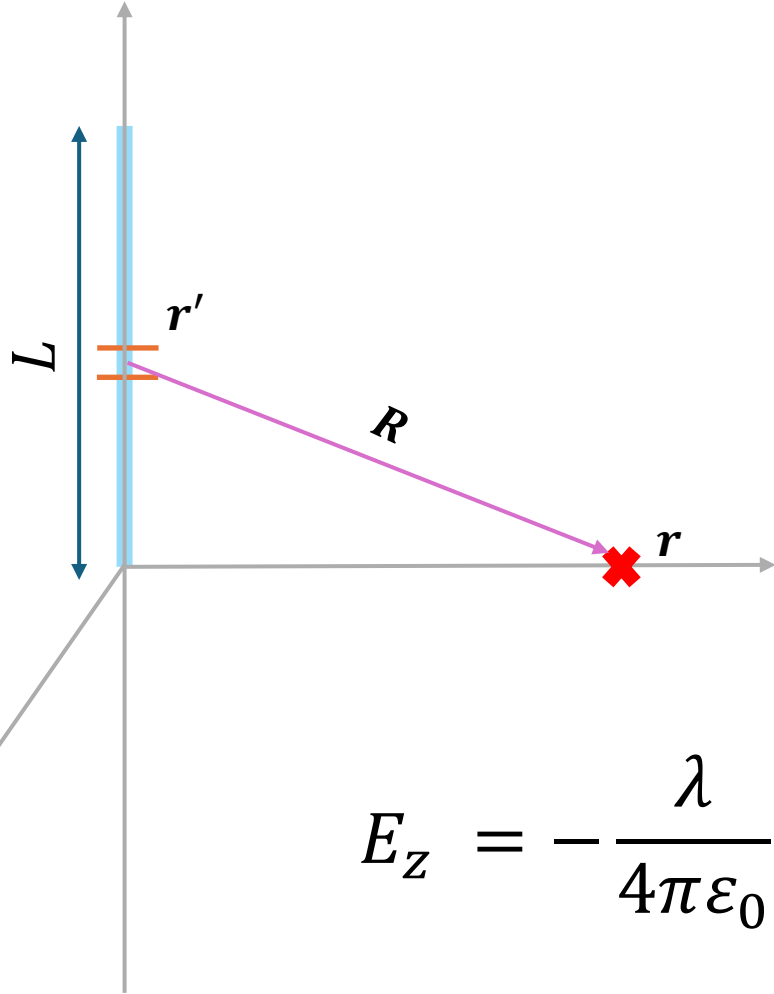


$$E_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \int_0^L \frac{dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

$$E_{\rho} = \frac{\lambda \rho}{4\pi\epsilon_0} \left[\frac{z'}{\rho^2 \sqrt{\rho^2 + z'^2}} \right]_0^L = \frac{\lambda}{4\pi\epsilon_0 \rho} \frac{L}{\sqrt{\rho^2 + L^2}}$$

Electrostatics

- Observation on xy-plane, asymmetric finite line



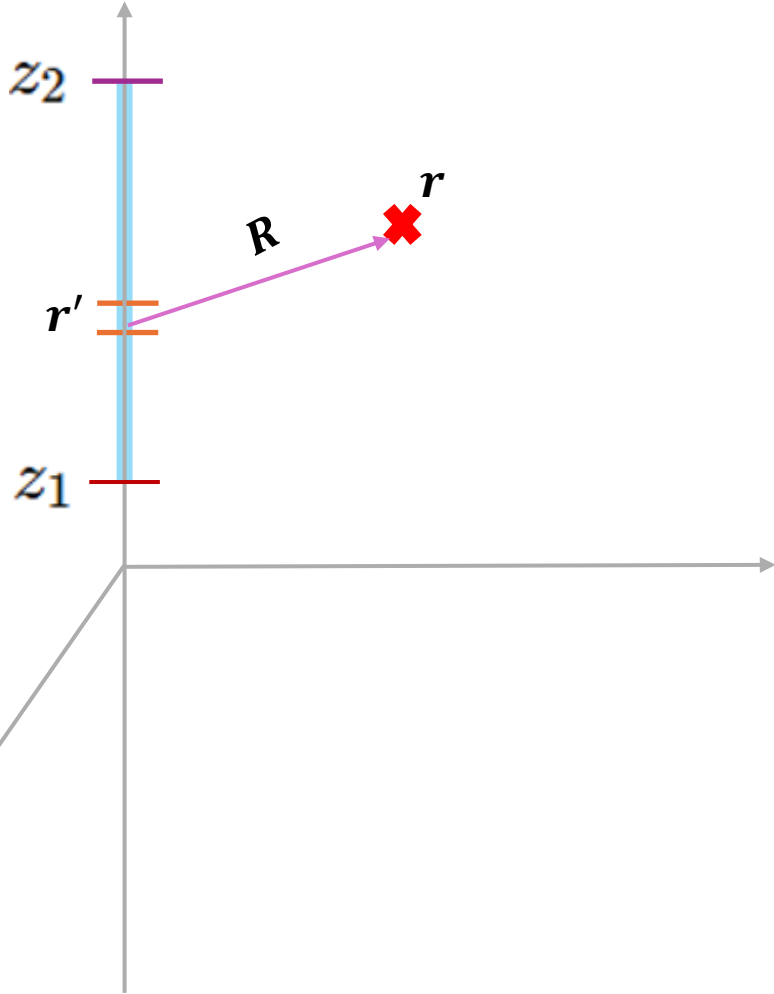
$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int_0^L \frac{-z' dz'}{(\rho^2 + (z - z')^2)^{3/2}}$$

$$\int \frac{z' dz'}{(\rho^2 + (z - z')^2)^{3/2}} = \frac{-1}{\sqrt{\rho^2 + z'^2}}$$

$$E_z = -\frac{\lambda}{4\pi\epsilon_0} \left[\frac{-1}{\sqrt{\rho^2 + z'^2}} \right]_0^L = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + L^2}} - \frac{1}{\rho} \right)$$

Electrostatics

- Random observation point (general 3D), finite line



$$E_\rho = \frac{\lambda}{4\pi\epsilon_0} \int_{z_1}^{z_2} \frac{\rho}{(\rho^2 + (z - z')^2)^{3/2}} dz'$$

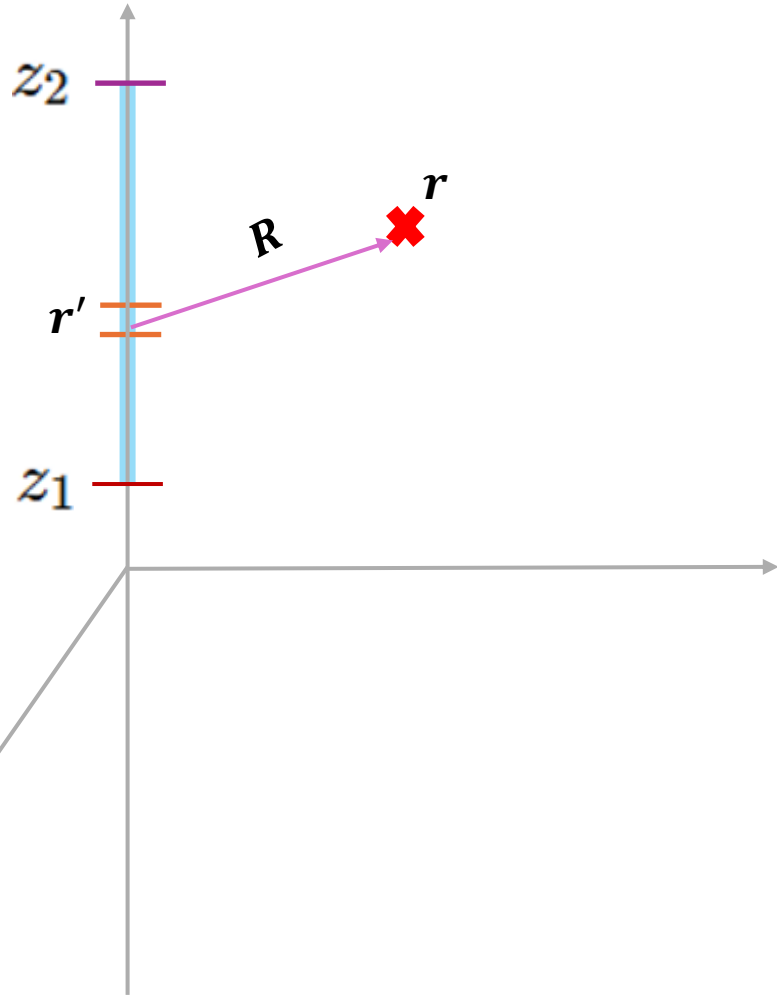
$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int_{z_1}^{z_2} \frac{(z - z')}{(\rho^2 + (z - z')^2)^{3/2}} dz'$$

$$u = z - z' \Rightarrow dz' = -du.$$

$$u_1 = z - z_1, \quad u_2 = z - z_2$$

Electrostatics

- Random observation point (general 3D), finite line



$$E_\rho = \frac{\lambda}{4\pi\epsilon_0} \rho \int_{u_2}^{u_1} \frac{du}{(\rho^2 + u^2)^{3/2}}$$

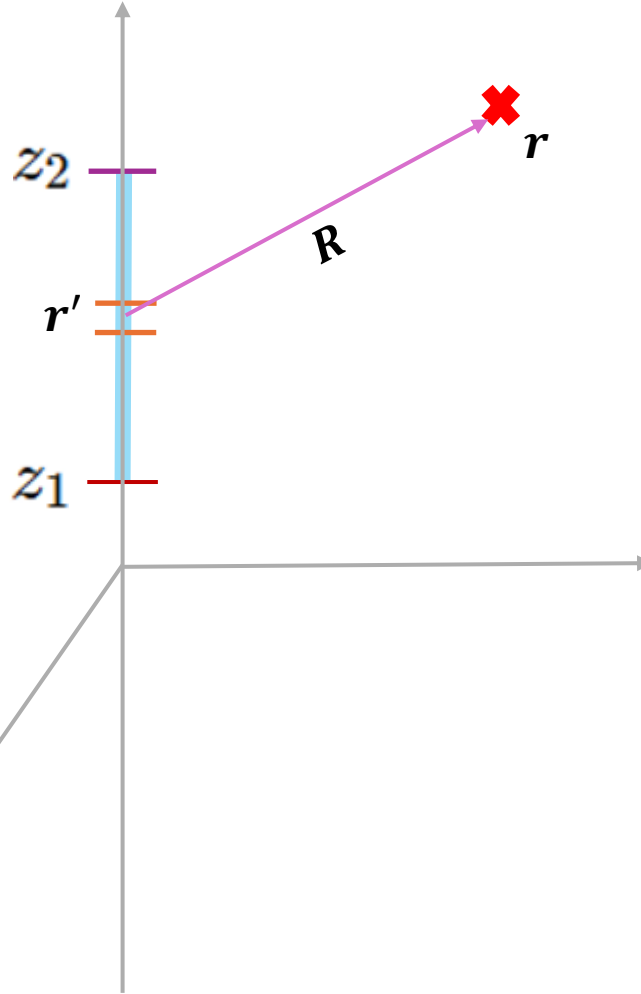
$$\int \frac{du}{(\rho^2 + u^2)^{3/2}} = \frac{u}{\rho^2 \sqrt{\rho^2 + u^2}}$$

$$E_\rho = \frac{\lambda}{4\pi\epsilon_0 \rho} \left(\frac{u_1}{\sqrt{\rho^2 + u_1^2}} - \frac{u_2}{\sqrt{\rho^2 + u_2^2}} \right)$$

$$E_\rho = \frac{\lambda}{4\pi\epsilon_0 \rho} \left(\frac{z - z_1}{\sqrt{\rho^2 + (z - z_1)^2}} - \frac{z - z_2}{\sqrt{\rho^2 + (z - z_2)^2}} \right)$$

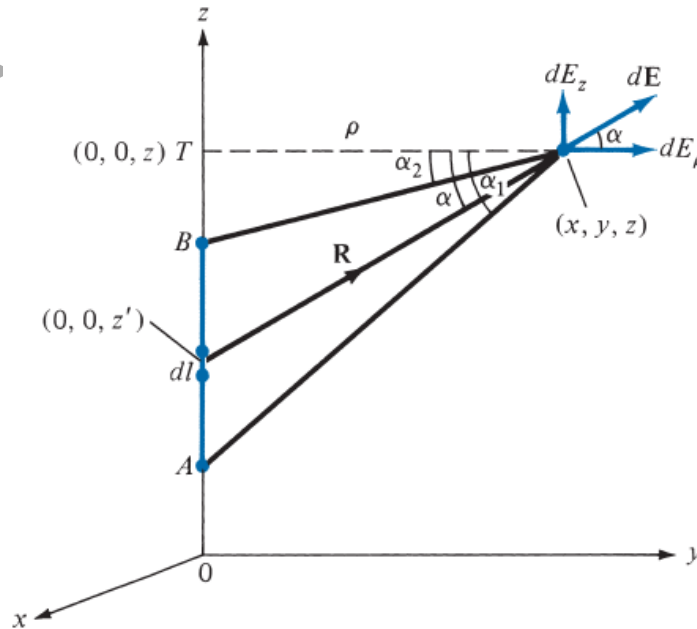
Electrostatics

- Random observation point (general 3D), finite line



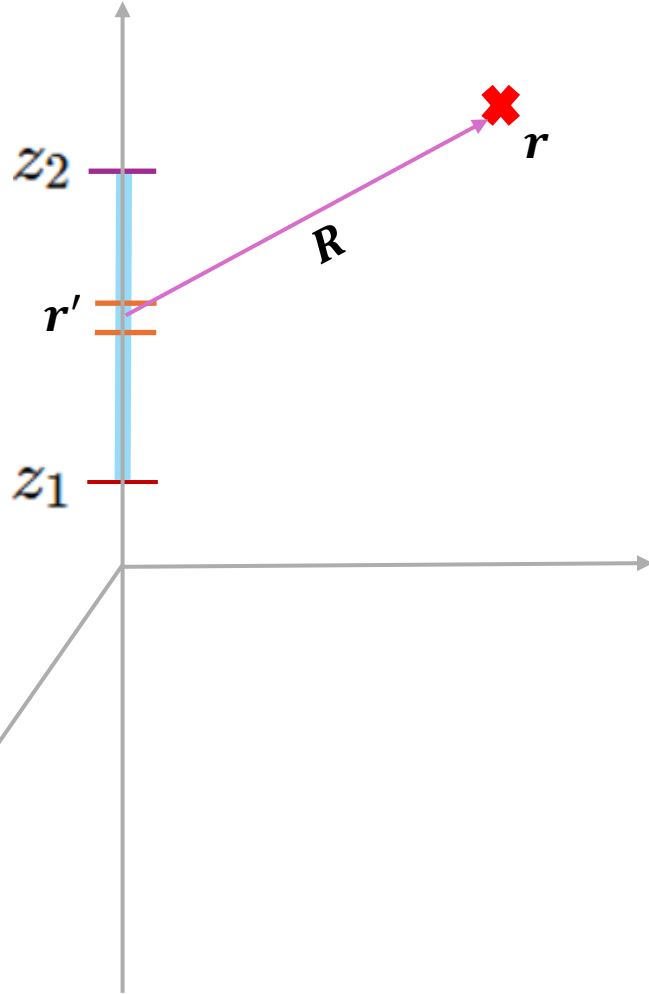
$$E_\rho = \frac{\lambda}{4\pi\epsilon_0\rho} \left(\frac{z - z_1}{\sqrt{\rho^2 + (z - z_1)^2}} - \frac{z - z_2}{\sqrt{\rho^2 + (z - z_2)^2}} \right)$$

$$\mathbf{E} = \frac{\rho_L}{4\pi\epsilon_0\rho} [-(\sin\alpha_2 - \sin\alpha_1)\mathbf{a}_\rho + (\cos\alpha_2 - \cos\alpha_1)\mathbf{a}_z]$$



Electrostatics

- Random observation point (general 3D), finite line



$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int_{z_1}^{z_2} \frac{(z - z')}{(\rho^2 + (z - z')^2)^{3/2}} dz'$$

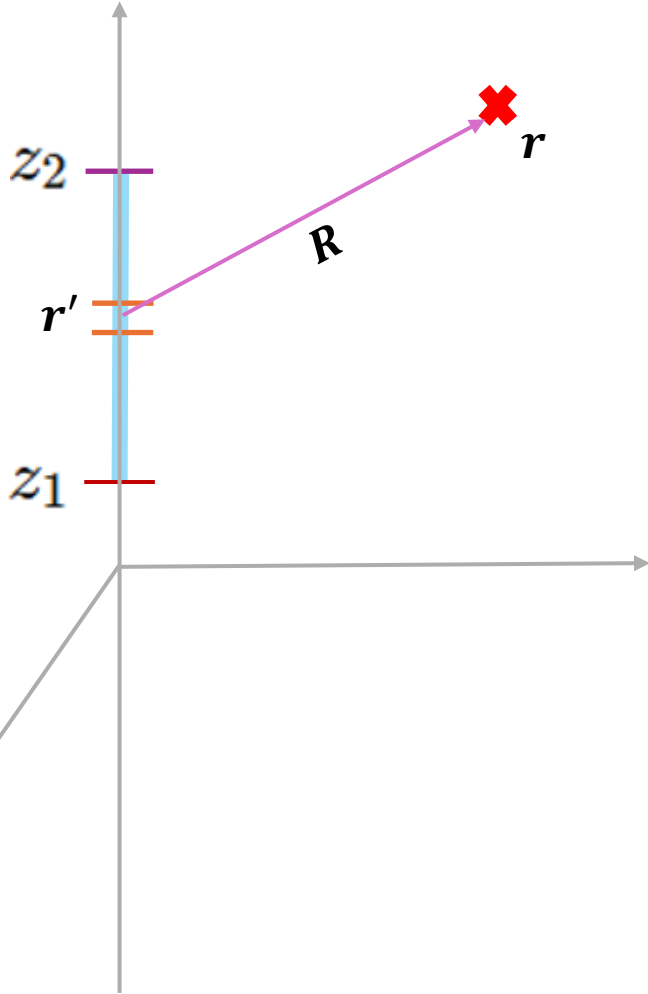
$$E_z = \frac{\lambda}{4\pi\epsilon_0} \int_{u_2}^{u_1} \frac{u}{(\rho^2 + u^2)^{3/2}} du$$

$$\int \frac{u du}{(\rho^2 + u^2)^{3/2}} = -\frac{1}{\sqrt{\rho^2 + u^2}}$$

$$E_z = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + u_2^2}} - \frac{1}{\sqrt{\rho^2 + u_1^2}} \right)$$

Electrostatics

- Random observation point (general 3D), finite line

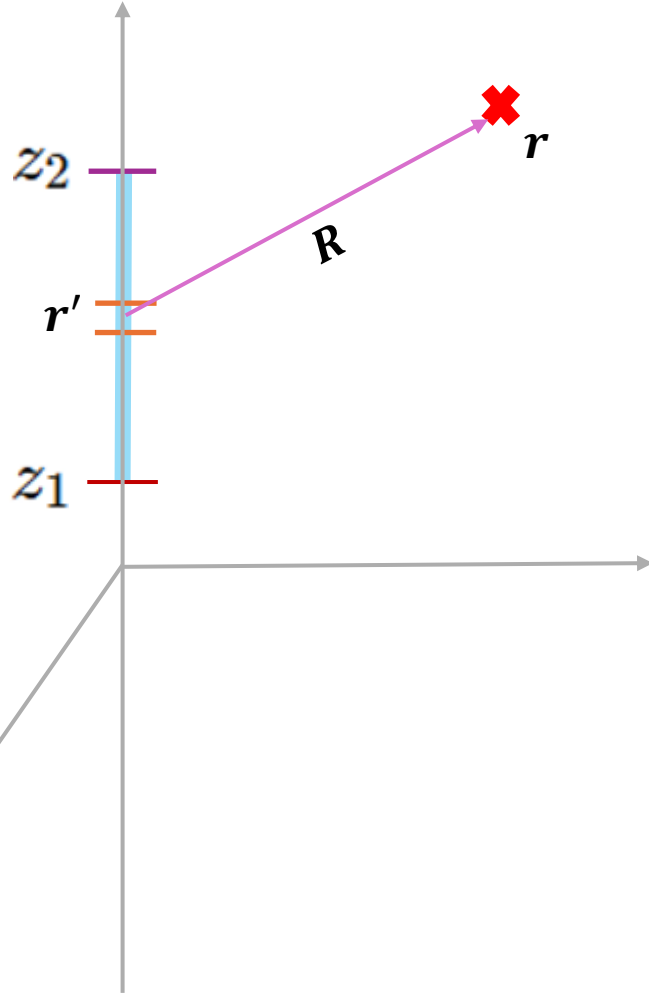


$$E_z = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + u_2^2}} - \frac{1}{\sqrt{\rho^2 + u_1^2}} \right)$$

$$E_z = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + (z - z_2)^2}} - \frac{1}{\sqrt{\rho^2 + (z - z_1)^2}} \right)$$

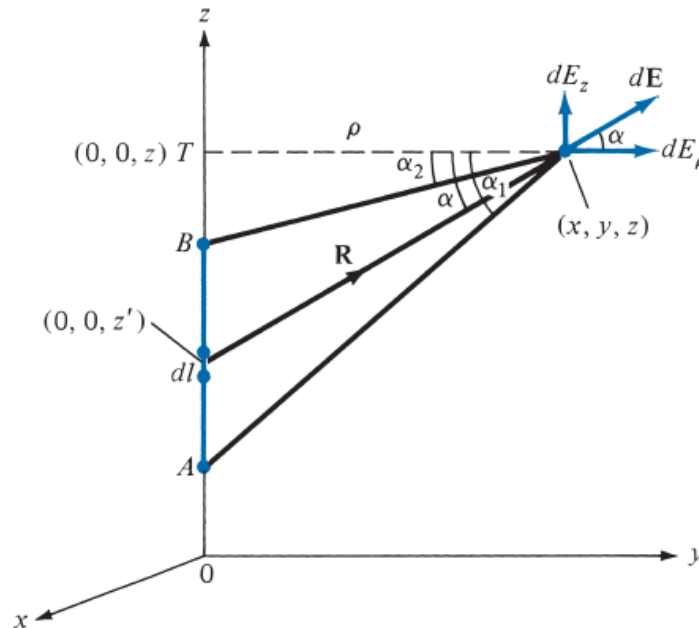
Electrostatics

- Random observation point (general 3D), finite line



$$E_z = \frac{\lambda}{4\pi\epsilon_0} \left(\frac{1}{\sqrt{\rho^2 + (z - z_2)^2}} - \frac{1}{\sqrt{\rho^2 + (z - z_1)^2}} \right)$$

$$\mathbf{E} = \frac{\rho L}{4\pi\epsilon_0\rho} [-(\sin \alpha_2 - \sin \alpha_1)\mathbf{a}_\rho + (\cos \alpha_2 - \cos \alpha_1)\mathbf{a}_z]$$



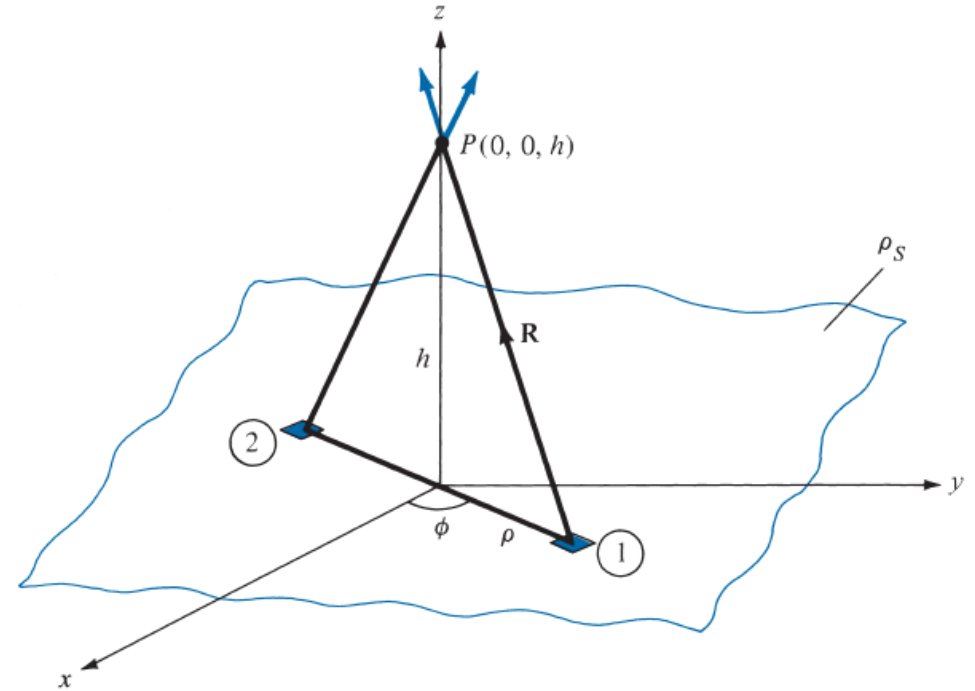
Electrostatics

- Infinite sheet of charge in the xy -plane

$$d\mathbf{E} = \frac{1}{4\pi\epsilon_0} \frac{dQ}{R^2} \mathbf{a}_R$$

$$dQ = \sigma dS$$

$$dS = \rho d\rho d\phi$$



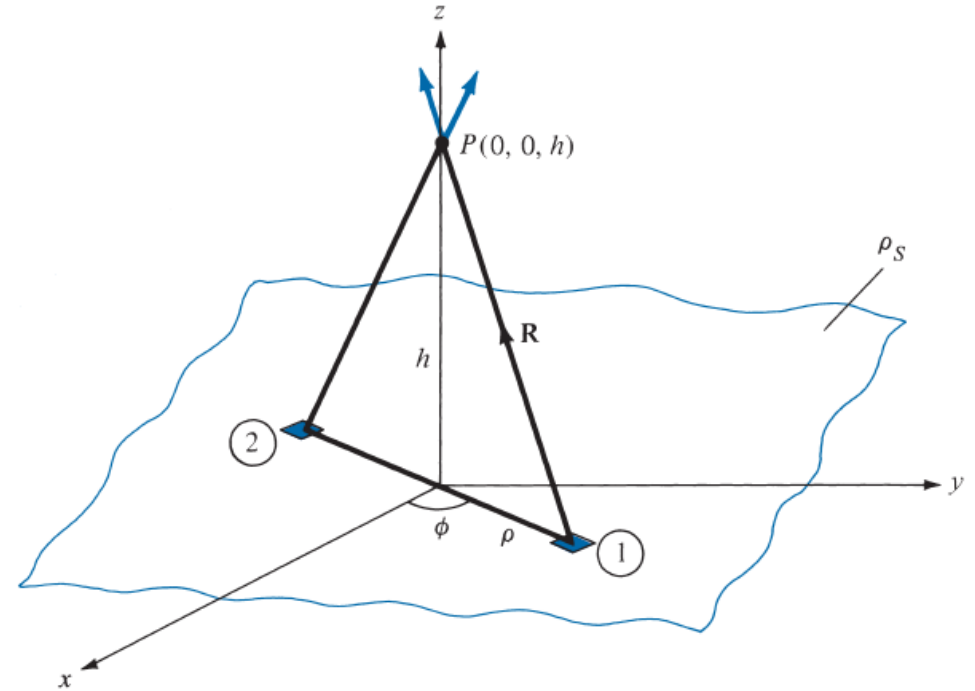
Electrostatics

- Infinite sheet of charge in the xy -plane

$$\mathbf{r}' = \rho \mathbf{a}_\rho$$

$$\mathbf{r} = h \mathbf{a}_z$$

$$\mathbf{R} = \mathbf{r} - \mathbf{r}' = -\rho \mathbf{a}_\rho + h \mathbf{a}_z \longrightarrow R = |\mathbf{R}| = \sqrt{\rho^2 + h^2}$$

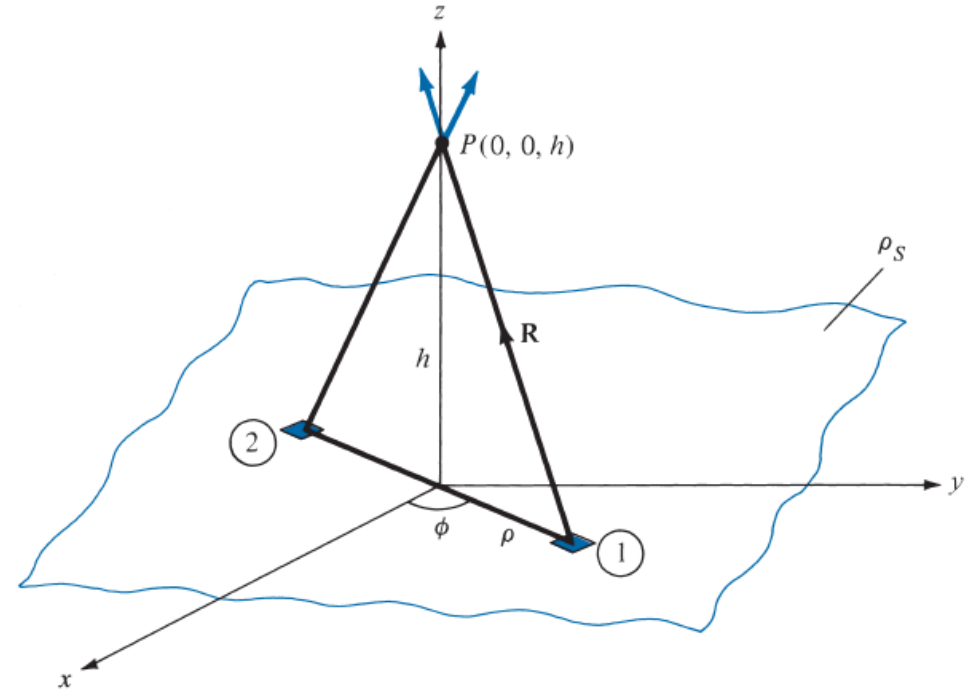


Electrostatics

- Infinite sheet of charge in the xy -plane

$$d\mathbf{E} = \frac{\sigma \rho d\rho d\phi}{4\pi\epsilon_0(\rho^2 + h^2)^{3/2}} (-\rho \mathbf{a}_\rho + h \mathbf{a}_z)$$

$$\mathbf{E} = \mathbf{a}_z \frac{\sigma}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\infty \frac{h \rho}{(\rho^2 + h^2)^{3/2}} d\rho d\phi$$

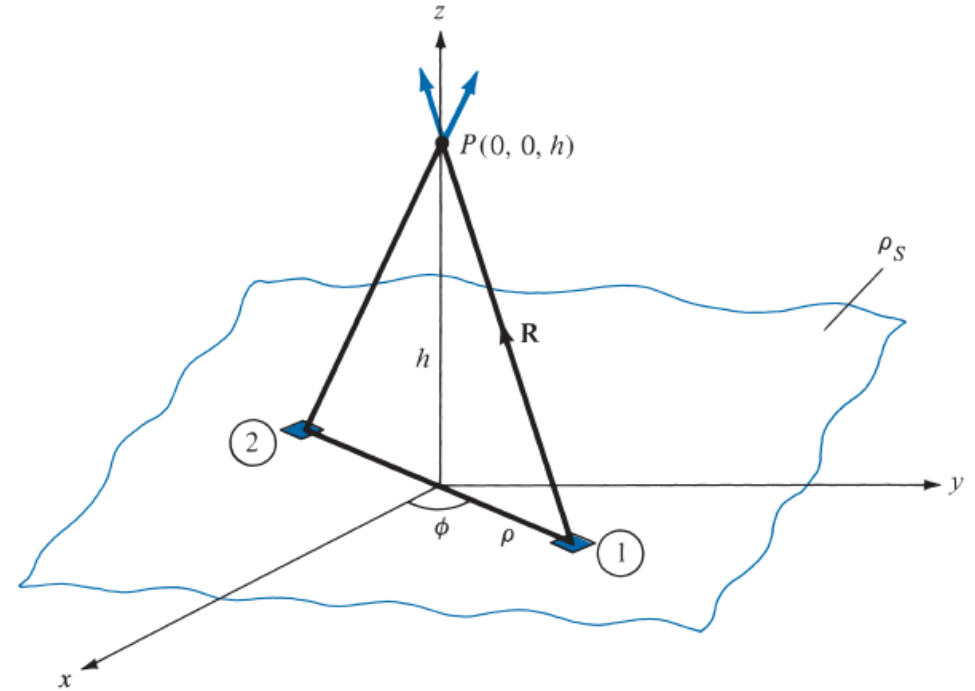


Electrostatics

- Infinite sheet of charge in the xy -plane

$$d\mathbf{E} = \frac{\sigma \rho d\rho d\phi}{4\pi\epsilon_0(\rho^2 + h^2)^{3/2}} (-\rho \mathbf{a}_\rho + h \mathbf{a}_z)$$

$$\mathbf{E} = \mathbf{a}_z \frac{\sigma}{4\pi\epsilon_0} \int_0^{2\pi} \int_0^\infty \frac{h \rho}{(\rho^2 + h^2)^{3/2}} d\rho d\phi$$



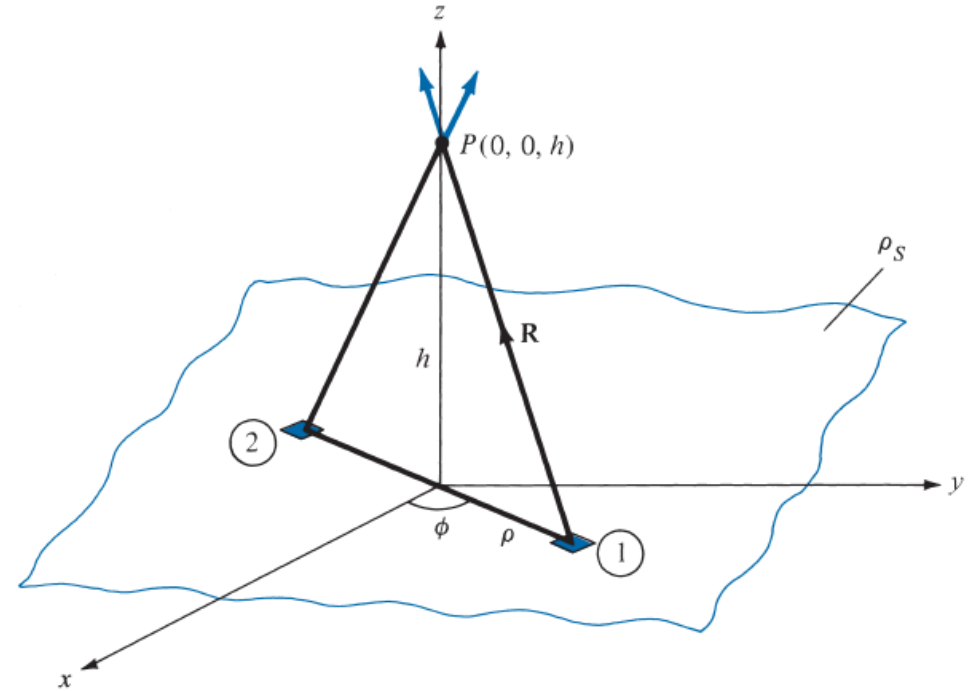
$$\int_0^\infty \frac{h \rho}{(\rho^2 + h^2)^{3/2}} d\rho = \frac{h}{2} \int_{h^2}^\infty u^{-3/2} du = \frac{h}{2} \left[-2u^{-1/2} \right]_{h^2}^\infty = 1$$

Electrostatics

- Infinite sheet of charge in the xy -plane

$$h > 0 \longrightarrow \mathbf{E} = \frac{\sigma}{2\epsilon_0} \mathbf{a}_z$$

$$h < 0 \longrightarrow \mathbf{E} = -\frac{\sigma}{2\epsilon_0} \mathbf{a}_z$$



Coulomb's Law

- * Q1. vector algebra $\rightarrow$ $\times 3$ product, projection
- Q2. i) divergence $\rightarrow$ similar today's lecture (next slide)
ii) curl $\rightarrow$ point charge
geometric shape, & field/ $\vec{F}$ at that point
- Q3.

Problem 1

Let

$$\mathbf{A} = xy \mathbf{a}_x + y^2 \mathbf{a}_y - xz \mathbf{a}_z$$

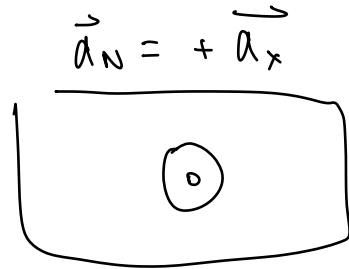
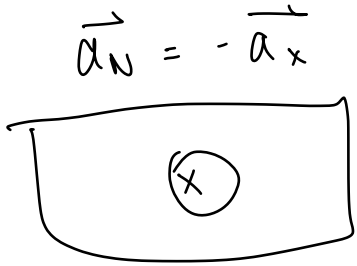
and consider the rectangular volume:

$$0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c$$

a) Calculate the total outward flux $\iint_S \mathbf{A} \cdot d\mathbf{S}$ through the entire closed surface S .

b) Verify your answer using the divergence theorem:

$$\iint_S \mathbf{A} \cdot d\mathbf{S} = \iiint_V (\nabla \cdot \mathbf{A}) dV$$



$$\phi_T = \phi_{x=0} + \phi_{x=a} + \dots + \phi_{z=c}$$

$$\begin{aligned}
 A \cdot d\vec{s} &= (A_x \vec{a}_x) \cdot (dy dz \vec{a}_n) \\
 &= \begin{cases} +\vec{a}_x \cdot \vec{a}_x dy dz \\ -\vec{a}_x \cdot \vec{a}_x dy dz \end{cases}
 \end{aligned}$$

$$d\phi_{x=0} = -xy dy dz$$

$$d\phi_{x=a} = +xy dy dz$$

$$\begin{aligned}
 \phi_{x=a} &= \int_0^b \int_0^c xy dy dz \\
 &= a \int_0^b \int_0^c y dz dy \\
 &= a \int_0^b y(z)|_0^c dy \\
 &= ac \int_0^b y dy \\
 &= ac \left[\frac{1}{2} y^2 \right]_0^b \\
 &= \frac{1}{2} ab^2 c
 \end{aligned}$$

$$\begin{aligned}
 \phi_{y=b} &= \int_0^a \int_0^c y^2 dx dz \\
 &= b^2 \int_0^a \int_0^c dx dz \\
 &= ab^2 c
 \end{aligned}$$

$$\begin{aligned}
 \phi_{y=0} &= 0 \\
 &\quad \uparrow \text{coeff} = 0
 \end{aligned}$$

$$\begin{aligned}
 \phi_{z=c} &= \int_0^a \int_0^b -xz dx dy \\
 &= -\frac{1}{2} a^2 bc
 \end{aligned}$$

$$\phi_T = \frac{1}{2}(abc)(3b-a)$$

$$\begin{aligned}
 \phi_{x=0} &= 0 \\
 &\quad \uparrow \text{coefficient} = 0
 \end{aligned}$$

$$\nabla \cdot \vec{A} = \left(\frac{\partial}{\partial x} \vec{a}_x + \frac{\partial}{\partial y} \vec{a}_y + \frac{\partial}{\partial z} \vec{a}_z \right) \cdot (xy \vec{a}_x + y^2 \vec{a}_y - xz \vec{a}_z)$$

$$= \frac{\partial}{\partial x} xy + \frac{\partial}{\partial y} y^2 - \frac{\partial}{\partial z} xz$$

$$= y + 2y - x$$

$$= 3y - x$$

$$\int_V \nabla \cdot \vec{A} \, dV = \oint_S \vec{A} \cdot d\vec{s}, \quad dV = dx \, dy \, dz$$

$$= \int_V (3y - x) \, dV = \int_0^a \int_0^b \int_0^c (3y - x) \, dx \, dy \, dz$$

$$\int_0^a (3y - x) \, dx = 3yx - \frac{x^2}{2} \Big|_0^a = 3ay - \frac{a^2}{2}$$

$$\int_0^b (3y - x) \, dy = \frac{3}{2}y^2a - \frac{a^2}{2}y \Big|_0^b = \frac{3}{2}b^2a - \frac{a^2b}{2}$$

$$\int_0^c (3y - x) \, dz = -\frac{3}{2}cb^2a - \frac{1}{2}ca^2b = \frac{1}{2}abc(3b - a)$$

Problem 2

Let

$$\mathbf{A} = x^2y \mathbf{a}_x + x \mathbf{a}_y + 2yz \mathbf{a}_z.$$

Consider the rectangular volume

$$0 \leq x \leq a, 0 \leq y \leq b, 0 \leq z \leq c.$$

- a) Calculate the divergence $\nabla \cdot \mathbf{A}$.
- b) Evaluate the total outward flux of $\mathbf{A}$ through the entire closed surface of the volume.
- c) Verify your result in part (b) using the divergence theorem.

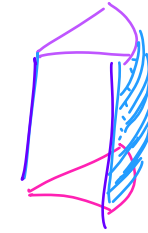
*Problem 3

Let

$$\mathbf{B} = 3\rho \sin \phi \mathbf{a}_\rho - 5\rho^2 z \mathbf{a}_\phi + 8z \cos^2 \phi \mathbf{a}_z.$$

Consider the cylindrical sector volume

$$\underline{0 \leq \rho \leq a}, \underline{0 \leq \phi \leq \frac{\pi}{2}}, \underline{0 \leq z \leq h}.$$



a) Calculate the divergence $\nabla \cdot \mathbf{B}$.

b) Evaluate the total outward flux of $\mathbf{B}$ through the entire closed surface of the volume.

c) Verify your result in part (b) using the divergence theorem.

$$\begin{aligned} a) \quad \nabla \cdot \vec{B} &= \frac{1}{\rho} \frac{\partial \rho B_\rho}{\partial \rho} \vec{a}_\rho + \frac{1}{\rho} \frac{\partial \rho B_\phi}{\partial \phi} \vec{a}_\phi + \frac{\partial B_z}{\partial z} \vec{a}_z \\ &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho B_\rho) = \cancel{\frac{\rho}{\rho}} \frac{\partial B_\rho}{\partial \rho} + \frac{B_\rho}{\rho} \cancel{\frac{\partial \rho}{\partial \rho}} \\ &= \frac{\partial (3 \rho \sin \phi)}{\partial \rho} + 3 \sin \phi \\ &= 3 \sin \phi + 3 \sin \phi \\ &= 6 \sin \phi \quad (1) \end{aligned}$$

$$\frac{1}{\rho} \frac{\partial B_\phi}{\partial \phi} = 0 \quad (2)$$

$$\frac{\partial}{\partial z} (8z \cos^2 \phi) = 8 \cos^2 \phi \quad (3)$$

$$\nabla \cdot \vec{B} = 6 \sin \phi + 8 \cos^2 \phi \quad (1) (2) (3)$$

$$\phi_T = \oint_{\rho=a} + \oint_{z=0} + \oint_{z=h} + \cancel{\oint_{\phi=\frac{\pi}{2}}} + \cancel{\oint_{\phi=0}}$$

← 5 surfaces

direction cancels each other out
 ↳ cw & ccw

Problem 4

Let

$$\mathbf{B} = \rho z^2 \mathbf{a}_\rho + \rho \sin^2 \phi \mathbf{a}_\phi + 2\rho z \sin^2 \phi \mathbf{a}_z$$

and consider the cylindrical sector volume:

$$1 \leq \rho \leq 2, 0 \leq \phi \leq \frac{\pi}{2}, 0 \leq z \leq h$$

- a) Calculate the total outward flux $\iint_S \mathbf{B} \cdot d\mathbf{S}$ through the entire closed surface S .
- b) Verify your answer using the divergence theorem.

Problem 5

Let

$$\mathbf{F} = \rho^2 \sin \phi \, \mathbf{a}_\rho + z \cos \phi \, \mathbf{a}_\phi + \rho z \, \mathbf{a}_z.$$

Consider the hollow cylindrical volume

$$2 \leq \rho \leq 3, 0 \leq z \leq 5, 0 \leq \phi \leq 2\pi.$$

- Calculate the total outward flux of $\mathbf{F}$ through the entire closed surface of the volume.
- Verify your answer in part (a) using the divergence theorem.

*Problem 6

Let

$$\mathbf{C} = r \mathbf{a}_r + r \cos^2 \theta \mathbf{a}_\phi$$

and consider the spherical sector volume:

$$0 \leq r \leq R, 0 \leq \theta \leq \frac{\pi}{2}, 0 \leq \phi \leq \frac{\pi}{2}$$

- a) Calculate the total outward flux $\iint_S \mathbf{C} \cdot d\mathbf{S}$ through the entire closed surface S .
- b) Verify your answer using the divergence theorem.

Problem 7

Let

$$\mathbf{C} = r^2 \cos \phi \, \mathbf{a}_r + 2r \, \mathbf{a}_\phi.$$

Consider the spherical sector volume

$$0 \leq r \leq R, 0 \leq \theta \leq \frac{\pi}{2}, 0 \leq \phi \leq \frac{\pi}{2}.$$

- a) Calculate the divergence $\nabla \cdot \mathbf{C}$.
- b) Evaluate the total outward flux of $\mathbf{C}$ through the entire closed surface of the volume.
- c) Verify your result in part (b) using the divergence theorem.

Problem 8

Let

$$\mathbf{A} = r^2 \mathbf{a}_r + r \sin \theta \cos \phi \mathbf{a}_\theta.$$

Consider the quarter-hemispherical volume defined by

$$0 \leq r \leq 3, 0 \leq \phi \leq \frac{\pi}{2}, 0 \leq \theta \leq \frac{\pi}{2}.$$

- a) Calculate the total outward flux of $\mathbf{A}$ through the entire closed surface of the volume.
- b) Verify your answer in part (a) using the divergence theorem.

Problem 9

Let

$$\mathbf{A} = xy \mathbf{a}_x + y^2 \mathbf{a}_y - xz \mathbf{a}_z.$$

Consider the open surface S in the plane $z = 0$ over the region

$$0 \leq x \leq a, 0 \leq y \leq b,$$

and let C be its boundary oriented counterclockwise when viewed from $+z$.

a) Calculate $\nabla \times \mathbf{A}$.

b) Evaluate the line integral $\oint_C \mathbf{A} \cdot d\boldsymbol{\ell}$.

c) Verify your answer in part (b) using Stokes' theorem:

$$\oint_C \mathbf{A} \cdot d\boldsymbol{\ell} = \iint_S (\nabla \times \mathbf{A}) \cdot d\mathbf{S}.$$

Problem 10

Let

$$\mathbf{B} = \rho z^2 \mathbf{a}_\rho + \rho \sin^2 \phi \mathbf{a}_\phi + 2\rho z \sin^2 \phi \mathbf{a}_z.$$

Consider the open surface S defined by

$$z = 0, 0 \leq \rho \leq a, 0 \leq \phi \leq \frac{\pi}{2},$$

and let C be its boundary oriented counterclockwise when viewed from $+z$.

a) Calculate $\nabla \times \mathbf{B}$.

b) Evaluate the line integral $\oint_C \mathbf{B} \cdot d\boldsymbol{\ell}$.

c) Verify your answer in part (b) using Stokes' theorem.

Problem 11

Let

$$\mathbf{C} = r \mathbf{a}_r + r \cos^2 \theta \mathbf{a}_\phi.$$

Consider the open surface S defined by

$$r = R, 0 \leq \theta \leq \frac{\pi}{2}, 0 \leq \phi \leq \frac{\pi}{2},$$

and let C be its boundary oriented according to the right-hand rule with the outward normal $\mathbf{a}_r$.

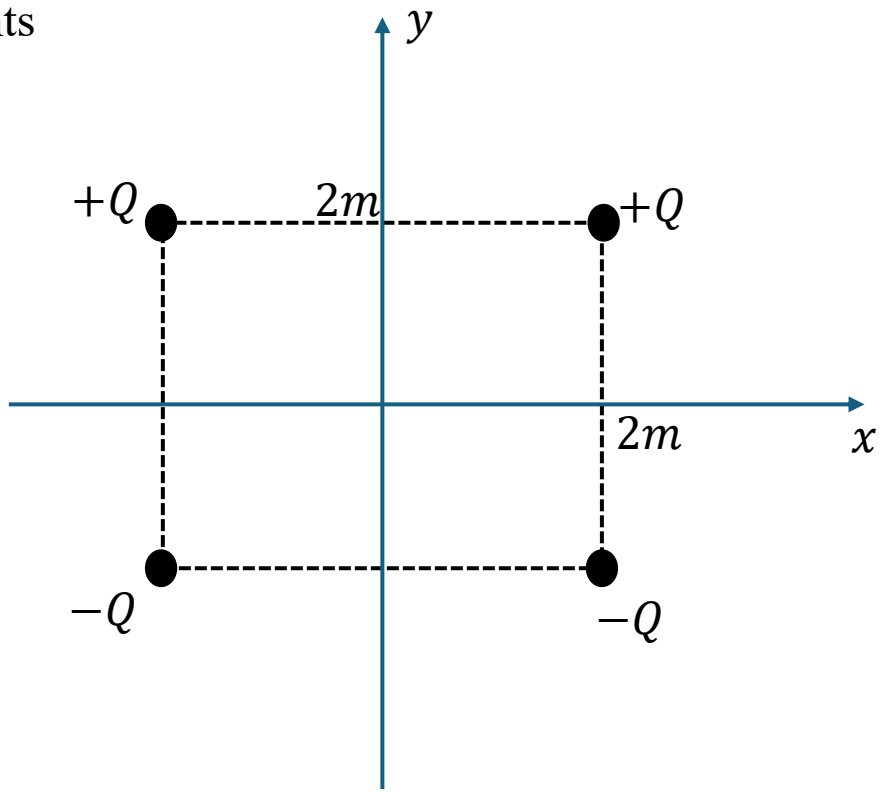
a) Calculate $\nabla \times \mathbf{C}$.

b) Evaluate the line integral $\oint_C \mathbf{C} \cdot d\boldsymbol{\ell}$.

c) Verify your answer in part (b) using Stokes' theorem.

Problem 12

Consider the following distribution of charges, find the electric field vector $\mathbf{E}$ at points at points $P(3,0,0)$ and $T(0,0,2)$. $|Q| = 6\mu C$.



Problem 13

Consider the following distribution of charges, choose the sign and magnitude of the charges such that the electric field becomes zero at $T(0, -4, 0)$.

